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# Computer Architecture (WBCS010-05)

## Lecture 1: Introduction

**Farhad Merchant**
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# Outline

- Information and rules
- Our group
- Motivation for computer architecture
- Basics of computer architecture

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# Outline

- **Information and rules**
- Our group
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# Learning Outcomes

- Describe the main concepts, design techniques, issues, and solutions across the abstraction layers of modern computer architectures
- Apply the acquired knowledge in order to solve basic design problems in computer architecture
- Describe the basic concepts of structured information storage and the techniques to get the information in and out of the computer

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# Useful Information

- Lectures
- Tutorials
- Practical
- Assignments

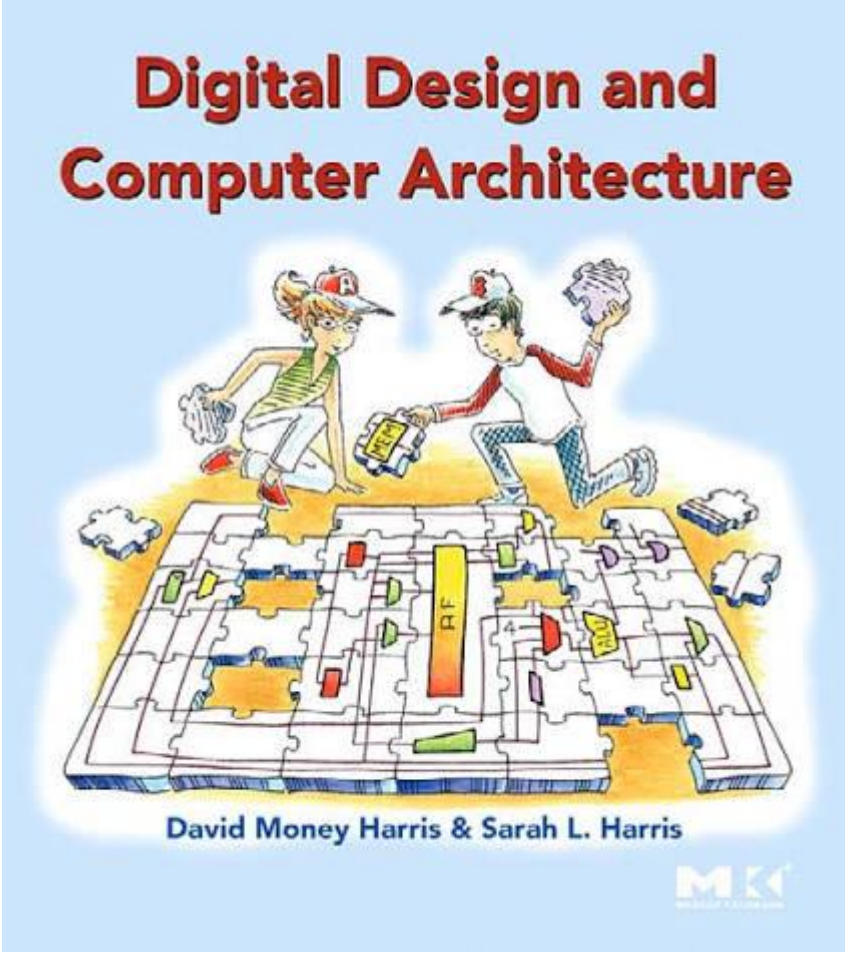
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# Course Plan

- Written exam
- Mock exams
- Grading scheme
- Bonus points

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# Text Book(s)



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# Outline

- Information and rules
- **Our group**
- Motivation for computer architecture
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# Our Group

## › Secure innovative computer architectures group

**Farhad Merchant, PhD**
Group leader
Security and computer architectures

**Simranjeet Singh, PhD**
Postdoc (Deep Tech Pioneering Fellow)
Co-founder: Ximplic Systems B.V.

**Lingyu Gong, MSc**
Security
Founding her own company

**Yuehai Chen, MSc**
Neuromrophic computing

**Yilin Li, MSc**
Security

**Devindu Arachchillage, MSc**
Neuromorphic computing

**Zhengkun Yu, MSc**
Neuromorphic computing

**Antonio Porisa, MSc**
Security, guest PhD student from Polito



**Projects & Funding (€6.7 mi. total)**

ERIC
tetramax

REACT
MSCA DOCTORAL NETWORK
NEUROMORPHIC SCIENTIFIC RESEARCH

**Deep Tech Pioneering Fellowship at RUG**

**AROP - Aachen**

BioNanoLock

**Dutch Japan Collaboration**

**Spin-offs**

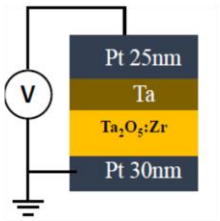
Ximplic Systems B.V.

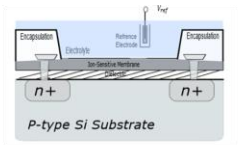
Homomorphix B.V.

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# Research Topics

## TECHNOLOGY



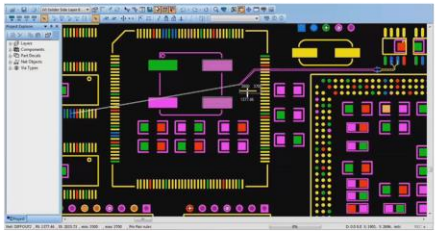

Emerging NVMs
ISFETs

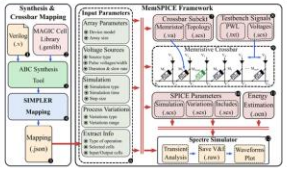


ReRAM Crossbars

Higher TRL

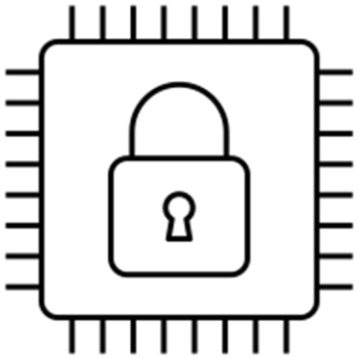
## TOOLS





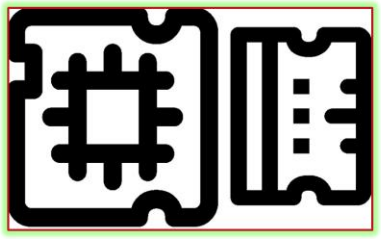
- Logic Synthesis
- Compilers
- Simulators
- Formal Verification

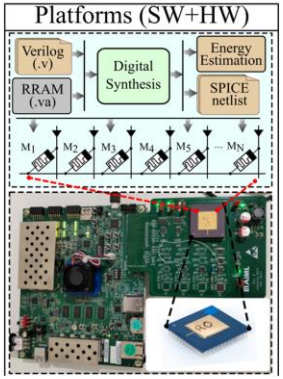
## HARDWARE SECURITY



- PUFs
- Side-Channel
- Supply-Chain
- Counterfeit ICs
- Homomorphic Encryption

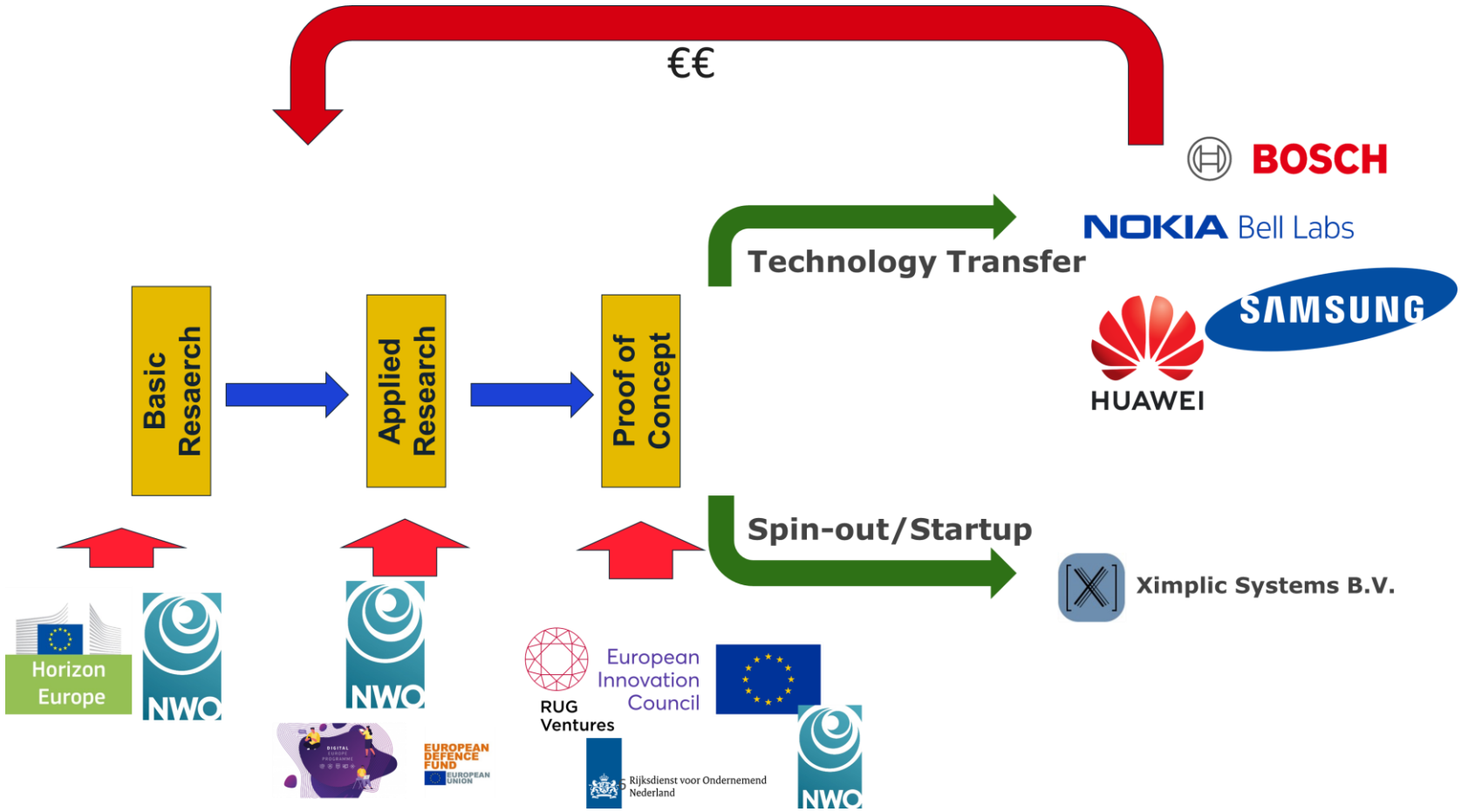
## Energy-Efficient and Secure Computer Architecture





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# Cash Flow



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# Tech Transfer History (Past)

- Double precision floating point unit
- Scientific computing
- Integrated to Morphing Machine's multicore architecture

MORPHING MACHINES

- Posit arithmetic unit
- Automotive applications
- Helped Bosch Corporate Research to evaluate performance

BOSCH

- Hardware Trojan-free RISC-V core
- Can boot seL4 operating system
- Project with Hensoldt Cyber GmbH



- Posit arithmetic unit
- Communication applications
- Helped Nokia Bell Labs to evaluate performance

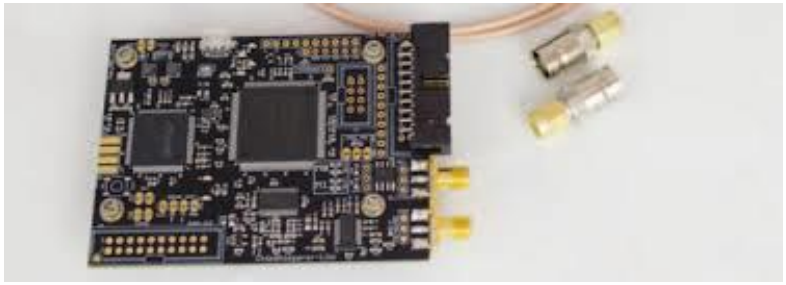
NOKIA BELL LABS

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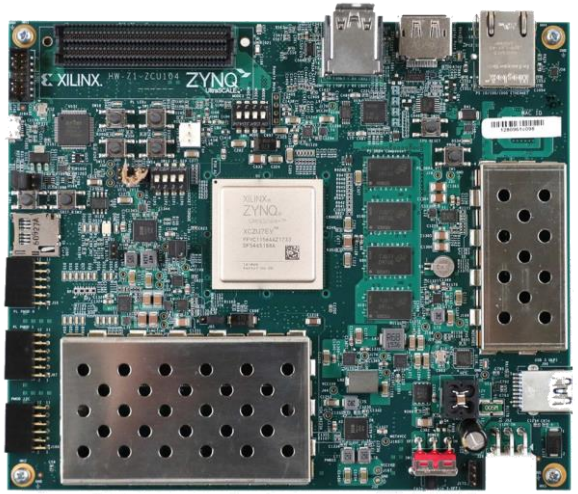
# Lab Equipment – Prototyping and Security



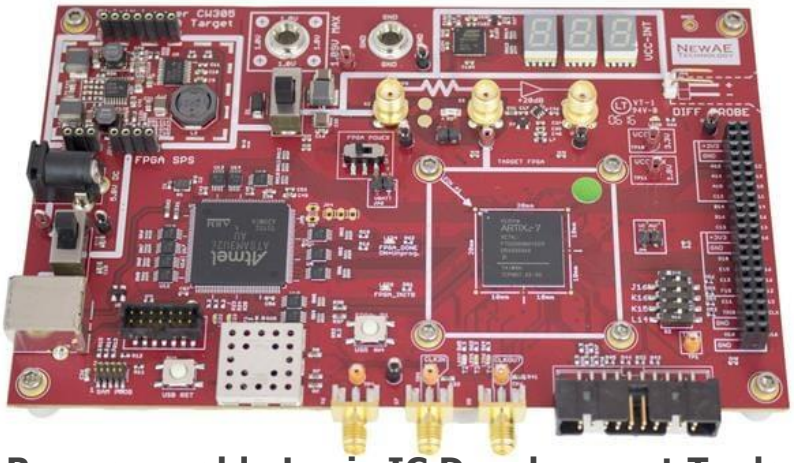

Digilent Xilinx Genesys 2 Kintex-7 FPG



ChipWhisperer Lite




Zynq Ultrascale+




Programmable Logic IC Development Tools
Artix A100 FPGA Target Board

**We offer exciting BSc and MSc projects**

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# How to Reach Out – Our Team



- Contact details
  - Farhad Merchant
  - Email: f.a.merchant@rug.nl
  - **Open door policy**
- Head TA & Team
  - Lingyu Gong
  - Lingyu.gong@rug.nl



For questions on Lectures contact me
For question on totorials,
assignments, etc contact TAs



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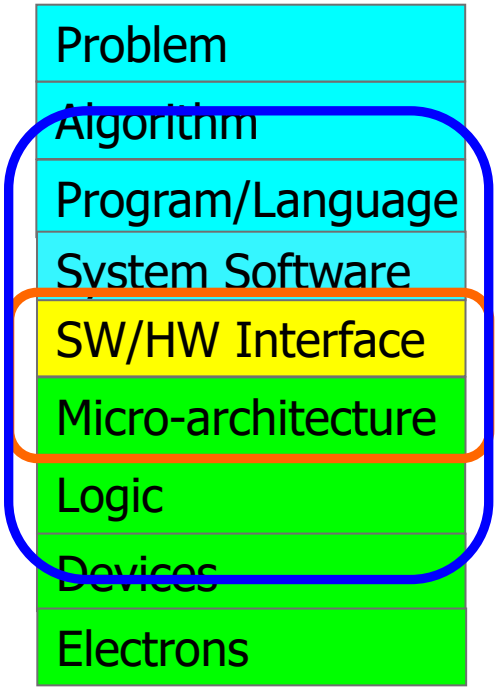
# Major Current Research Topics

- Fundamentally Robust (Secure/Reliable/Safe) Architectures
- Fundamentally Energy-Efficient Architectures
  - Memory-centric (Data-centric) Architectures
- Fundamentally Low-Latency and Predictable Architectures
- Fundamentally Intelligent and Evolving Architectures
  - ML/AI-Assisted (Data-driven) and Data-aware Architectures
- Architectures for ML/AI, Genomics, Medicine, Health, …

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# The Transformation Hierarchy

**Computer Architecture (expanded view)**



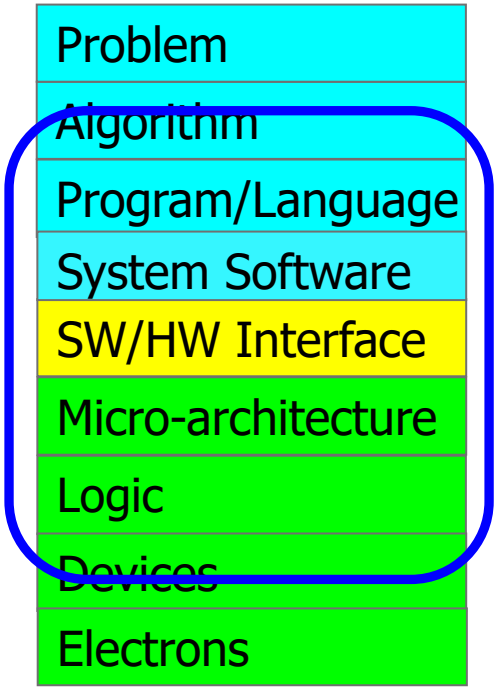

**Computer Architecture (narrow view)**

# The Expanded View

› To achieve the highest efficiency, performance, robustness:

**we must take the expanded view**

of computer architecture




**Co-design across the hierarchy: Algorithms to devices**

**Specialize as much as possible within the design goals**

**Broad research**
**spanning applications, systems, software, logic, circuits**
**with architecture at the center**

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# Teaching & Research

...

Teaching drives Research

Research drives Teaching

...

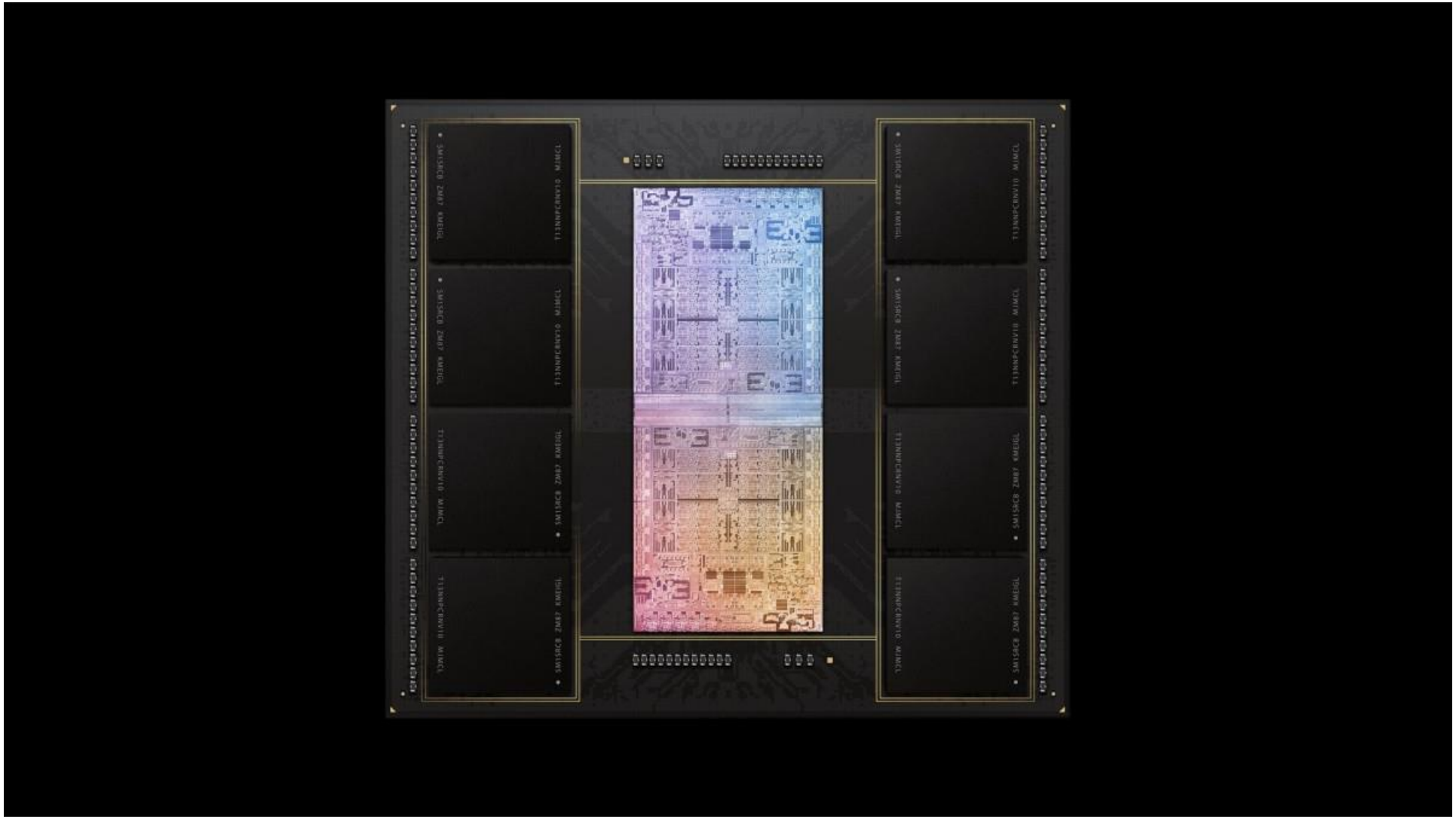
## What Will We Learn in This Course?

# How Computers Work

(from the ground up)

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# We Will Study How Something Like This Works



Apple M1 Ultra System (2022)

https://www.gsmarena.com/apple_announces_m1_ultra_with_20core_cpu_and_64core_gpu-news-53481.php

# High Level Goals

- Understand the basics
- Understand the principles (of design)
- Understand the precedents
- Based on such understanding:
  - learn how a modern computer works underneath
  - evaluate tradeoffs of different designs and ideas
  - implement a principled design (a simple microprocessor)
  - learn to systematically debug increasingly complex systems
  - Hopefully enable you to develop novel, out-of-the-box designs
- The focus is on basics, principles, precedents, and how to use them to create/implement good designs

# Computer Science?

- Because you are here for a Computer Science degree!
- Regardless of your future direction, learning the principles of computer architecture will be useful to
  - design better hardware
  - design better software
  - design better systems
  - make better tradeoffs in design
  - understand why computers behave the way they do
  - solve problems better
  - think “in parallel”
  - think critically
  - …

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# Learning & Exam

- We will enable you to learn + prepare you for the exam
- My suggestions:
  - focus on understanding, learning, mastering the material
  - reinforce problem solving skills with homeworks/assignments
  - do **not** worry about the exam while listening to lectures
    - most of you will pass this course
- We will release a lot of material to help you with the exam
  - Problem solving sessions - tutorials
  - Exam guidance
  - Mock exams will be made available for you

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# In Conclusion…

- Learning is for life (never ends)
- Exam study is until you pass (ends, hopefully January 2027)

**Focus on**

**learning and scholarship**

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# How to Approach this Course

Learning experience

Long-term tradeoff analysis

Critical thinking & decision making

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# Learning Style

Find and choose the learning style that works best for you

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# What Will We Learn in This Course?

# How Computers Work

(from the ground up)

# What Will We Learn in This Course?

# And Why We Care

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# Critical Questions

Why Do We Have Computers?

Why Do We Do Computing?



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# Why Do We Do Computing?

## To Solve Problems

# Why Do We Do Computing?

## To Gain Insight

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More..

# To Enable a Better Life & Future

## How Does a Computer Solve Problems?



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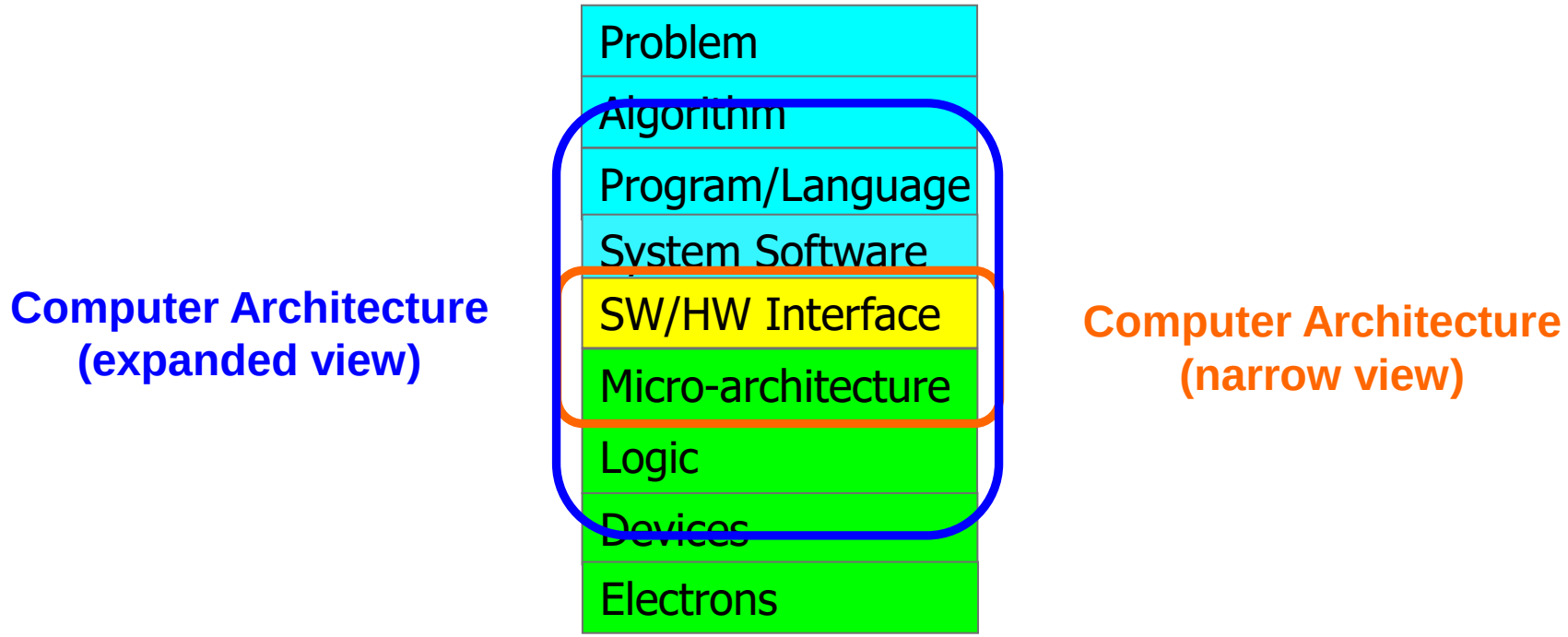
How Does a Computer Solve Problems?

# Orchestrating Electrons

How Do Problems Get Solved by Electrons?

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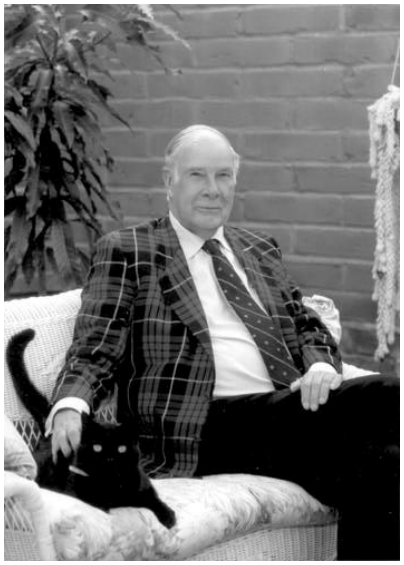
# The Transformation Hierarchy



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# Levels of Transformation

"The purpose of computing is [to gain] insight" (*Richard Hamming*)
*We gain and generate insight by solving problems*
*How do we ensure problems are solved by electrons?*

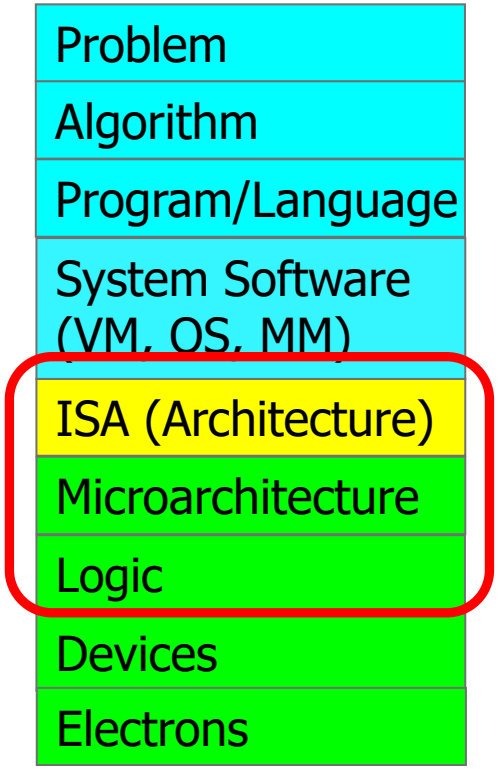


Algorithm

Step-by-step procedure that is **guaranteed to terminate** where **each step is precisely stated** and **can be carried out by a computer**

- **Finiteness**
- **Definiteness**
- **Effective computability**

Many algorithms for the same problem




ISA (Instruction Set Architecture)

Interface/contract between SW and HW.

What the programmer assumes hardware will satisfy.

Microarchitecture
An implementation of the ISA

Digital logic circuits
Building blocks of micro-arch (e.g., gates)

# Computer Architecture

- is the science and art of designing computing platforms (hardware, interface, system SW, and programming model)
- to achieve a set of design goals
  - E.g., highest performance on earth on workloads X, Y, Z
  - E.g., longest battery life at a form factor that fits in your pocket with cost < $$$ CHF  Euros
  - E.g., best average performance across all known workloads at the best performance/cost ratio
  - …
  - Designing a supercomputer is different from designing a smartphone → But, many fundamental principles are similar

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# Different Platforms, Different Goals






Source: http://www.sia-online.org (semiconductor industry association)

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# Different Platforms, Different Goals



Source: https://iq.intel.com/5-awesome-uses-for-drone-technology/

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# Different Platforms, Different Goals




Source: https://taxistartup.com/wp-content/uploads/2015/03/UK-Self-Driving-Cars.jpg

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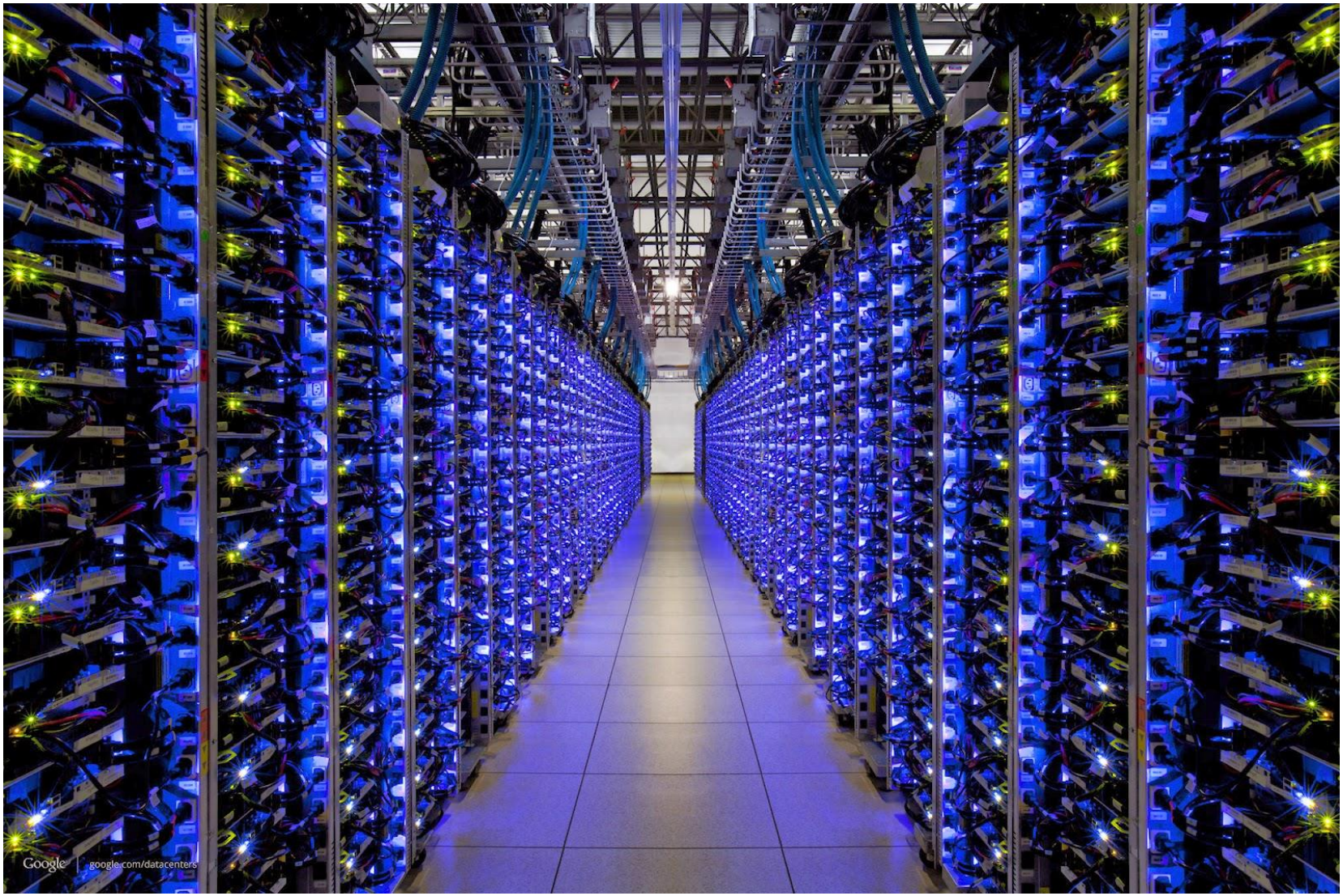
# Different Platforms, Different Goals



Source: http://sm.pcmag.com/pcmag_uk/photo/g/google-self-driving-car-the-guts/google-self-driving-car-the-guts_dwx8.jpg

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# Different Platforms, Different Goals




Source: http://datacentervoice.com/wp-content/uploads/2015/10/data-center.jpg

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# Different Platforms, Different Goals




Source: https://fossbytes.com/wp-content/uploads/2015/06/Supercomputer-TIANHE2-china.jpg

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# Different Platforms, Different Goals



Source: https://www.itmagazine.ch/artikel/72401/Fugaku_Der_schnellste_Supercomputer_der_Welt.html

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# Different Platforms, Different Goals

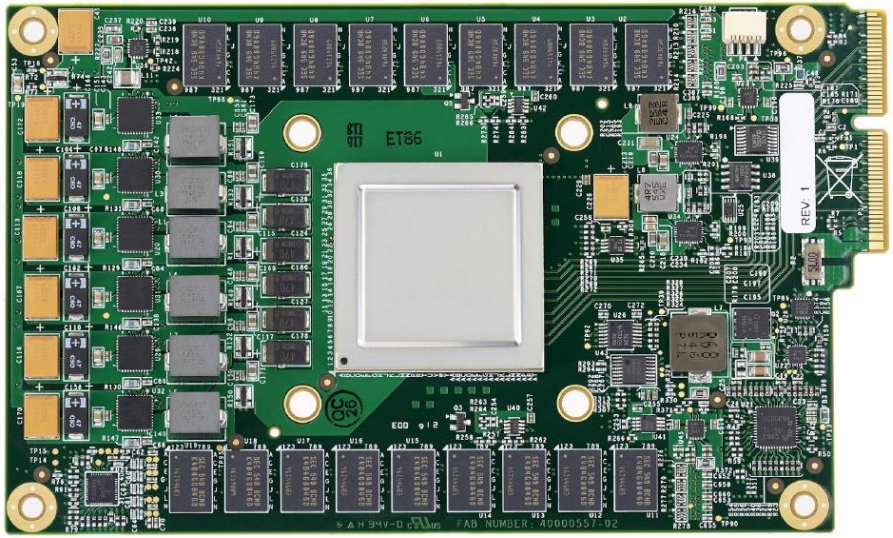


**Figure 3.** TPU Printed Circuit Board. It can be inserted in the slot for an SATA disk in a server, but the card uses PCIe Gen3 x16.

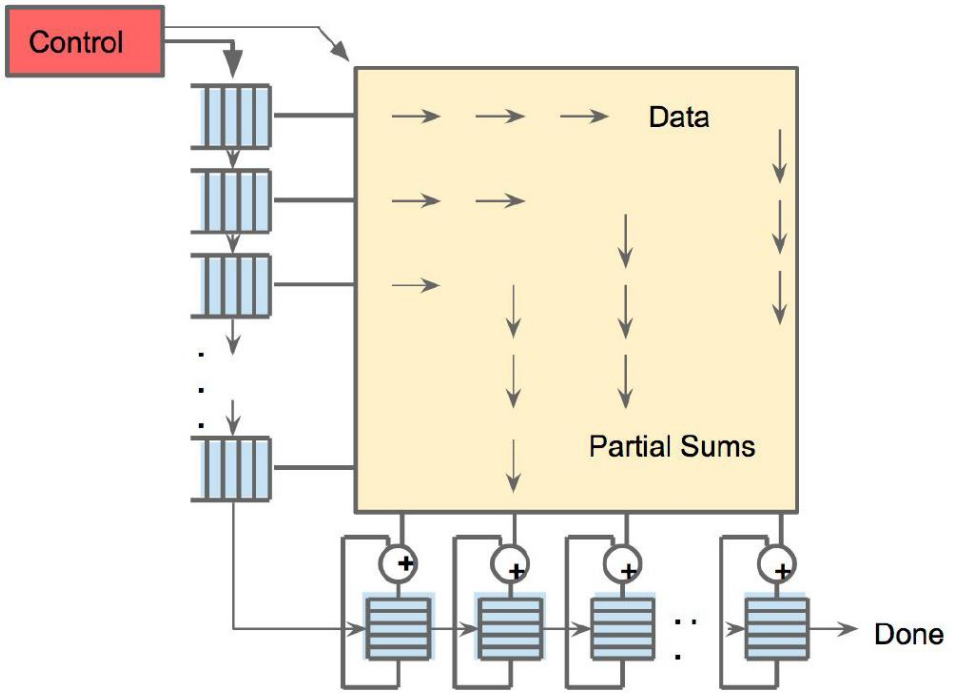
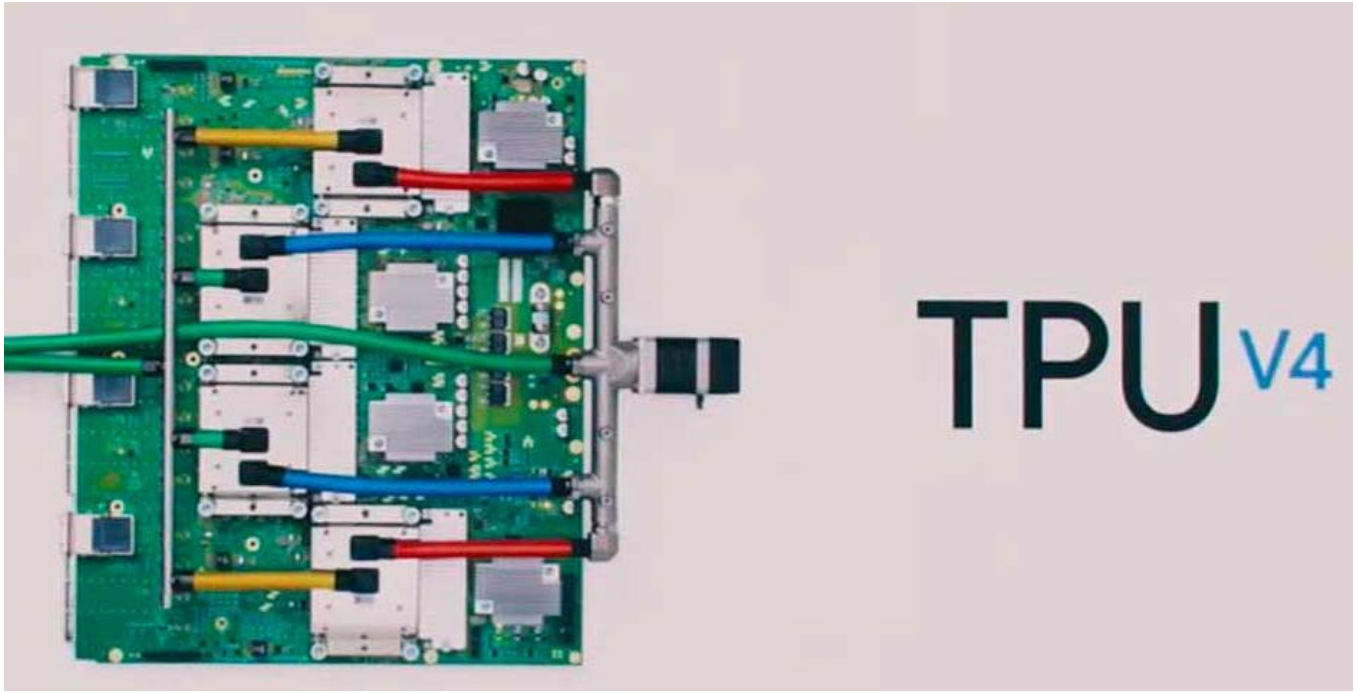



**Figure 4.** Systolic data flow of the Matrix Multiply Unit. Software has the illusion that each 256B input is read at once, and they instantly update one location of each of 256 accumulator RAMs.

Jouppi et al., "In-Datacenter Performance Analysis of a Tensor Processing Unit", ISCA 2017.

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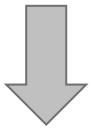
# Different Platforms, Different Goals




New ML applications (vs. TPU3):

- Computer vision
- Natural Language Processing (NLP)
- Recommender system
- Reinforcement learning that plays Go

250 TFLOPS per chip in 2021
vs 90 TFLOPS in TPU3



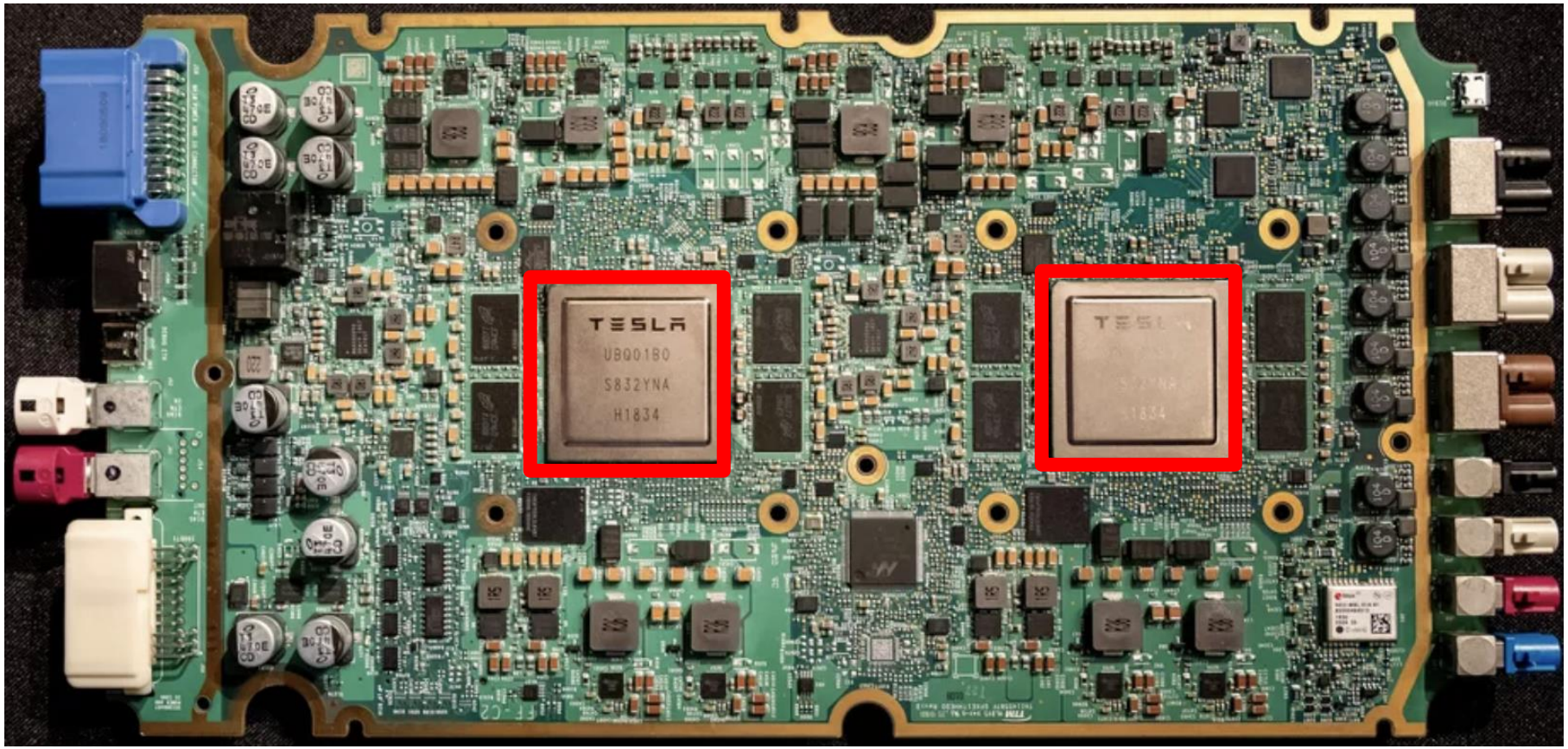
1 ExaFLOPS per board

https://spectrum.ieee.org/tech-talk/computing/hardware/heres-how-googles-tpu-v4-ai-chip-stacked-up-in-training-tests

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# Different Platforms, Different Goals

- ML accelerator: 260 mm$^2$, 6 billion transistors, 600 GFLOPS GPU, 12 ARM 2.2 GHz CPUs.
- Two redundant chips for better safety.

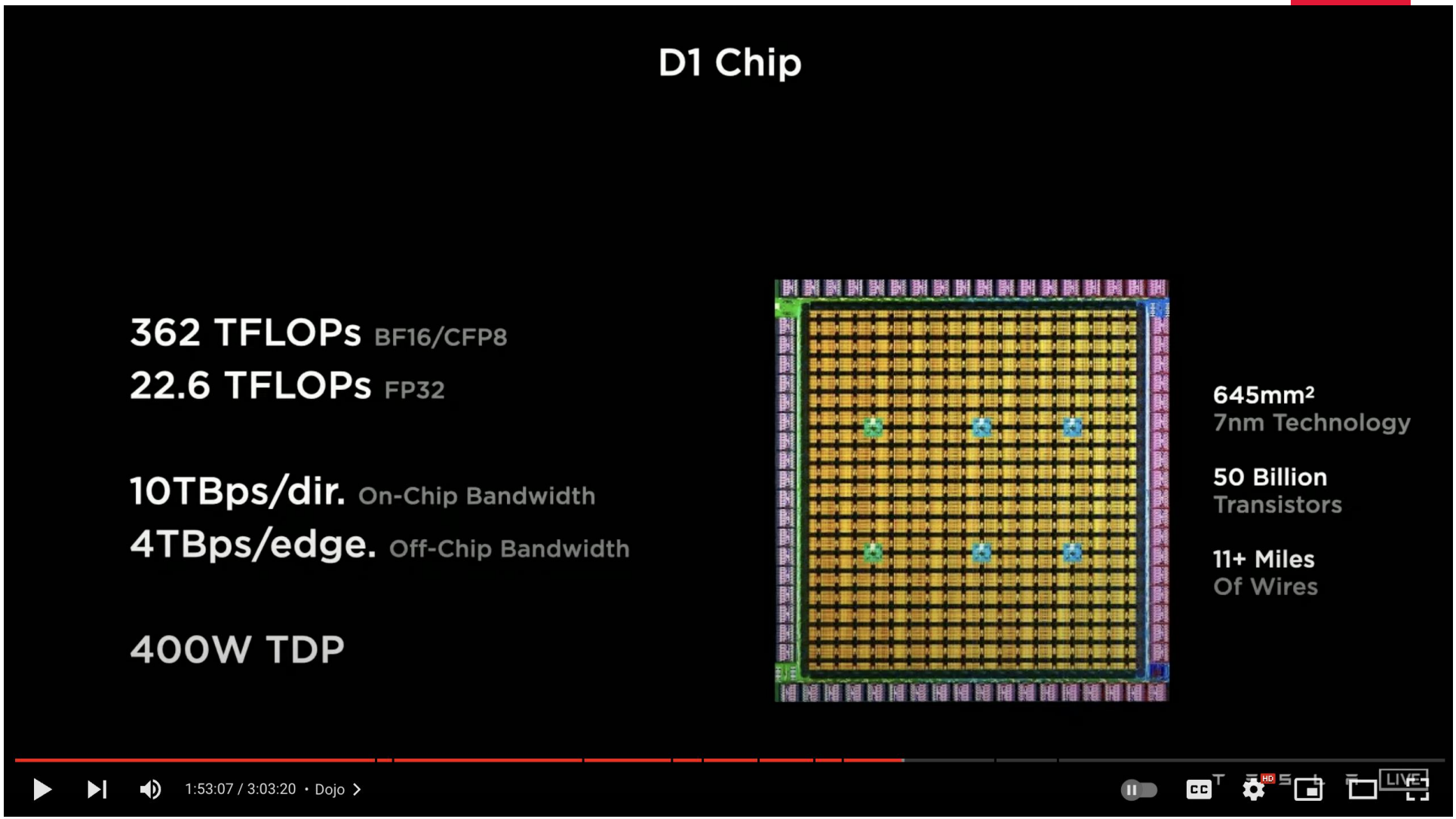



https://www.youtube.com/watch?v=joz4FweCy4M

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# Different Platforms, Different Goals

- Tesla Dojo Chip & System

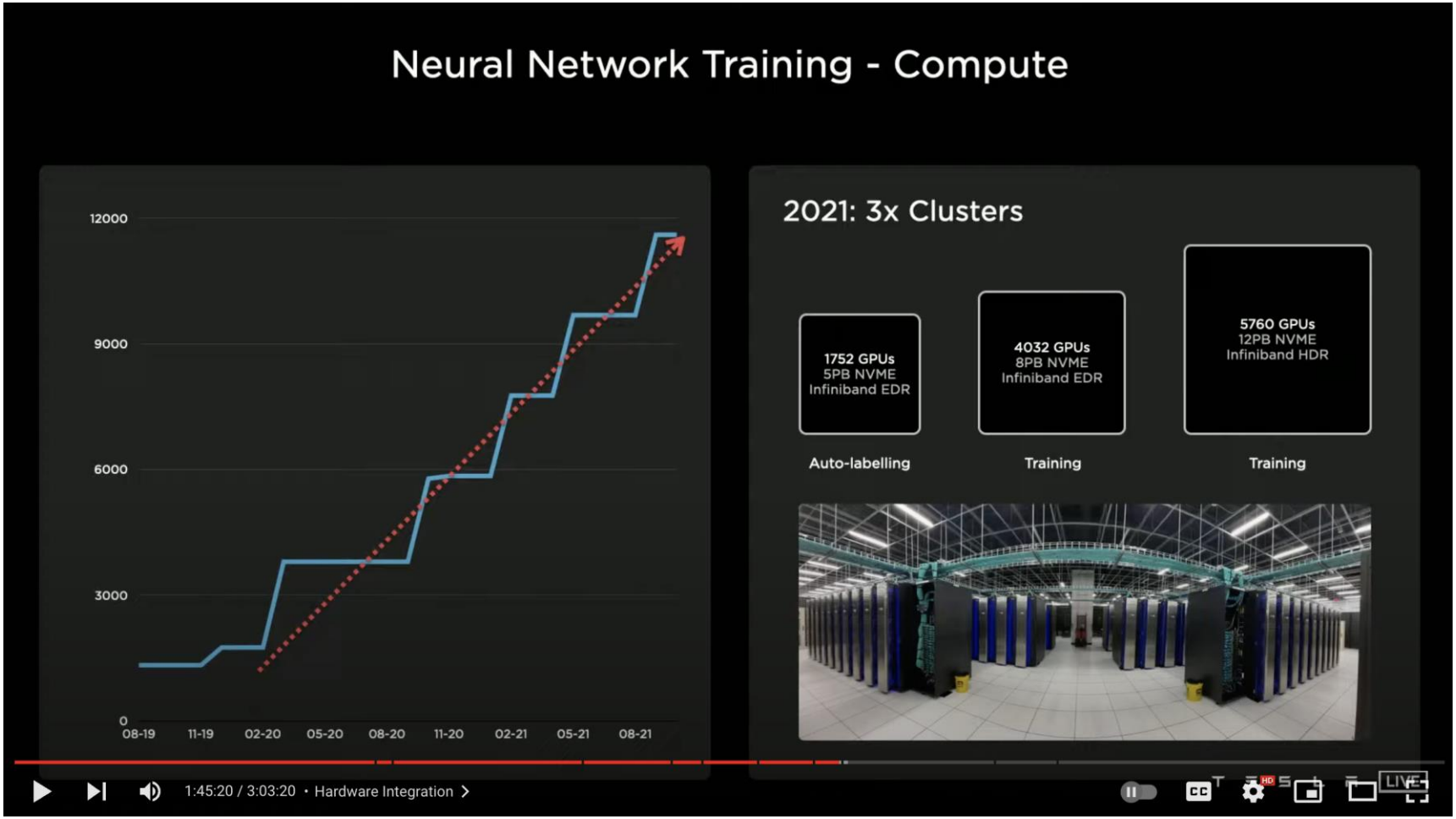



https://www.youtube.com/watch?v=j0z4FweCy4M&t=6340s

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# Different Platforms, Different Goals

- Tesla Dojo Chip & System

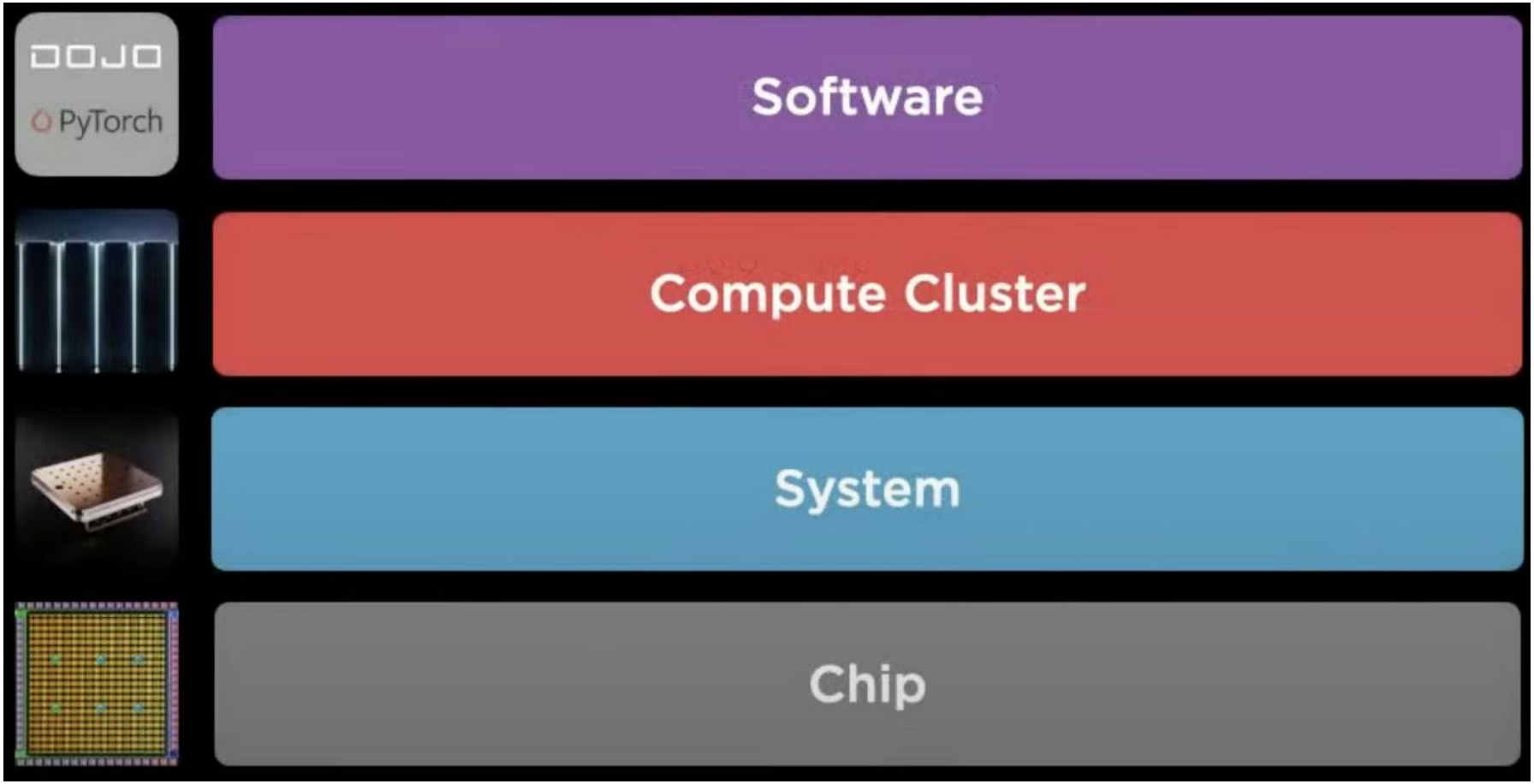



https://www.youtube.com/watch?v=j0z4FweCy4M&t=6340s

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# Different Platforms, Different Goals

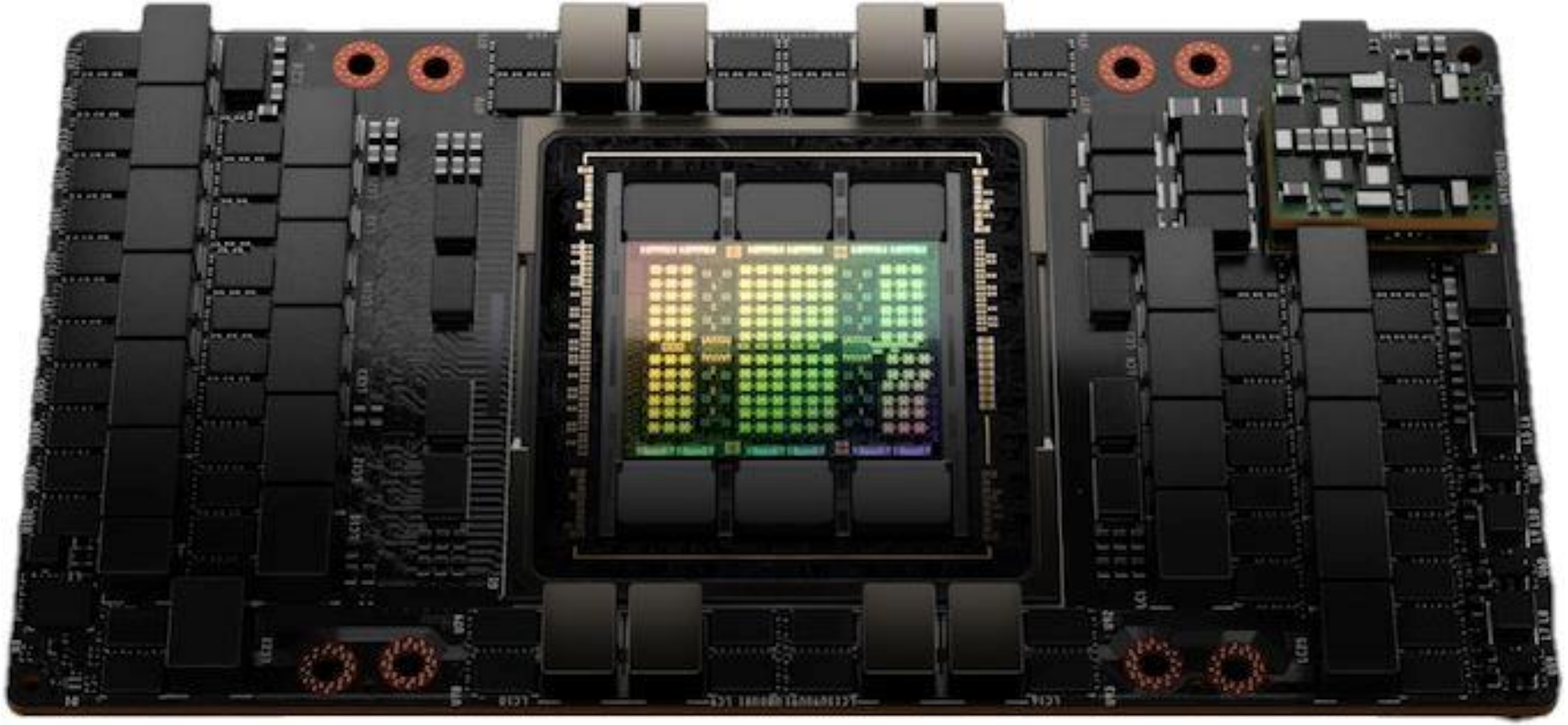
- Tesla Dojo Chip & System



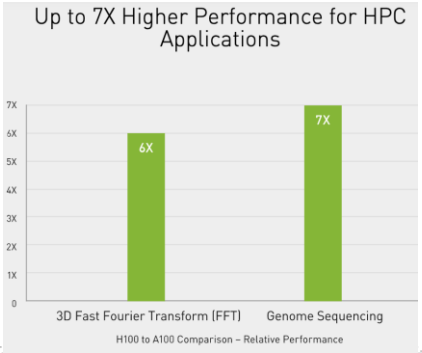

https://www.youtube.com/watch?v=j0z4FweCy4M&t=6340s

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# Different Platforms, Different Goals



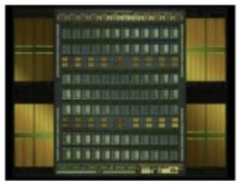
NVIDIA is claiming a **7x improvement** in dynamic programming algorithm (**DPX instructions**) performance on a single H100 versus naïve execution on an A100.




https://www.nvidia.com/en-us/data-center/h100/

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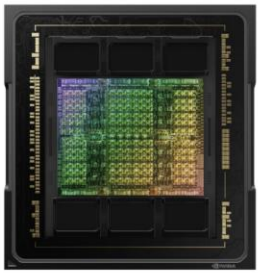
# Evolution of Recent GPUs (I)



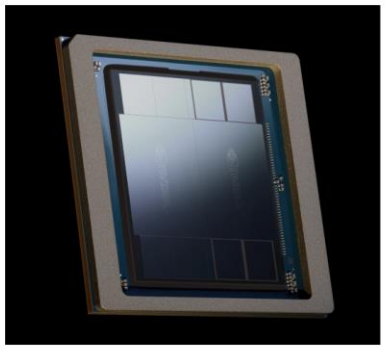
**Volta**
>21 billion transistors
815mm^2
TSMC 12nm FFN



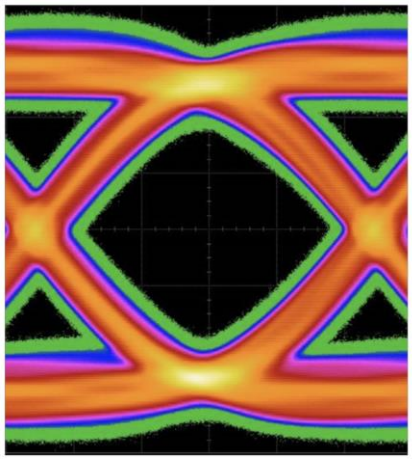
**Ampere**
>54 billion transistors
826 mm^2
TSMC N7



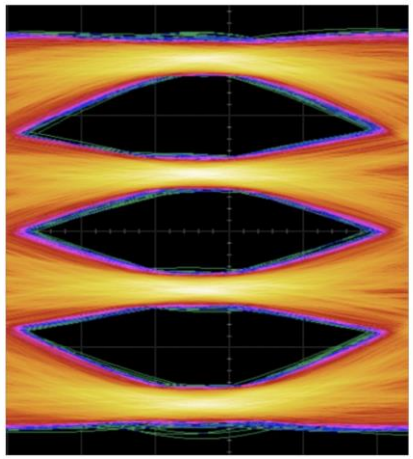
**Hopper**
>80 billion transistors
814 mm^2
TSMC 4N



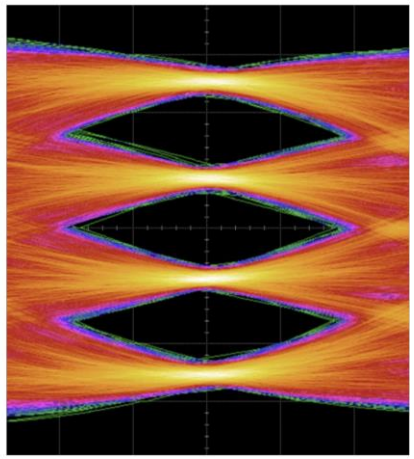
**Blackwell**
>208 billion transistors
>1600 mm^2
TSMC 4NP



**Ampere | NVLink3**
12 NVLinks | 50GB/s each
x4@50Gbps-NRZ
600GB/s total



**Hopper | NVLink4**
18 NVLinks | 50GB/s each
x2@100Gbps-PAM4
900GB/s total



**Blackwell | NVLink5**
18 NVLinks | 100GB/s each
x2@200Gbps-PAM4
1800GB/s total

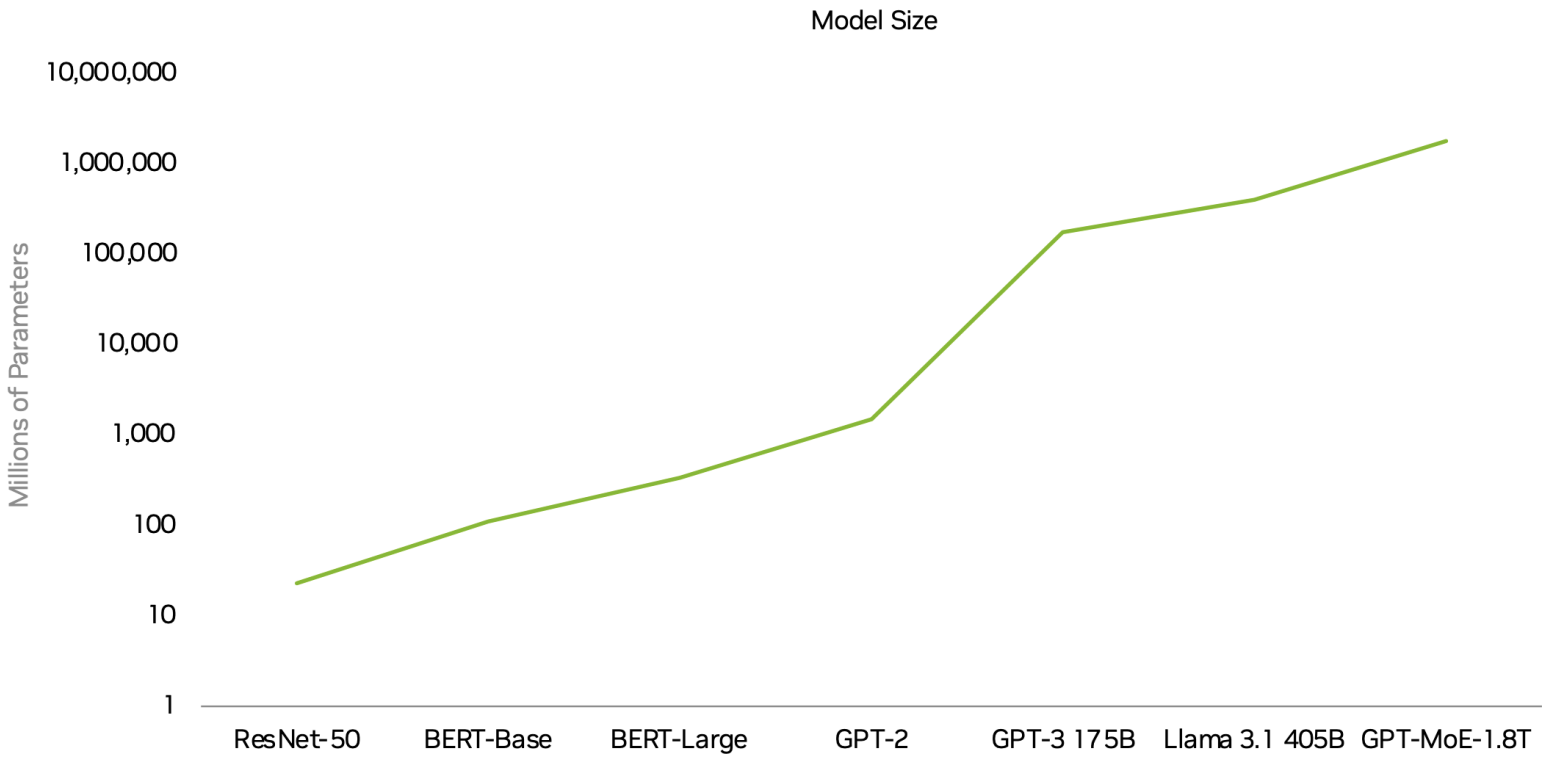
NVIDIA Blackwell Platform: Advancing Generative AI and Accelerated Computing, Hot Chips 2024

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# Multiple GPUs to Tackle Large Workloads

## AI Models Growing Exponentially

Need for multi-GPU inference at scale

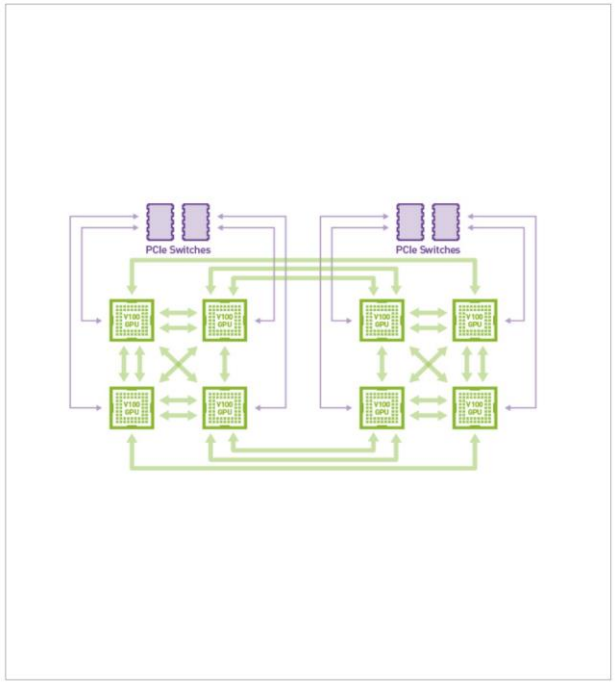



**New Capabilities | Trillions of Parameters | 70,000X Growth in a Decade**

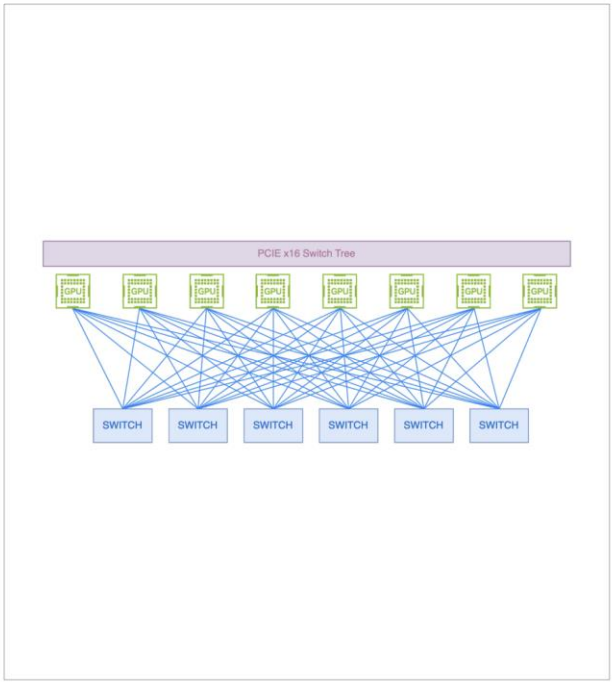
NVIDIA Blackwell Platform: Advancing Generative AI and Accelerated Computing, Hot Chips 2024

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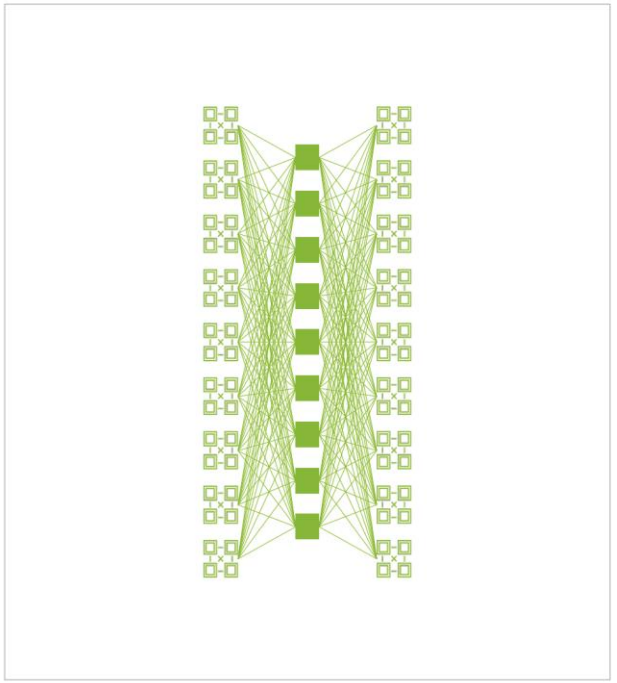
# Evolution of Recent GPUs (II)




**2016**
Hybrid Cube Mesh NVLink technology




**2022**
3rd Gen NVLink Switch
All-to-all connection among NVLink domain of 8 GPU

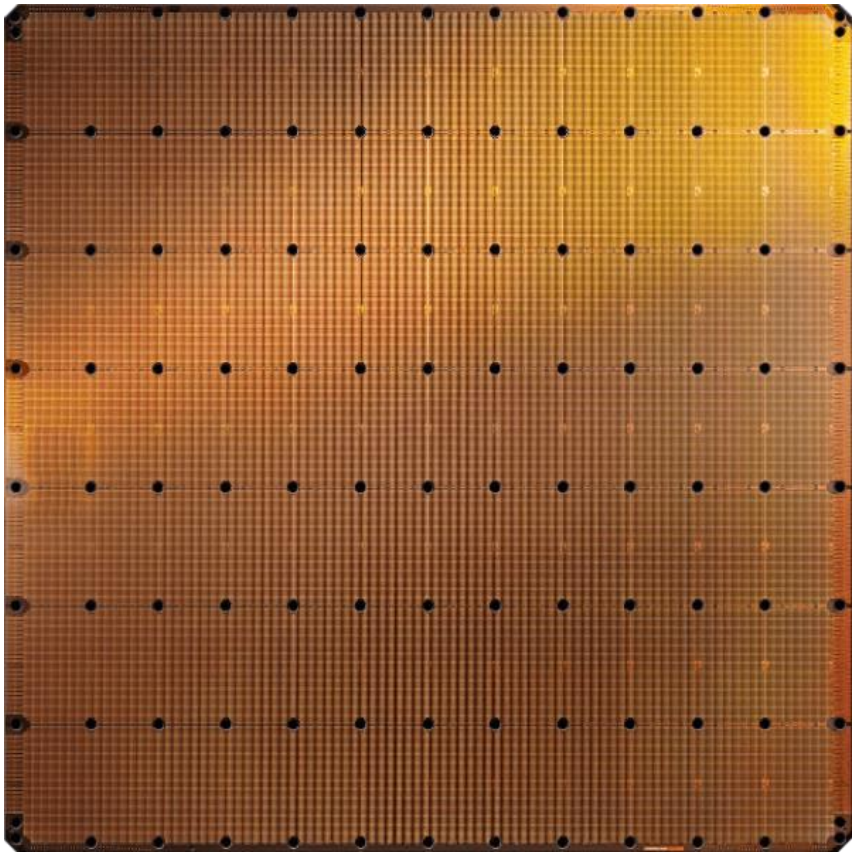


**2024**
4th Gen NVLink Switch Chip
All-to-all connection among NVLink domain of 72 GPU

NVIDIA Blackwell Platform: Advancing Generative AI and Accelerated Computing, Hot Chips 2024

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# Different Platforms, Different Goals



- The largest ML accelerator chip (2021)
- 850,000 cores




**Cerebras WSE-2**
2.6 Trillion transistors
46,225 mm$^2$

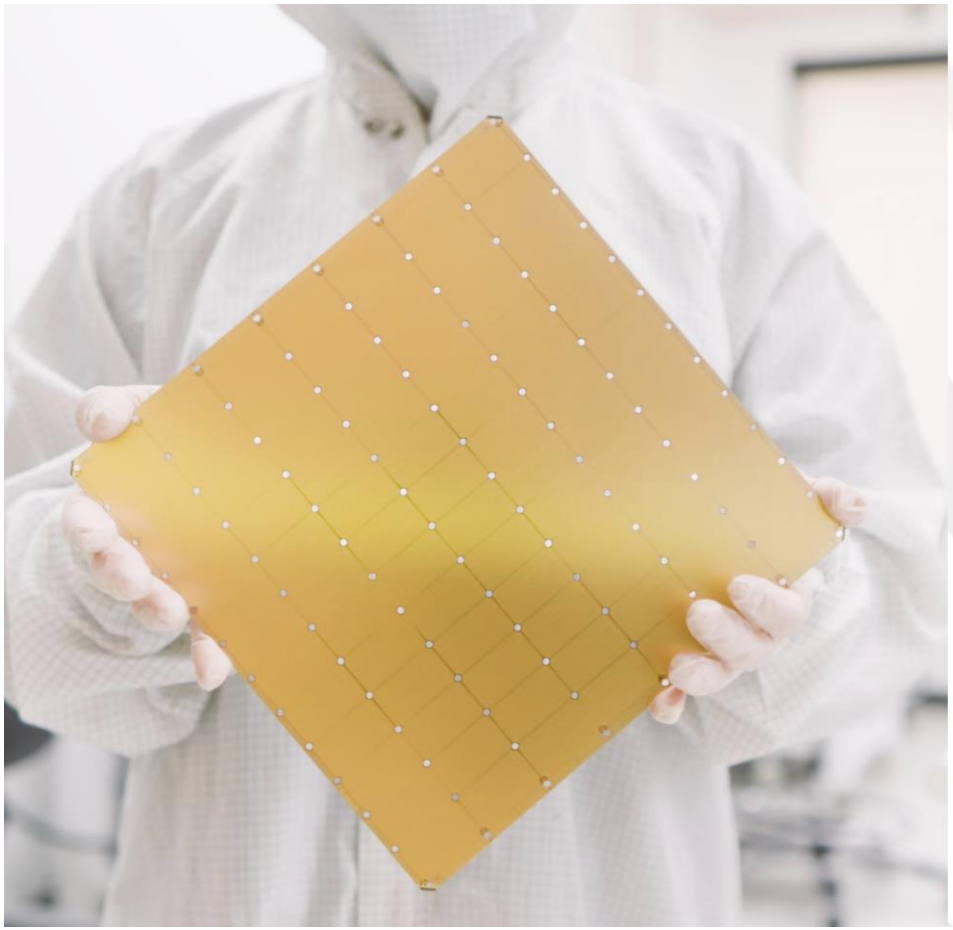
**Largest GPU**
54.2 Billion transistors
826 mm$^2$

https://www.anandtech.com/show/14758/hot-chips-31-live-blogs-cerebras-wafer-scale-deep-learning

https://www.cerebras.net/cerebras-wafer-scale-engine-why-we-need-big-chips-for-deep-learning/

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# Cerebras's Wafer Scale Engine-3 (2023)



**Cerebras Wafer-Scale Engine**

**The largest chip ever produced**

**46,225 mm²** silicon

**4 trillion** transistors

**900,000** AI cores

**125 Petaflops** of AI compute

**44 Gigabytes** of on-chip memory

**21 PByte/s** memory bandwidth

**214 Pbit/s** fabric bandwidth

**5nm** TSMC process

Wafer-Scale AI: Enabling Unprecedented AI Compute Performance, Hot Chips 2024

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# Different Platforms, Different Goals

Mohammed Alser, Zülal Bingöl, Damla Senol Cali, Jeremie Kim, Saugata Ghose, Can Alkan, Onur Mutlu
**"Accelerating Genome Analysis: A Primer on an Ongoing Journey"** IEEE Micro, August 2020.



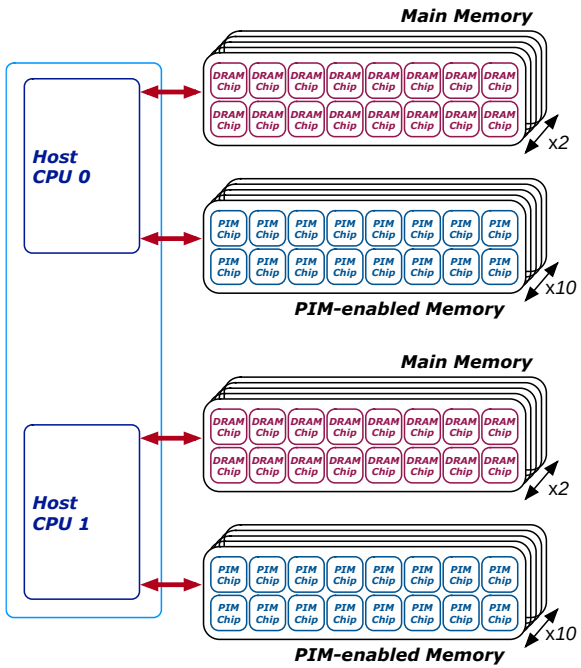
**Accelerating Genome Analysis: A Primer on an Ongoing Journey**
Sept.-Oct. 2020, pp. 65-75, vol. 40
DOI Bookmark: 10.1109/MM.2020.3013728

**FPGA-Based Near-Memory Acceleration of Modern Data-Intensive Applications**
July-Aug. 2021, pp. 39-48, vol. 41
DOI Bookmark: 10.1109/MM.2021.3088396



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# Different Platforms, Different Goals



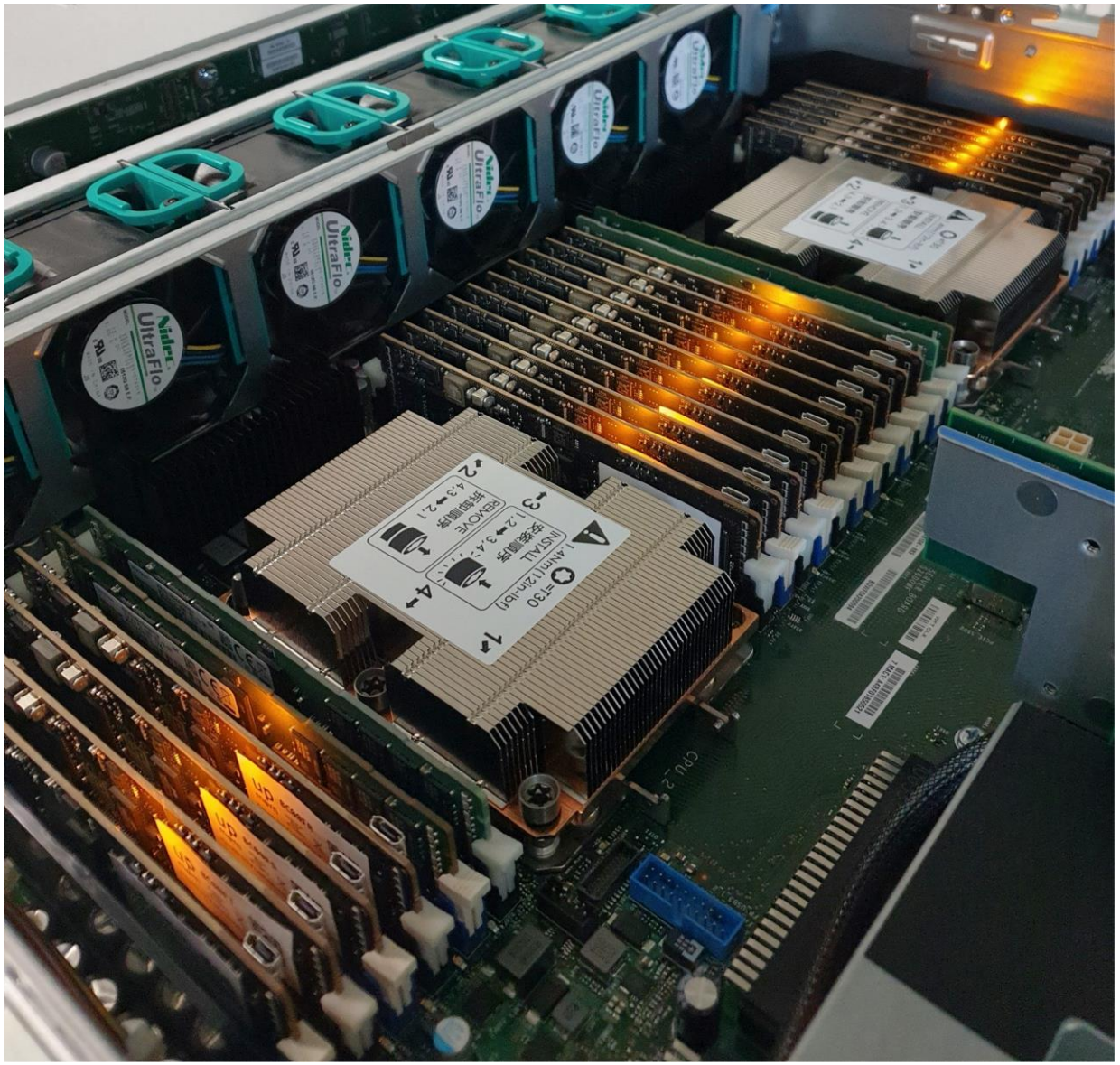

**Benchmarking a New Paradigm: An Experimental Analysis of a Real Processing-in-Memory Architecture**

JUAN GÓMEZ-LUNA, ETH Zürich, Switzerland
IZZAT EL HAJJ, American University of Beirut, Lebanon
IVAN FERNANDEZ, ETH Zürich, Switzerland and University of Malaga, Spain
CHRISTINA GIANNOULA, ETH Zürich, Switzerland and NTUA, Greece
GERALDO F. OLIVEIRA, ETH Zürich, Switzerland
ONUR MUTLU, ETH Zürich, Switzerland

Many modern workloads, such as neural networks, databases, and graph processing, are fundamentally memory-bound. For such workloads, the data movement between main memory and CPU cores imposes a significant overhead in terms of both latency and energy. A major reason is that this communication happens through a narrow bus with high latency and limited bandwidth, and the low data reuse in memory-bound workloads is insufficient to amortize the cost of main memory access. Fundamentally addressing this *data movement bottleneck* requires a paradigm where the memory system assumes an active role in computing by integrating processing capabilities. This paradigm is known as *processing-in-memory (PIM)*.

Recent research explores different forms of PIM architectures, motivated by the emergence of new 3D-stacked memory technologies that integrate memory with a logic layer where processing elements can be easily placed. Past works evaluate these architectures in simulation or, at best, with simplified hardware prototypes. In contrast, the UPMEM company has designed and manufactured the first publicly-available real-world PIM architecture. The UPMEM PIM architecture combines traditional DRAM memory arrays with general-purpose in-order cores, called *DRAM Processing Units (DPUs)*, integrated in the same chip.

This paper provides the first comprehensive analysis of the first publicly-available real-world PIM architecture. We make two key contributions. First, we conduct an experimental characterization of the UPMEM-based PIM system using microbenchmarks to assess various architecture limits such as compute throughput and memory bandwidth, yielding new insights. Second, we present *PrIM (Processing-In-Memory benchmarks)*, a benchmark suite of 16 workloads from different application domains (e.g., dense/sparse linear algebra, databases, data analytics, graph processing, neural networks, bioinformatics, image processing), which we identify as memory-bound. We evaluate the performance and scaling characteristics of PrIM benchmarks on the UPMEM PIM architecture, and compare their performance and energy consumption to their state-of-the-art CPU and GPU counterparts. Our extensive evaluation conducted on two real UPMEM-based PIM systems with 640 and 2,556 DPUs provides new insights about suitability of different workloads to the PIM system, programming recommendations for software designers, and suggestions and hints for hardware and architecture designers of future PIM systems.

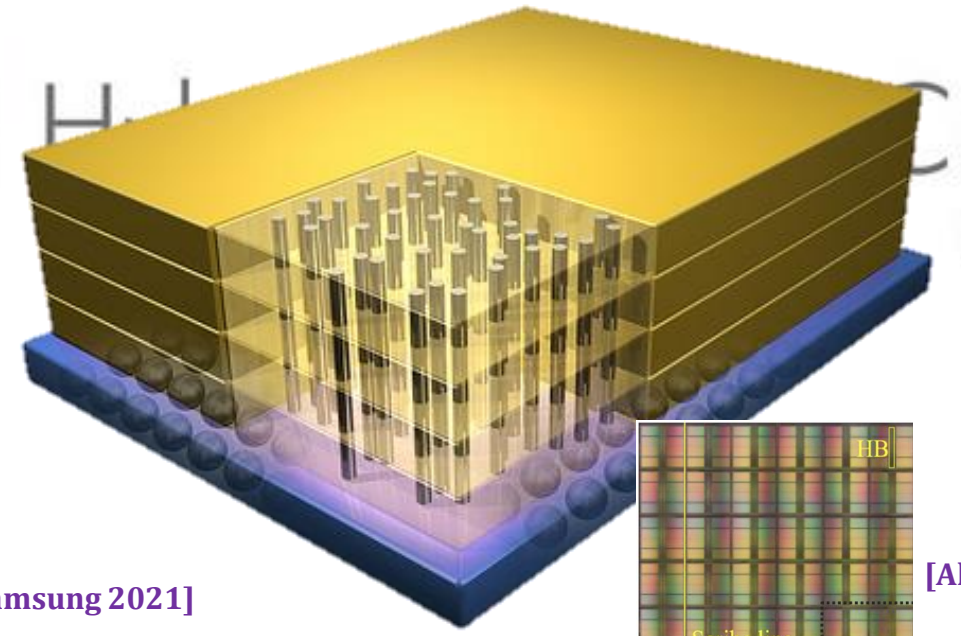



https://arxiv.org/pdf/2105.03814.pdf

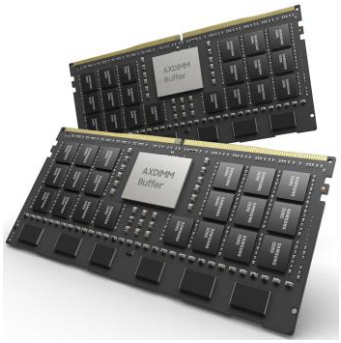
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# Processing-in-Memory Systems (2022)

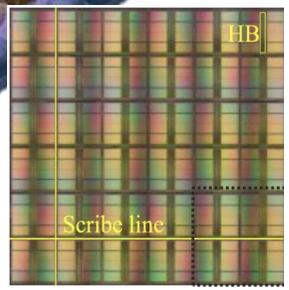




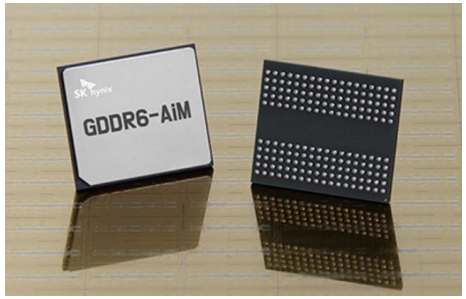


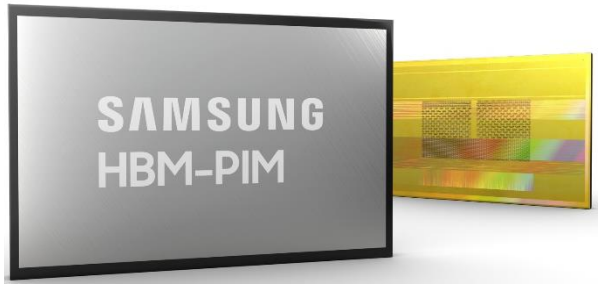
[Samsung 2021]



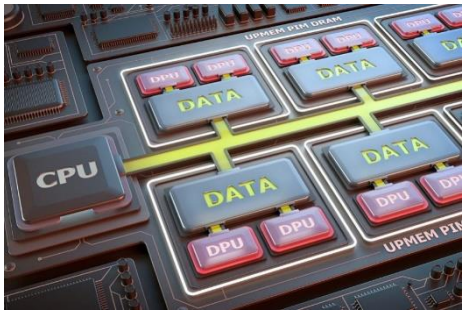

[Alibaba 2022]




[SK Hynix 2022]




[Samsung 2021]




[UPMEM 2019]

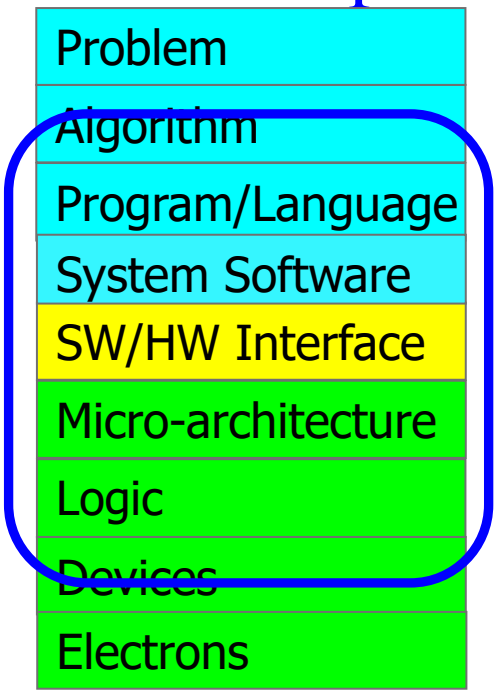
And, many other experimental chips and startups

# Axiom

To achieve the highest energy efficiency and performance:

**we must take the expanded view**

of computer architecture




**Co-design across the hierarchy: Algorithms to devices**

**Specialize as much as possible within the design goals**

# Topics

- What is the Computer Architecture course about?
  - What does a computer do?
  - How does a computer work?
  - What is a computer system?
  - What are the main components of a computer system?
- What will you learn?
  - The hardware of a computer
  - The relationships with other courses

# Why Study Computer Architecture?

- What is Computer Architecture?
  - The science and art of designing, selecting, and interconnecting hardware components and designing the hardware/software interface to create a computing system that meets functional, performance, energy consumption, cost, and other specific goals.

# Why Study Computer Architecture?

- Enable better systems: make computers faster, cheaper, smaller, more reliable, …
  - By exploiting advances and changes in underlying technology/circuits
- Enable new applications
  - Life-like 3D visualization 20 years ago? Virtual reality?
  - Self-driving cars?
  - Personalized genomics? Personalized medicine?
- Enable better solutions to problems
  - Software innovation is built on trends and changes in computer architecture
    - > 50% performance improvement per year has enabled this innovation
- Understand why computers work the way they do

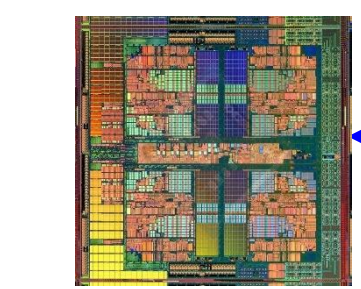
# Computer Architecture Today (I)

- Today is a very exciting time to study computer architecture
- Industry is in a large paradigm shift (to novel architectures) – many different potential system designs possible
- Many difficult problems *motivating* and *caused by* the shift
  - Huge hunger for data and new data-intensive applications
  - Power/energy/thermal constraints
  - Complexity of design
  - Difficulties in technology scaling
  - Memory bottleneck
  - Reliability problems
  - Programmability problems
  - Security and privacy issues
- No clear, definitive answers to these problems

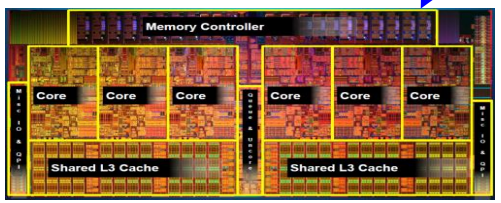
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# Computer Architecture Today (II)

- Computing landscape is very different from 10-20 years ago
- Applications and technology both demand novel architectures



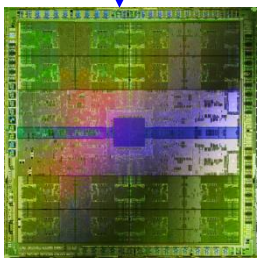
Heterogeneous Processors and Accelerators



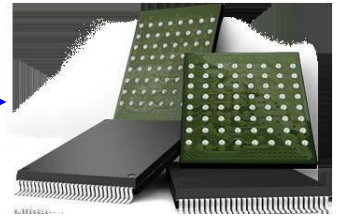



Hybrid Main Memory





General Purpose GPUs

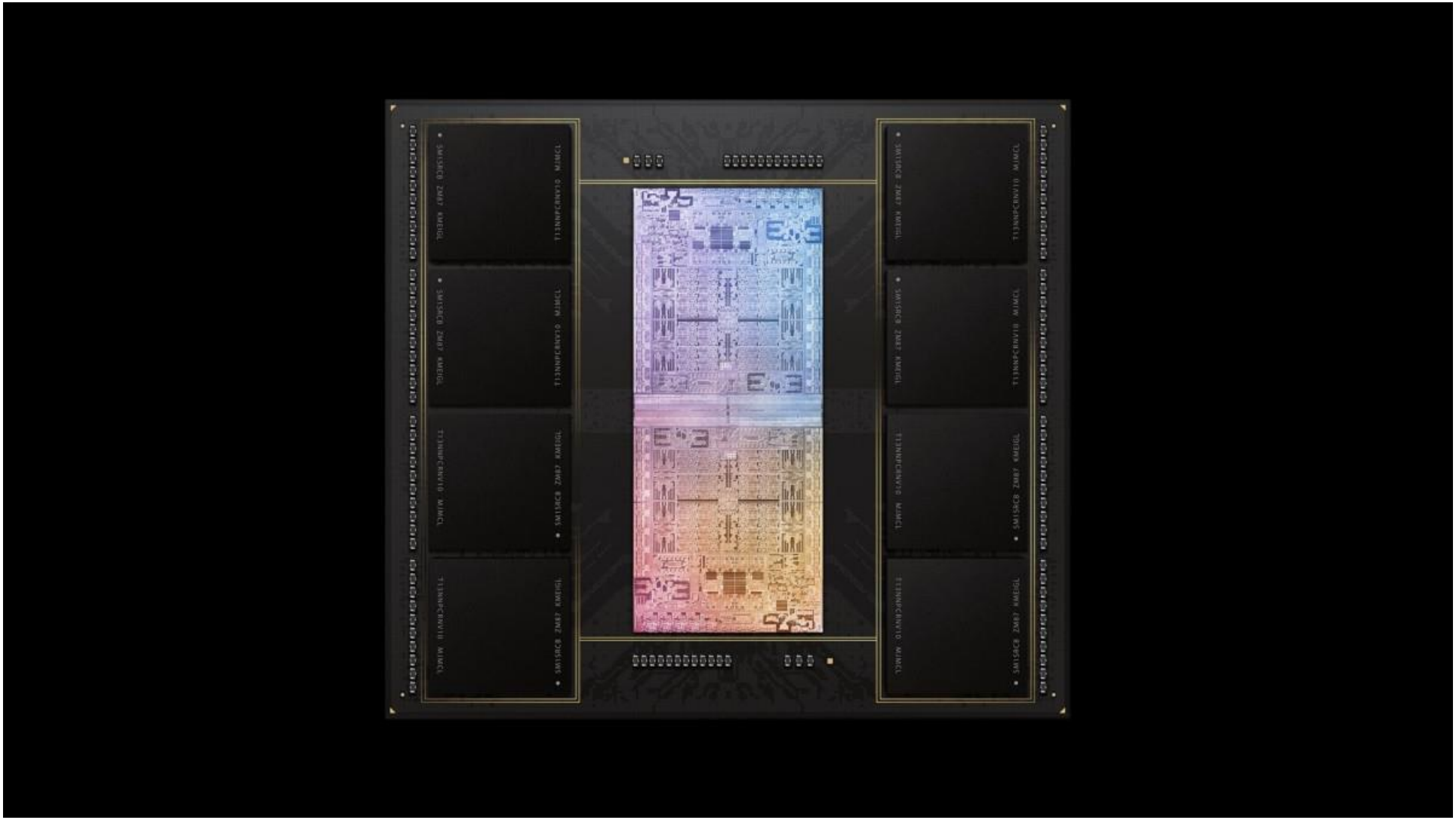


Persistent Memory/Storage

**Every component and its interfaces, as well as entire system designs are being re-examined**

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# An Example System in Your Pocket



Apple M1 Ultra System (2022)

https://www.gsmarena.com/apple_announces_m1_ultra_with_20core_cpu_and_64core_gpu-news-53481.php

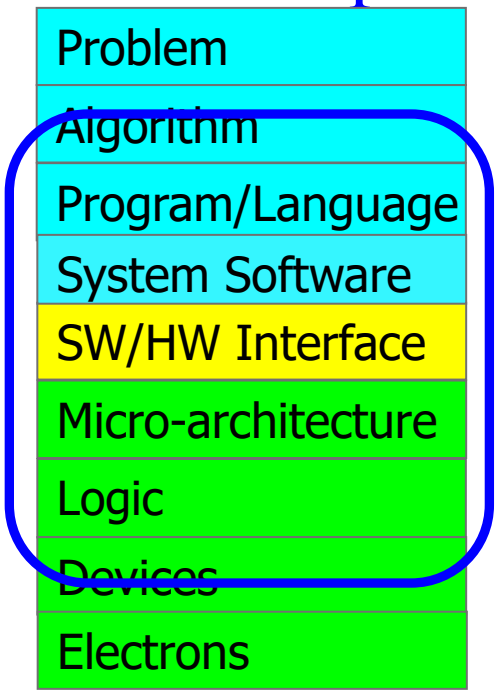
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# Historical: Opportunities at the Bottom

## There's Plenty of Room at the Bottom

From Wikipedia, the free encyclopedia

"**There's Plenty of Room at the Bottom: An Invitation to Enter a New Field of Physics**" was a lecture given by physicist Richard Feynman at the annual American Physical Society meeting at Caltech on December 29, 1959.[1] Feynman considered the possibility of direct manipulation of individual atoms as a more powerful form of synthetic chemistry than those used at the time. Although versions of the talk were reprinted in a few popular magazines, it went largely unnoticed and did not inspire the conceptual beginnings of the field. Beginning in the 1980s, nanotechnology advocates cited it to establish the scientific credibility of their work.

https://en.wikipedia.org/wiki/There%27s_Plenty_of_Room_at_the_Bottom

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# Historical: Opportunities at the Bottom (II)

## There's Plenty of Room at the Bottom

From Wikipedia, the free encyclopedia

Feynman considered some ramifications of a general ability to manipulate matter on an atomic scale. He was particularly interested in the possibilities of denser computer circuitry, and microscopes that could see things much smaller than is possible with scanning electron microscopes. These ideas were later realized by the use of the scanning tunneling microscope, the atomic force microscope and other examples of scanning probe microscopy and storage systems such as Millipede, created by researchers at IBM.

Feynman also suggested that it should be possible, in principle, to make nanoscale machines that "arrange the atoms the way we want", and do chemical synthesis by mechanical manipulation.

He also presented the possibility of "swallowing the doctor", an idea that he credited in the essay to his friend and graduate student Albert Hibbs. This concept involved building a tiny, swallowable surgical robot.

https://en.wikipedia.org/wiki/There%27s_Plenty_of_Room_at_the_Bottom

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# Historical: Opportunities at the Top

REVIEW

## There's plenty of room at the Top: What will drive computer performance after Moore's law?

Charles E. Leiserson[1], Neil C. Thompson[1,2,*], Joel S. Emer[1,3], Bradley C. Kuszmaul[1,†], Butler W. Lampson[1,4], ...
+ See all authors and affiliations

*Science* 05 Jun 2020:
Vol. 368, Issue 6495, eaam9744
DOI: 10.1126/science.aam9744

Much of the improvement in computer performance comes from decades of miniaturization of computer components, a trend that was foreseen by the Nobel Prize–winning physicist Richard Feynman in his 1959 address, "There's Plenty of Room at the Bottom," to the American Physical Society. In 1975, Intel founder Gordon Moore predicted the regularity of this miniaturization trend, now called Moore's law, which, until recently, doubled the number of transistors on computer chips every 2 years.

Unfortunately, semiconductor miniaturization is running out of steam as a viable way to grow computer performance—there isn't much more room at the "Bottom." If growth in computing power stalls, practically all industries will face challenges to their productivity. Nevertheless, opportunities for growth in computing performance will still be available, especially at the "Top" of the computing-technology stack: software, algorithms, and hardware architecture.

https://www.science.org/doi/10.1126/science.aam9744

# Axiom, Revisited

There is plenty of room both at the top and at the bottom

but **much more so**

when you

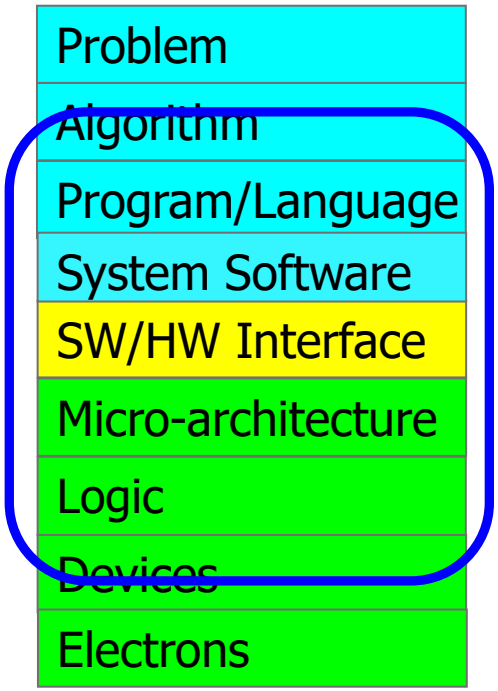
**communicate well between and optimize across**

**the top and the bottom**

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# Hence the Expanded View

**Computer Architecture (expanded view)**



# A Note on Hardware vs. Software

- This course might seem like it is only “Computer Hardware”
- However, you will be much more capable if you master both hardware and software (and the interface between them)
  - Can develop better software if you understand the hardware
  - Can design better hardware if you understand the software
  - Can design a better computing system if you understand both
- This course covers the HW/SW interface and microarchitecture
  - We will focus on tradeoffs and how they affect software

# Fundamental Concepts

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# First …

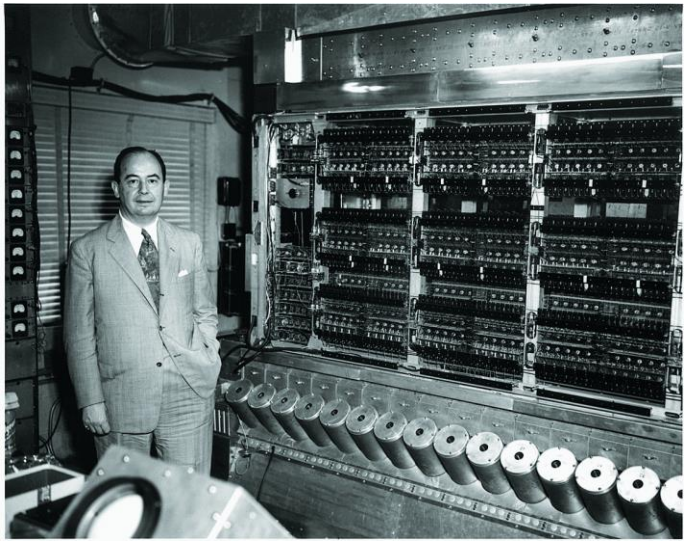
› Let's understand the fundamentals…

› You can change the world only if you understand it well enough…
  - Especially the basics (fundamentals)
  - Past and present dominant paradigms
  - And, their advantages and shortcomings – tradeoffs
  - And, what remains fundamental across generations
  - And, what techniques you can use and develop to solve problems

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# What is A Computer?

- Three key components
- Computation
- Communication
- Storage/memory



Burks, Goldstein, von Neumann, "Preliminary discussion of the [illegible] ument," 1946.

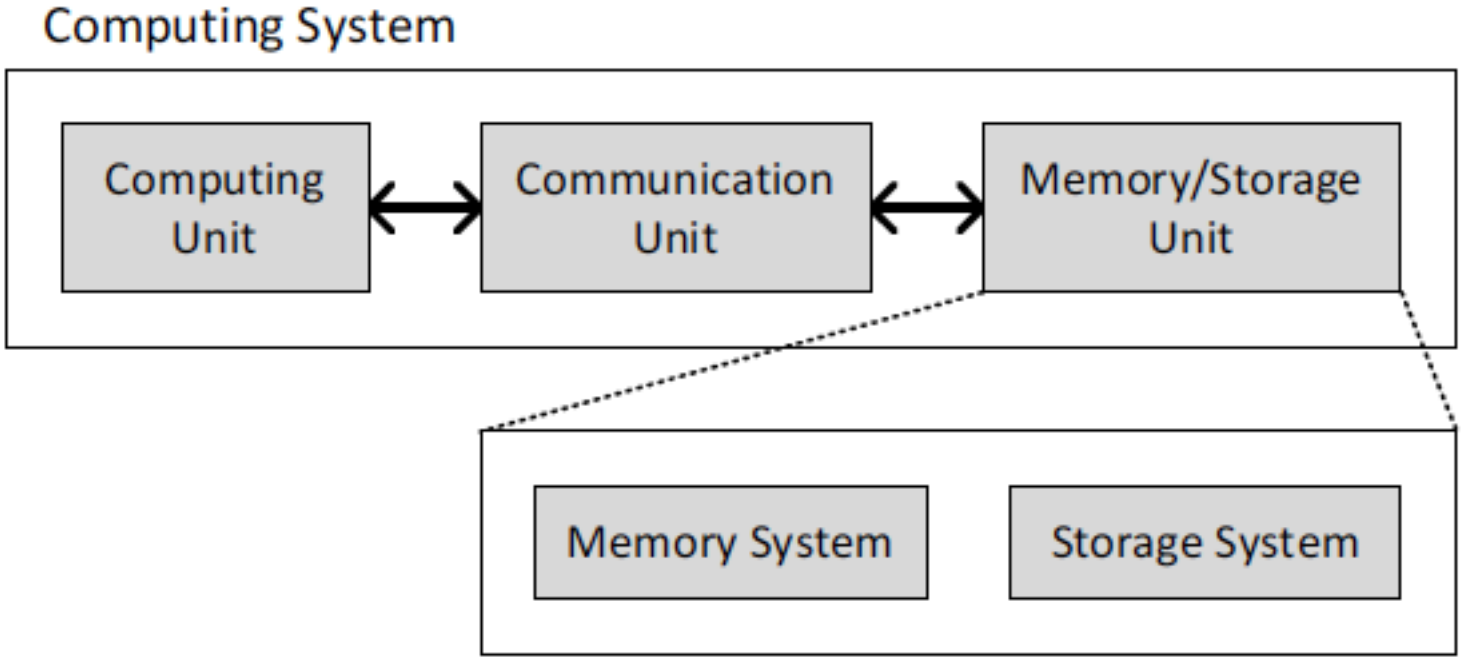
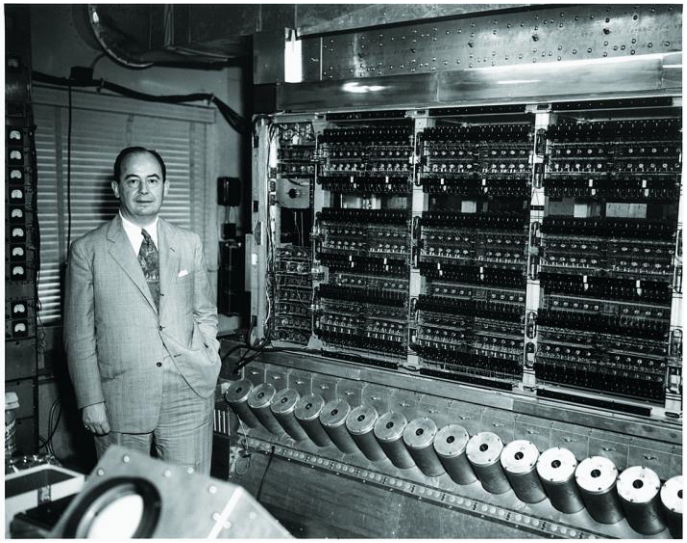



Image source: https://lbsitbytes2010.wordpress.com/2013/03/29/john-von-neumann-roll-no-15/

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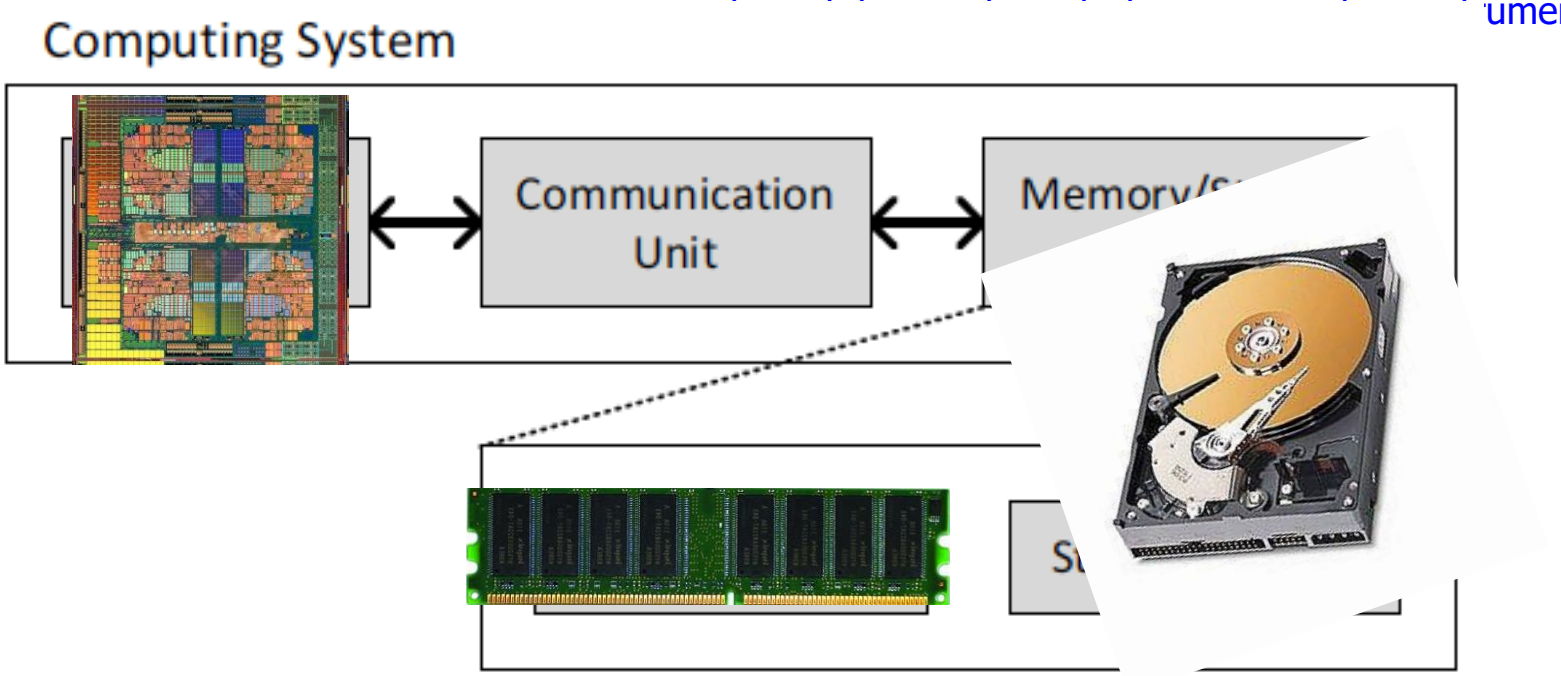
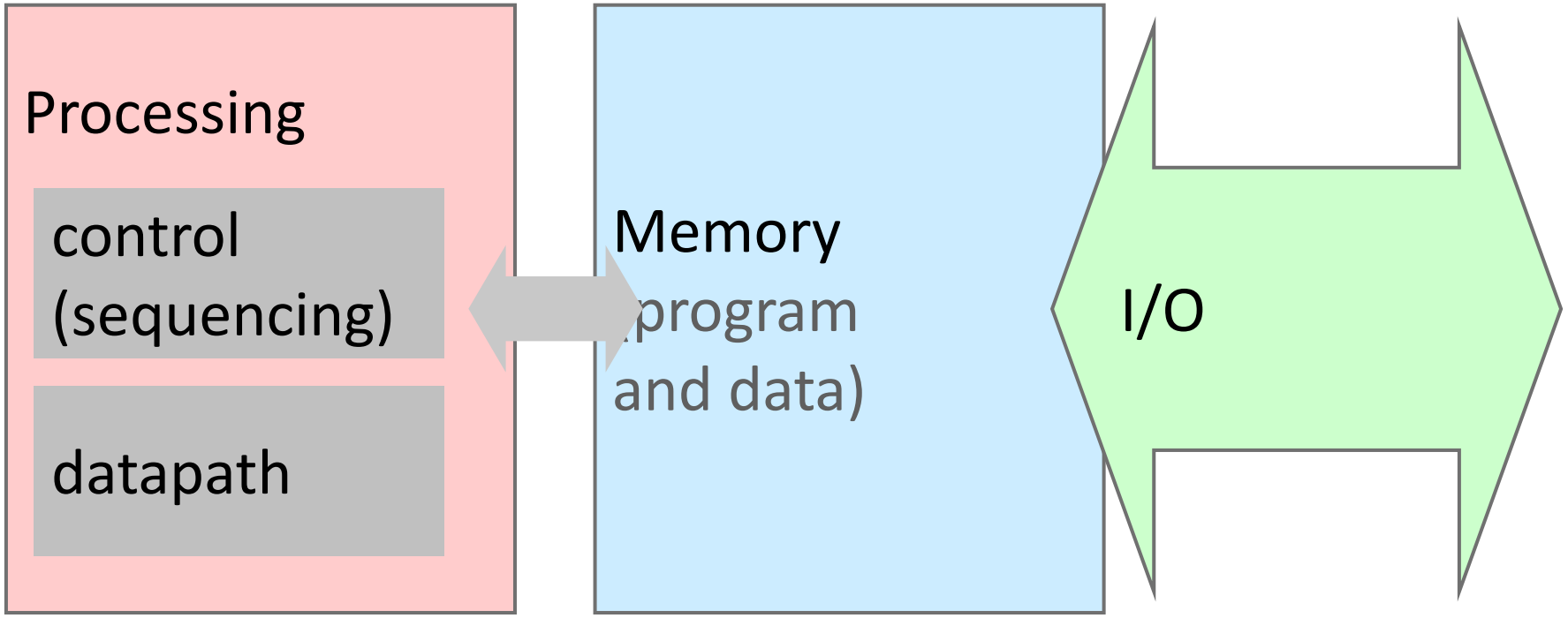



Image source: https://lbsitbytes2010.wordpress.com/2013/03/29/john-von-neumann-roll-no-15/

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# What is A Computer?

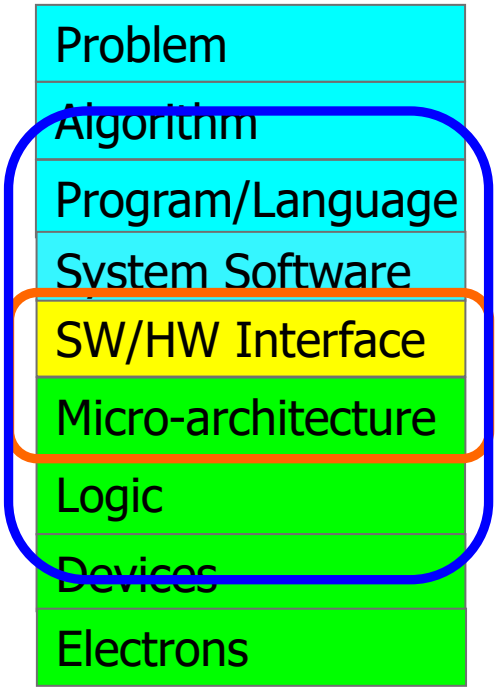
› We will cover all three components



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# Recall: The Transformation Hierarchy

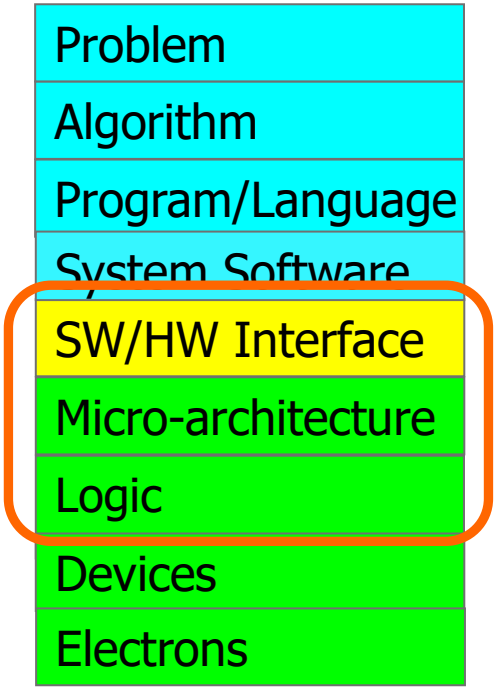
**Computer Architecture (expanded view)**



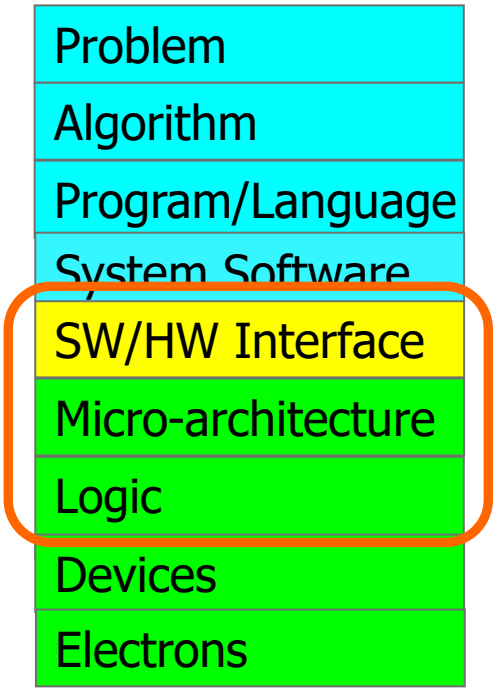

**Computer Architecture (narrow view)**

# What We Will Cover (I)

- Combinational Logic Design
- Hardware Description Languages (Verilog)
- Sequential Logic Design
- Timing and Verification
- ISA (MIPS and LC3b as examples)
- Assembly Programming



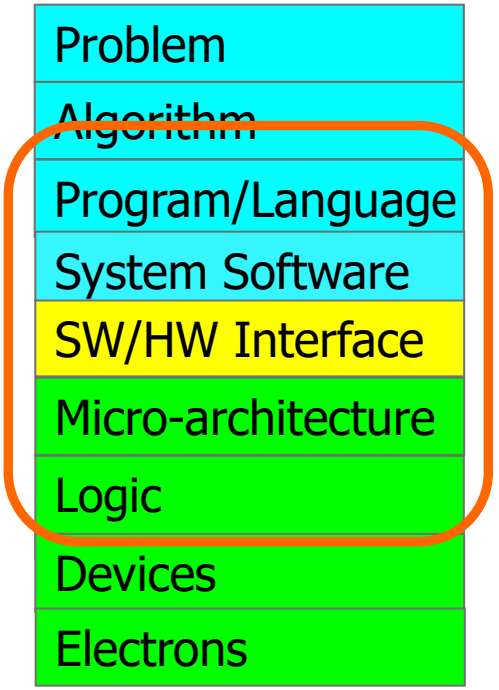
# What We Will Cover (II)




- Microarchitecture Fundamentals
- Single-cycle Microarchitectures
- Multi-cycle and Microprogrammed Microarchitectures
- Pipelining
- Issues in Pipelining: Dependence Handling, State Maintenance and Recovery, …
- Branch Prediction
- Out-of-Order Execution
- Superscalar Execution
- Other Paradigms: Dataflow, VLIW, Systolic, SIMD/GPUs, …

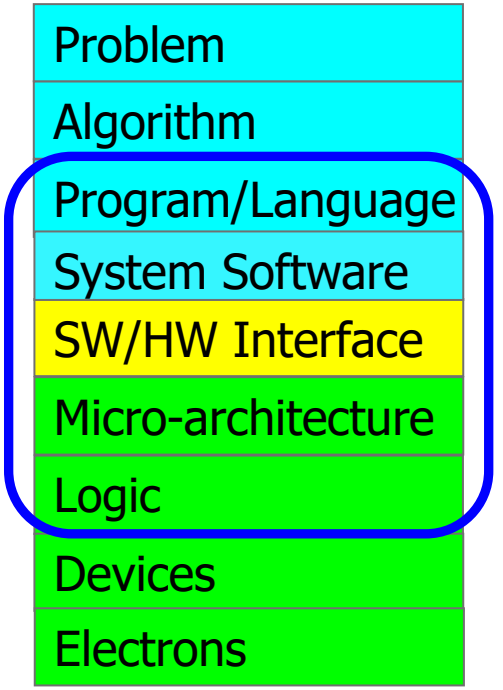
# What We Will Cover (III)

- Memory Technology and Organization
- Memory Hierarchy
- Caches
- Multi-Core Caches
- Prefetching
- Virtual Memory



# Processing Paradigms We Will Cover

- Pipelining
- Out-of-order execution
- Dataflow (at the ISA level)
- Superscalar Execution
- VLIW
- Decoupled Access-Execute
- Systolic Arrays
- SIMD Processing (Vector & Array)
- GPUs





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# Combinational Logic Circuits and Design

# What Will We Learn Today?

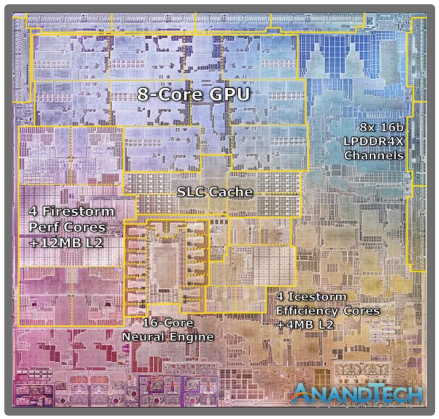
- Basic building blocks of modern computers
  - Transistors
  - Logic gates
- Boolean algebra
- Combinational logic circuits
- How to use Boolean algebra to represent combinational circuits
- Minimizing logic circuits (if time permits)

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# General Purpose vs. Special Purpose Systems

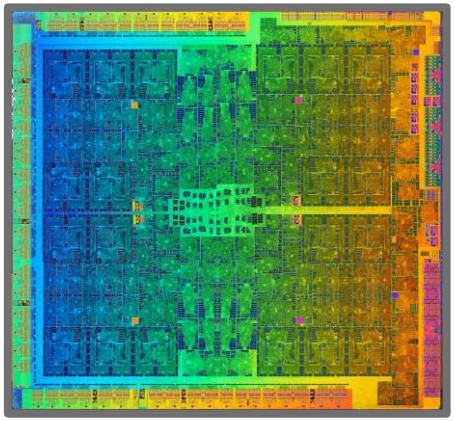
**General Purpose**

**CPUs**



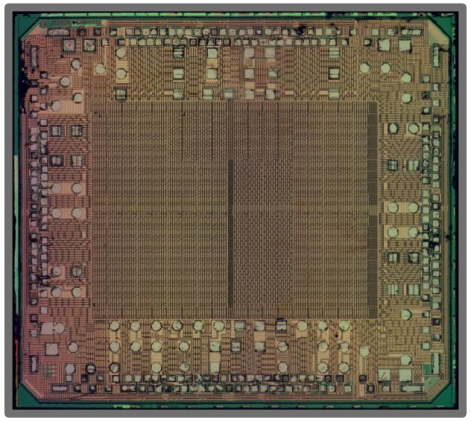

Apple M1

**GPUs**



Nvidia GTX 1070

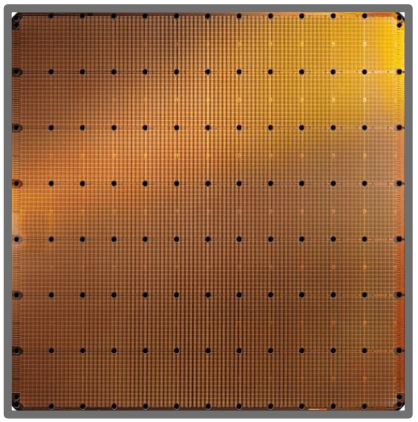
**FPGAs**



Xilinx Spartan

**Special Purpose**

**ASICs**



Cerebras WSE-2



**Flexible: Can execute any program**

**Easy to program & use**

**Not the best performance & efficiency**

**Efficient & High performance**

**(Usually) Difficult to program & use**

**Inflexible: Limited set of programs**

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# General Purpose vs. Special Purpose Systems

## General Purpose




**Flexible: Can work with any bolt**
**Easy to use**
**Not the best fit, results or efficiency**

https://www.ubuy.vn/en/product/2MBUW8M-crescent-8-adjustable-wrench-carded-ac28vs
https://capritools.com/shop/bolt-extractor-wrench-set-metric-8-19-mm/

## Special Purpose



**Efficient & High performance**
**(Usually) Difficult to use**
**Inflexible: Works only for fitting bolts**

https://www.vigor-equipment.com/en/combination-wrench-4971

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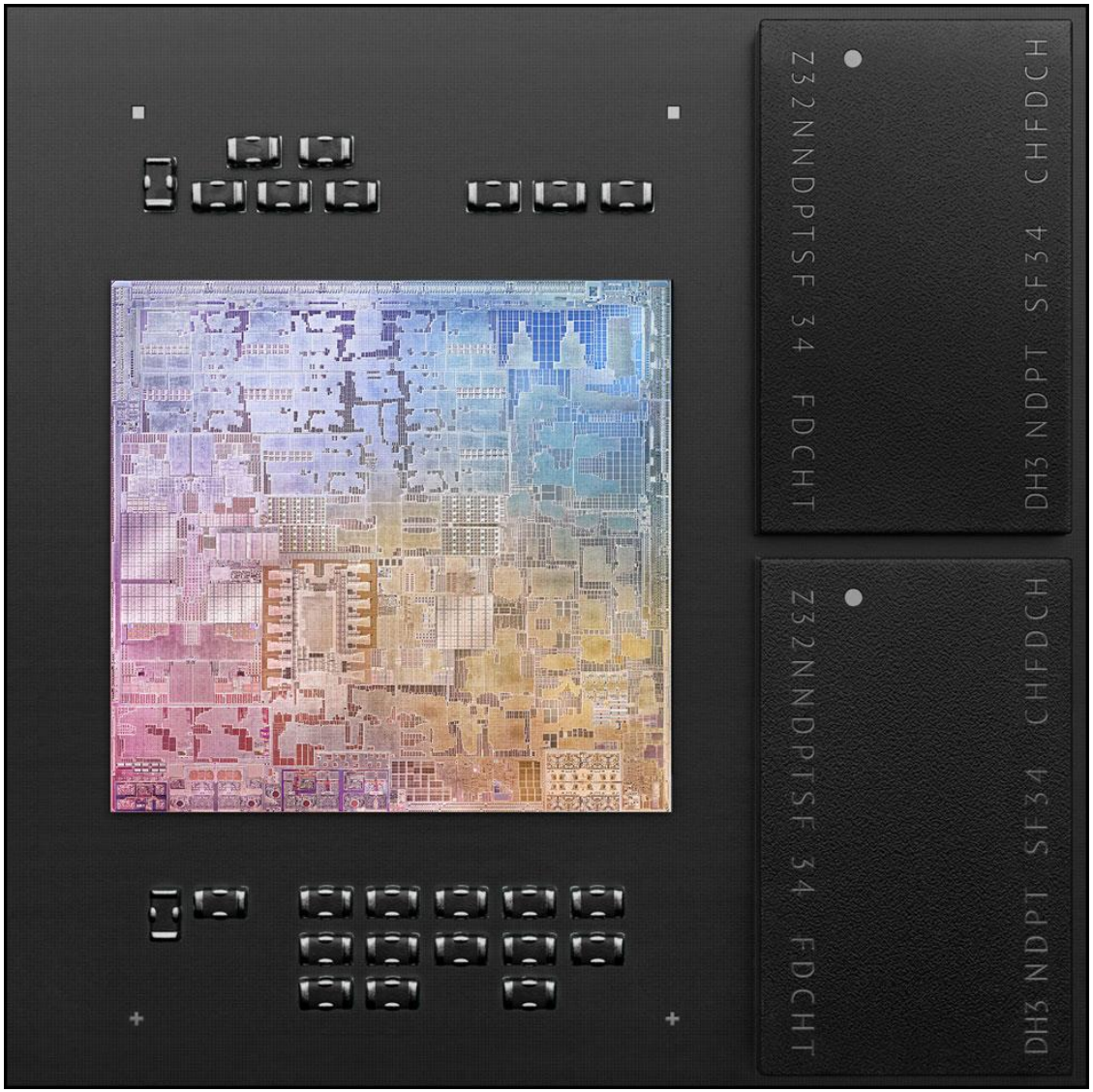
# Modern General-Purpose Microprocessors

**5-nanometer process**

The first personal computer chip built with this cutting-edge technology.

**16 billion transistors**

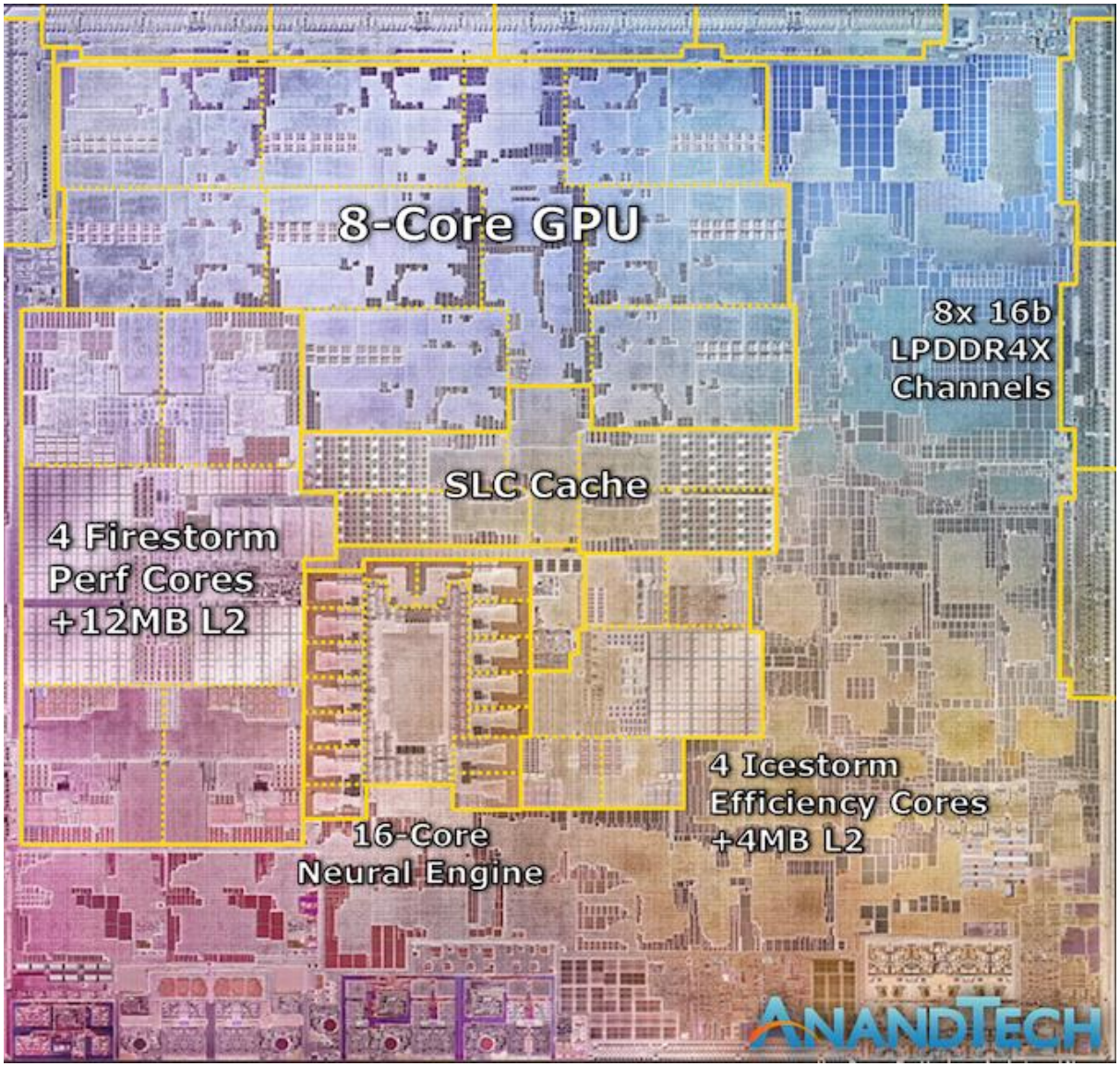
The most we've ever put into a single chip.




Source: https://www.apple.com/mac/m1/

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# Modern General-Purpose Microprocessors

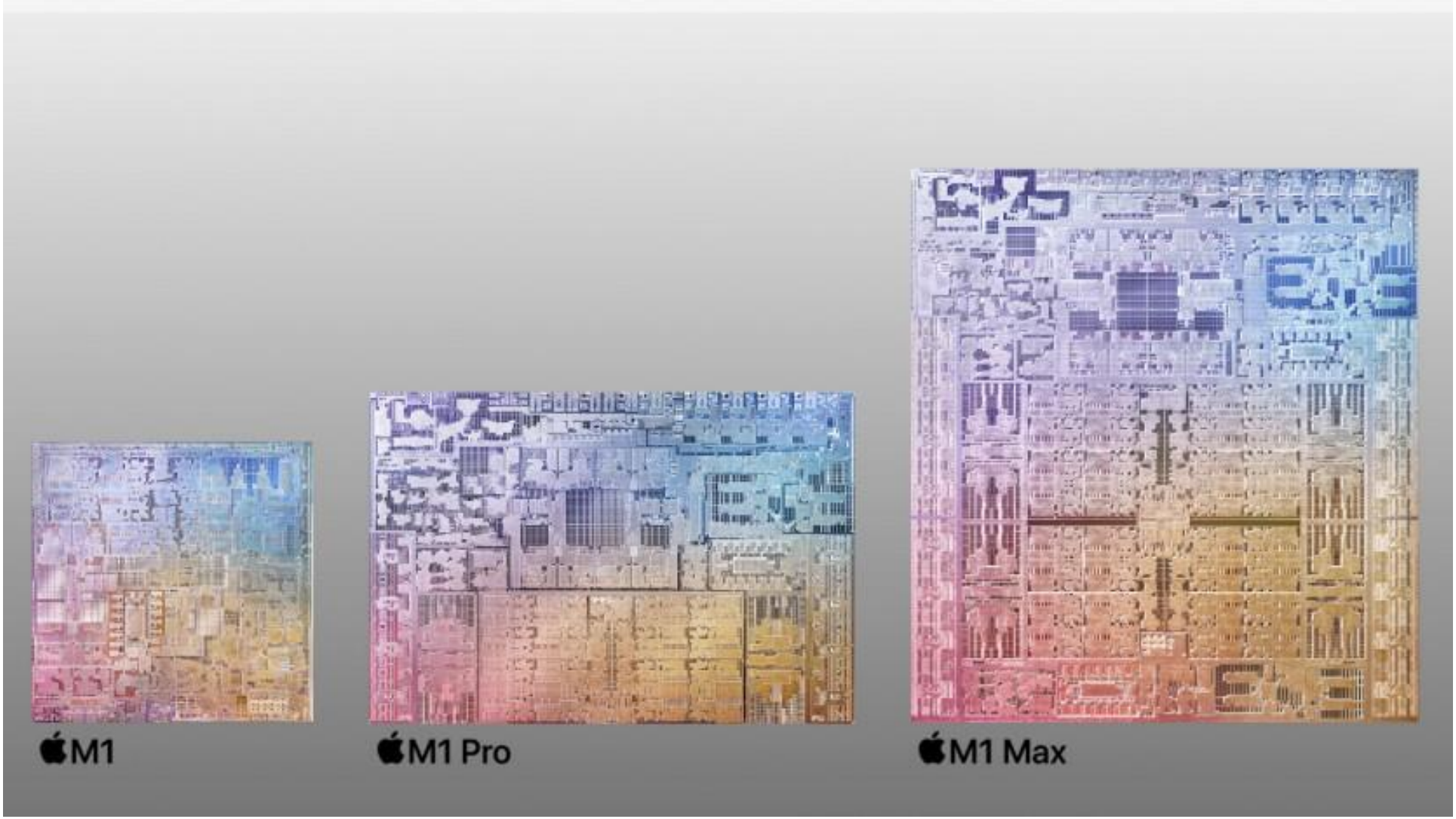



Apple M1, 2021

Source: https://www.anandtech.com/show/16252/mac-mini-apple-m1-tested

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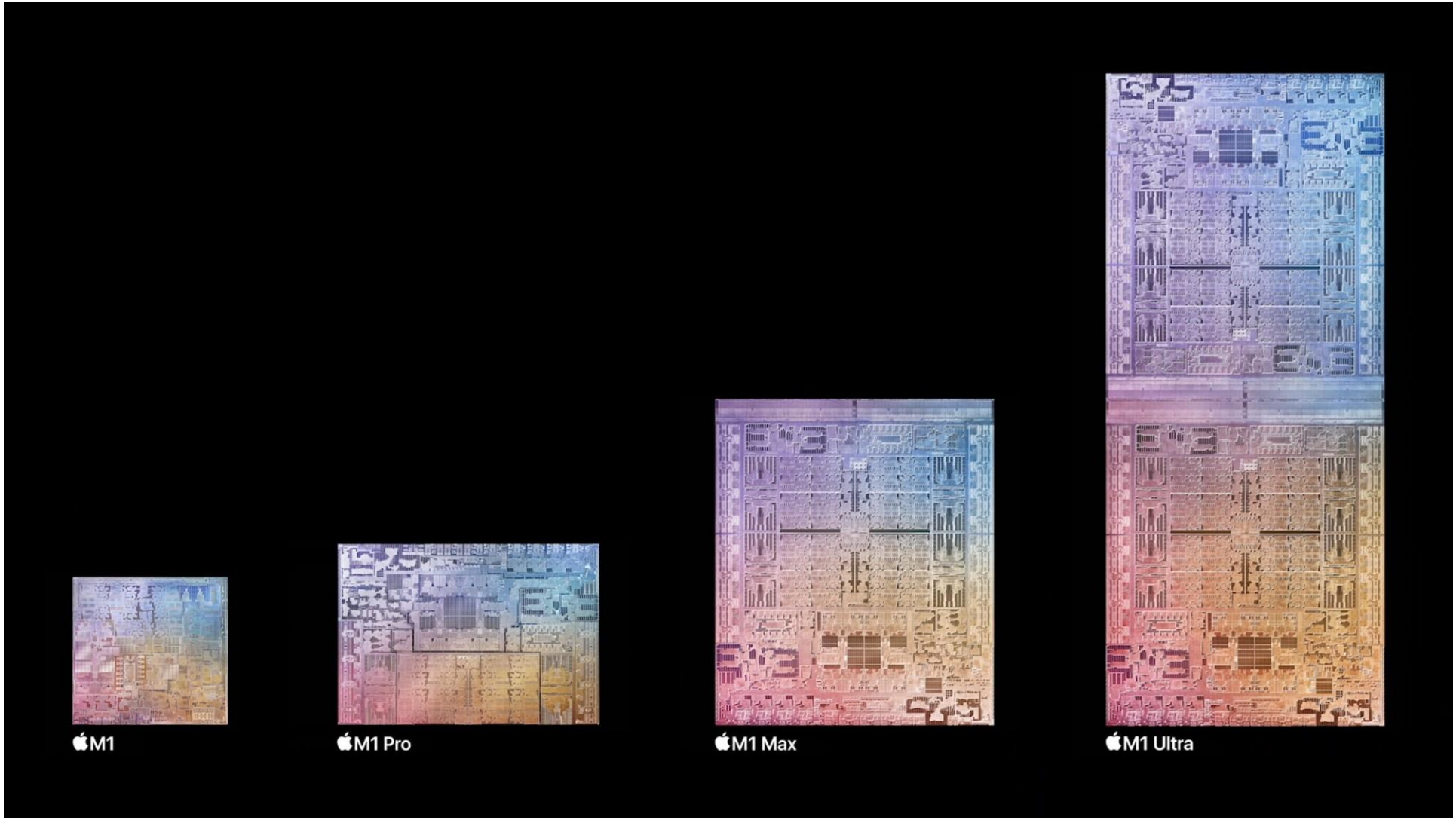
# Modern General-Purpose Microprocessors




Source: https://www.golem.de/news/m1-pro-max-dieses-apple-silicon-ist-gigantisch-2110-160415.html

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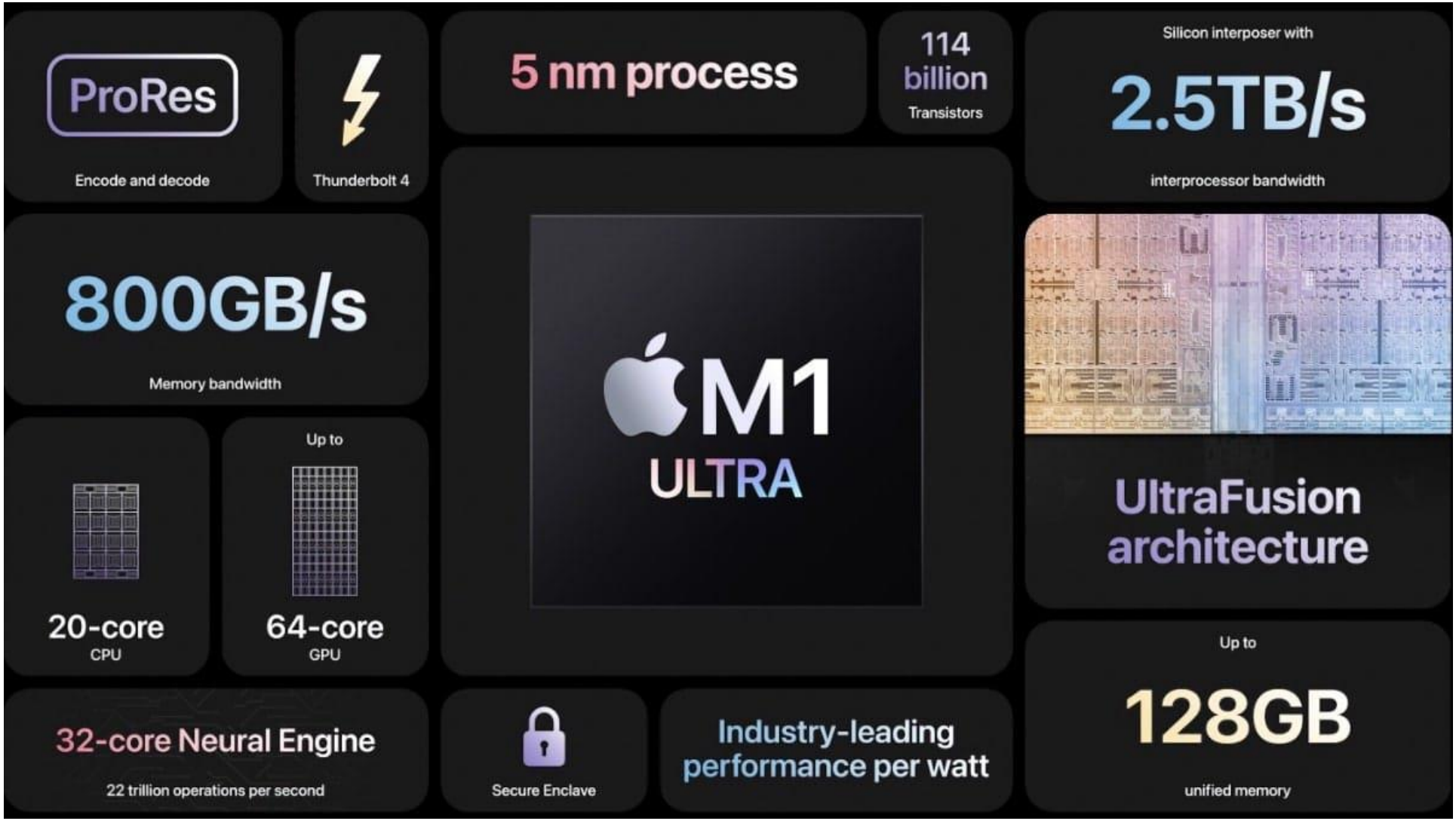
# Apple M1 Ultra (2022)




https://www.theverge.com/2022/3/9/22968611/apple-m1-ultra-gpu-nvidia-rtx-3090-comparison

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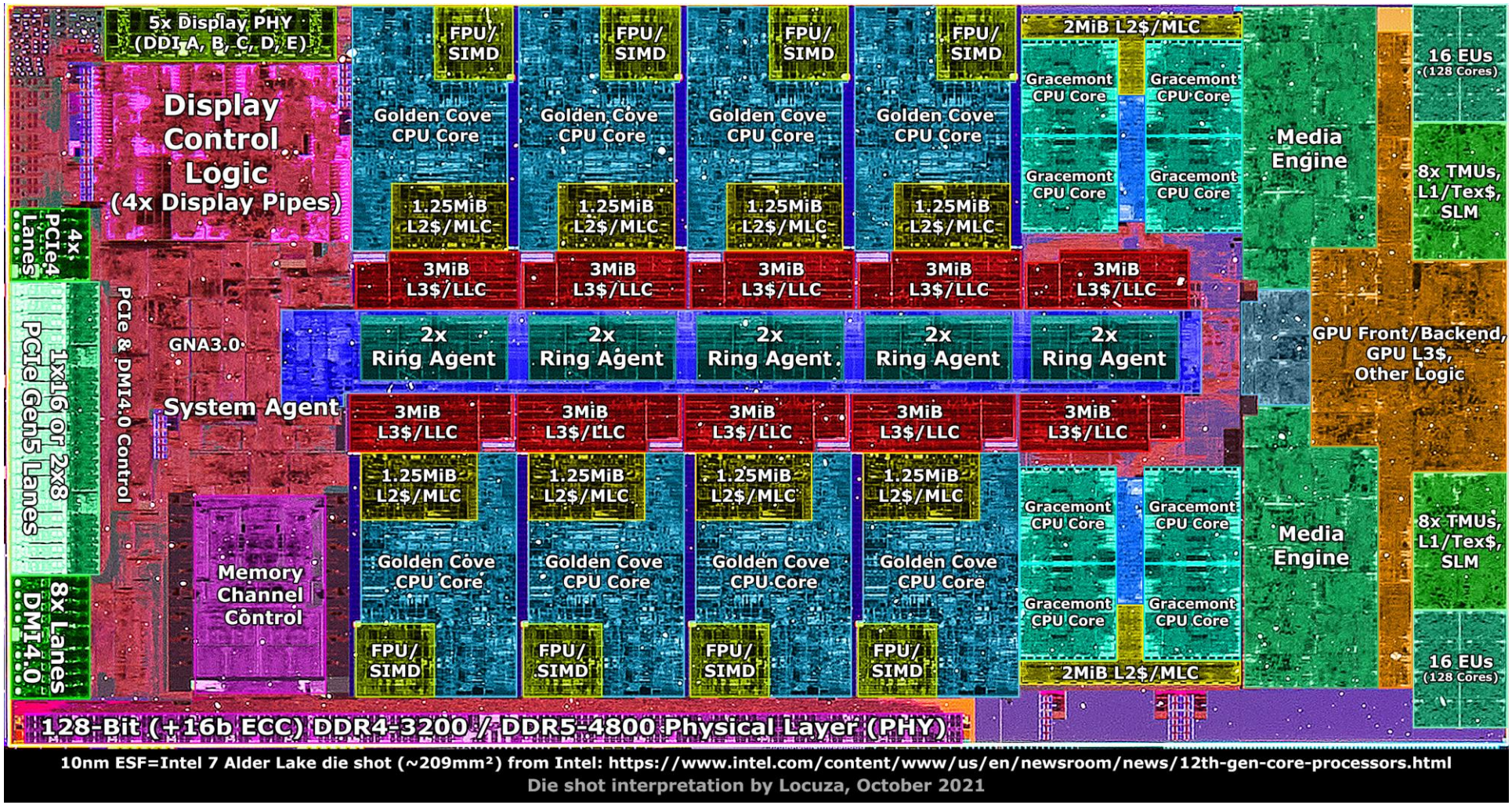
# Apple M1 Ultra (2022)




https://stadt-bremerhaven.de/apple-neuer-m1-ultra-chip-ist-offiziell/

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# Modern General-Purpose Microprocessors

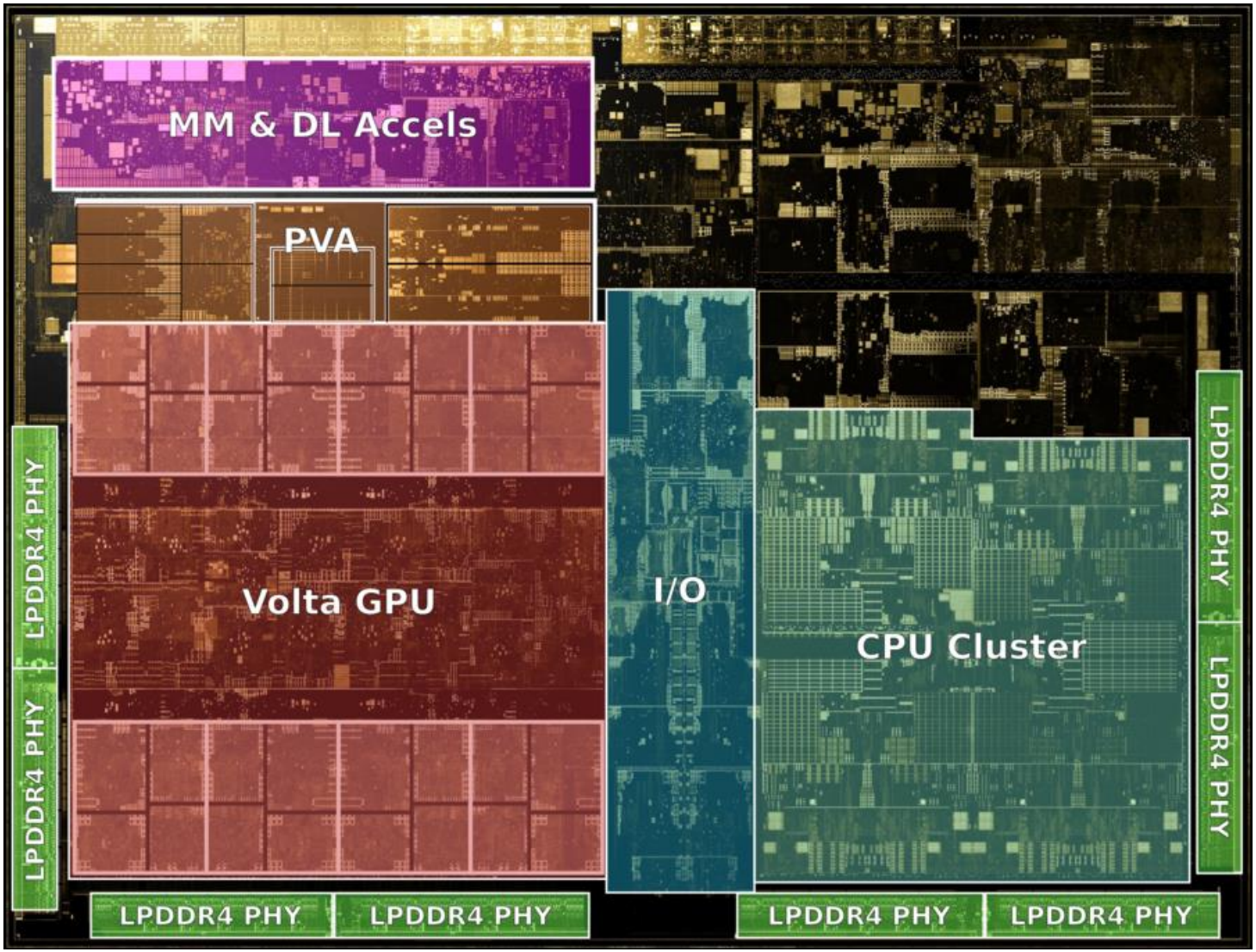



10nm ESF=Intel 7 Alder Lake die shot (~209mm²) from Intel: https://www.intel.com/content/www/us/en/newsroom/news/12th-gen-core-processors.html
Die shot interpretation by Locuza, October 2021

Source: https://twitter.com/Locuza_/status/1454152714930331652

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# Modern GPUs




Source: https://en.wikichip.org/wiki/nvidia/tegra/xavier

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# FPGAs



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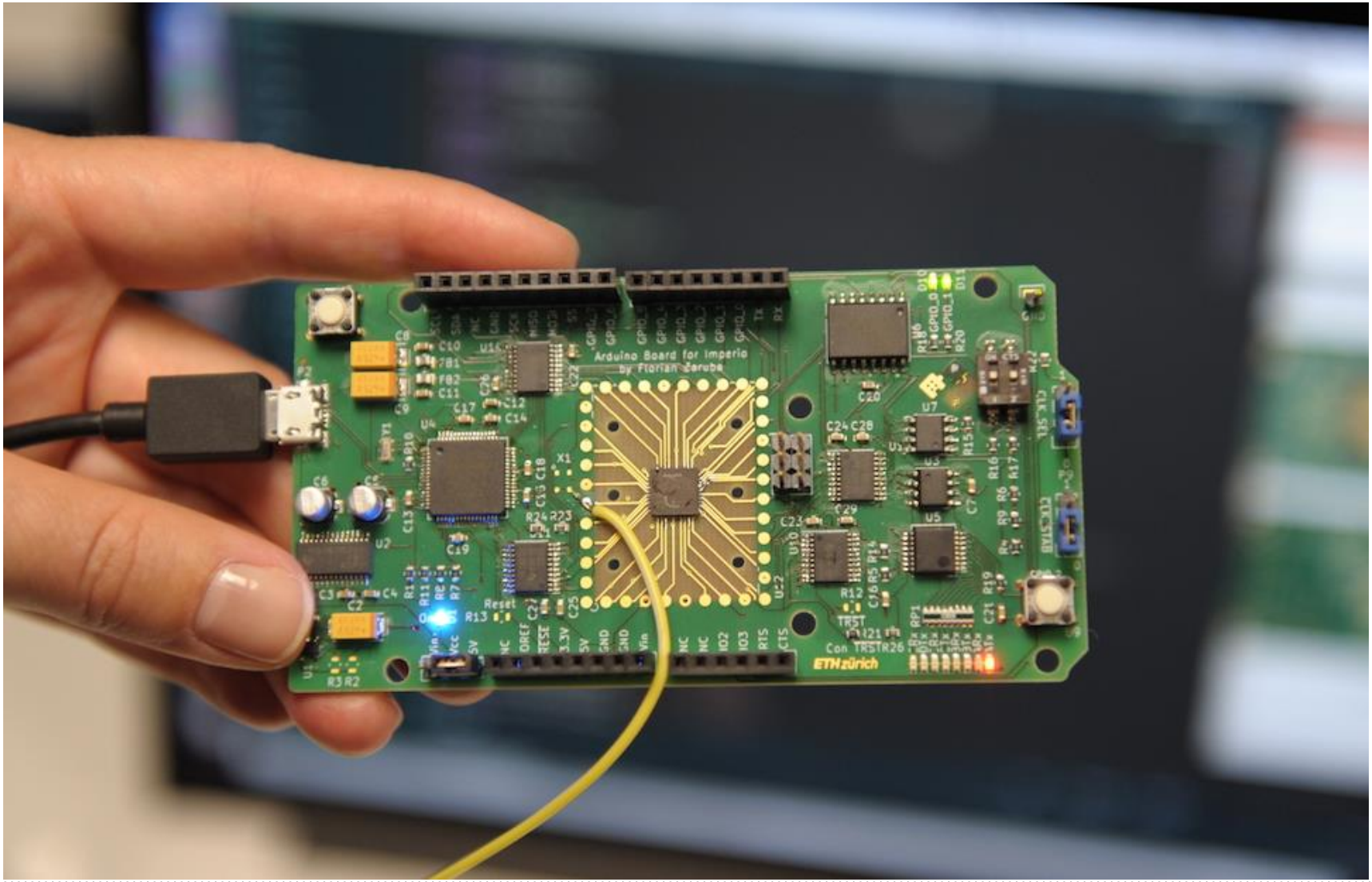
# Modern FPGAs




Source: https://www.mouser.ch/new/xilinx/xilinx-zynq-7000-zc702-eval-kit/

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# Special-Purpose ASICs (App-Specific Integrated Circuits)



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# Modern Special-Purpose ASICs

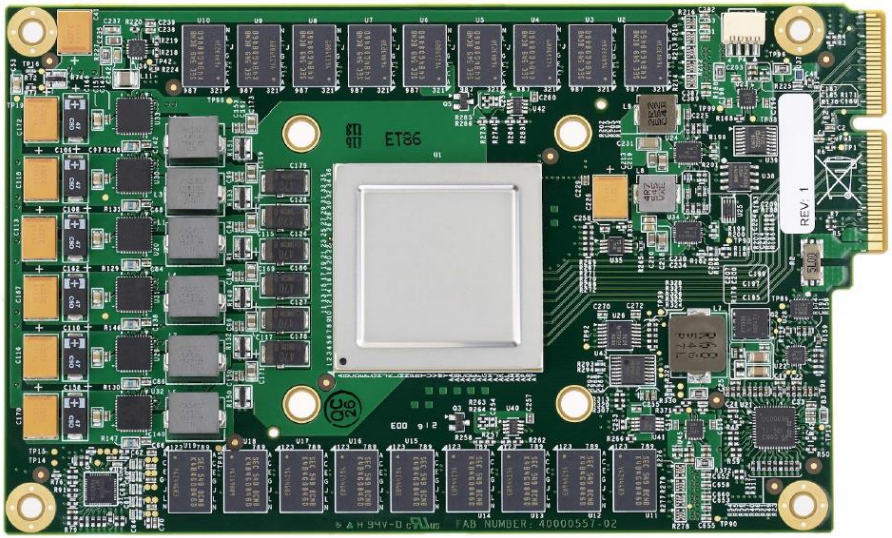


**Figure 3.** TPU Printed Circuit Board. It can be inserted in the slot for an SATA disk in a server, but the card uses PCIe Gen3 x16.

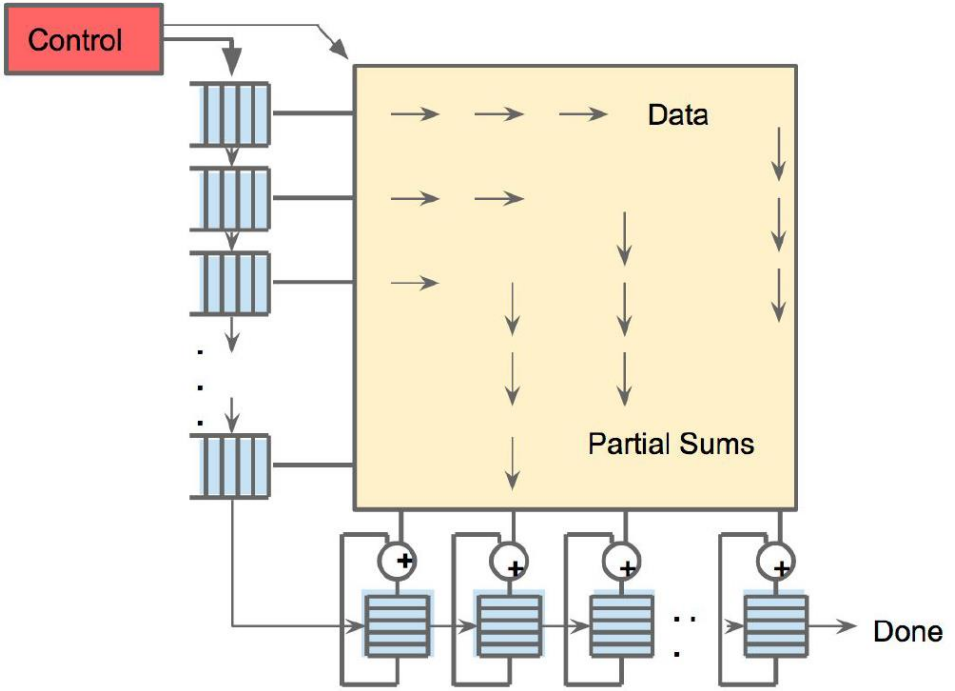
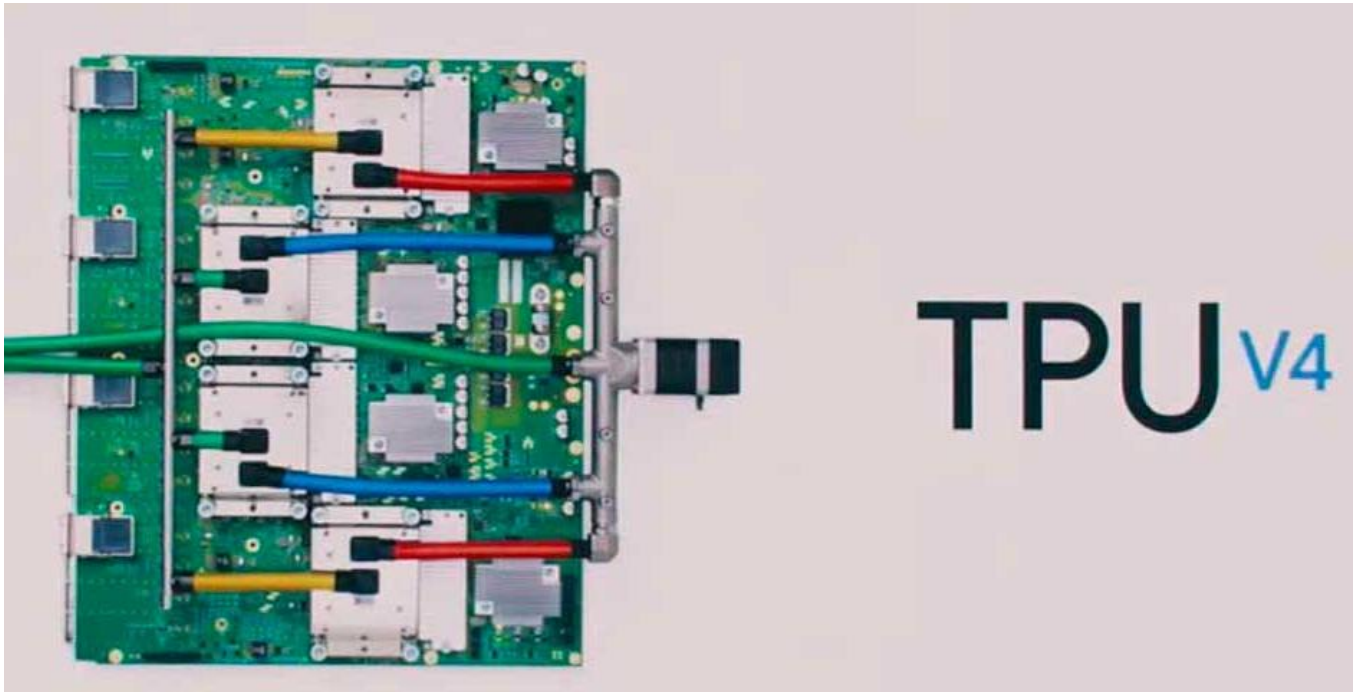



**Figure 4.** Systolic data flow of the Matrix Multiply Unit. Software has the illusion that each 256B input is read at once, and they instantly update one location of each of 256 accumulator RAMs.

Jouppi et al., "In-Datacenter Performance Analysis of a Tensor Processing Unit", ISCA 2017.

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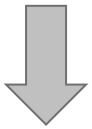
# Modern Special-Purpose ASICs




New ML applications (vs. TPU3):

- Computer vision
- Natural Language Processing (NLP)
- Recommender system
- Reinforcement learning that plays Go

250 TFLOPS per chip in 2021
vs 90 TFLOPS in TPU3

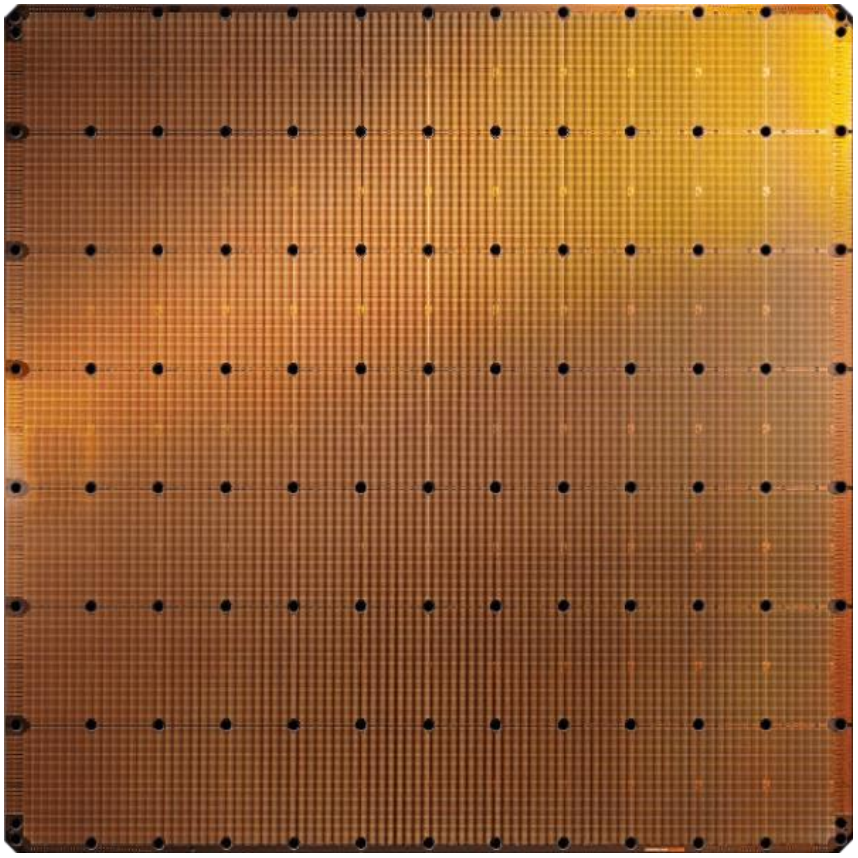


1 ExaFLOPS per board

https://spectrum.ieee.org/tech-talk/computing/hardware/heres-how-googles-tpu-v4-ai-chip-stacked-up-in-training-tests

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# Modern Special-Purpose ASICs



- The largest ML accelerator chip (2021)
- 850,000 cores




**Cerebras WSE-2**
2.6 Trillion transistors
46,225 $mm^2$

**Largest GPU**
54.2 Billion transistors
826 $mm^2$
**NVIDIA** Ampere GA100

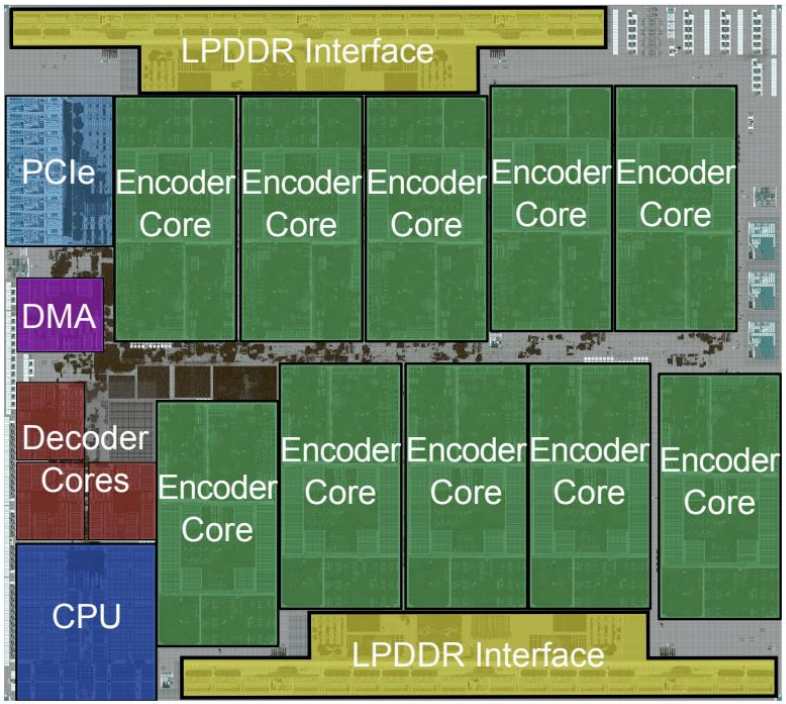
https://www.anandtech.com/show/14758/hot-chips-31-live-blogs-cerebras-wafer-scale-deep-learning

https://www.cerebras.net/cerebras-wafer-scale-engine-why-we-need-big-chips-for-deep-learning/

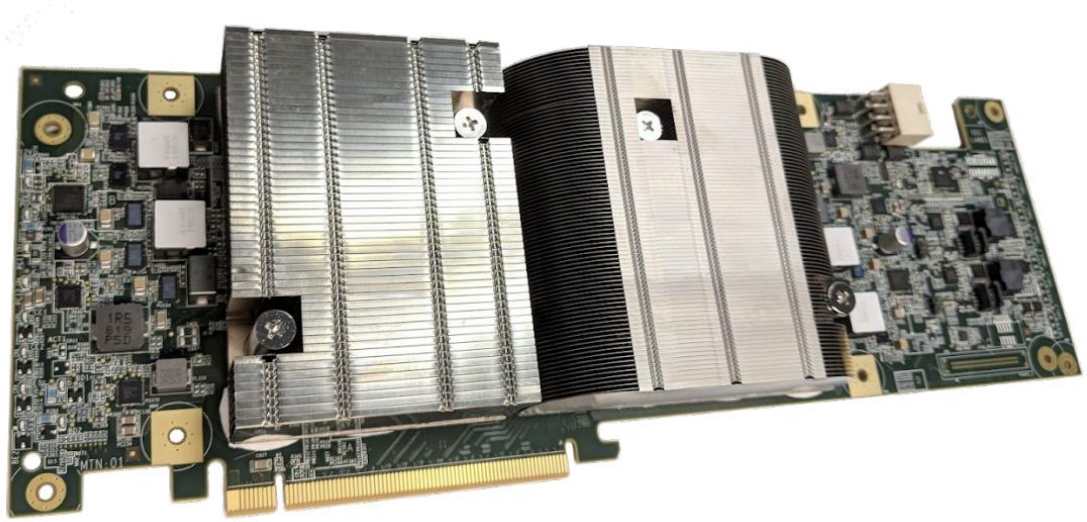
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# Modern Special-Purpose ASICs

## Warehouse-Scale Video Acceleration: Co-design and Deployment in the Wild




(a) Chip floorplan



(b) Two chips on a PCBA

**Figure 5: Pictures of the VCU**

Source: https://dl.acm.org/doi/pdf/10.1145/3445814.3446723

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# They All Look the Same

| | Microprocessors | FPGAs | ASICs |
|---|---|---|---|
| |  |  |  |
| **In short:** | Common building block of computers | Reconfigurable hardware, flexible | You customize everything |
| | | | |
| | | | |
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| | Microprocessors | FPGAs | ASICs |
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| **Program Development Time** | minutes | days | months |
| | | | |
| | | | |
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# They All Look the Same

| | Microprocessors | FPGAs | ASICs |
|---|---|---|---|
| |  |  |  |
| **In short:** | Common building block of computers | Reconfigurable hardware, flexible | You customize everything |
| **Program Development Time** | minutes | days | months |
| **Performance** | 0 | + | ++ |
| | | | |
| | | | |
| | | | |

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# They All Look the Same

| | Microprocessors | FPGAs | ASICs |
|---|---|---|---|
| |  |  |  |
| **In short:** | Common building block of computers | Reconfigurable hardware, flexible | You customize everything |
| **Program Development Time** | minutes | days | months |
| **Performance** | o | + | ++ |
| **Good for** | Ubiquitous<br>Simple to use | Prototyping<br>Small volume | Mass production,<br>Max performance |
| | | | |
| | | | |
| | | | |

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# They All Look (Kind of) the Same

| | Microprocessors | FPGAs | ASICs |
|---|---|---|---|
| |  |  |  |
| **In short:** | Common building block of computers | Reconfigurable hardware, flexible | You customize everything |
| **Program Development Time** | minutes | days | months |
| **Performance** | o | + | ++ |
| **Good for** | Ubiquitous<br>Simple to use | Prototyping<br>Small volume | Mass production,<br>Max performance |
| **Programming** | Executable file | Bit file | Design masks |
| **Languages** | C/C++/Java/… | Verilog/VHDL | Verilog/VHDL |
| **Main Companies** | Intel, ARM, AMD, Apple, NVIDIA | Xilinx, Altera | TSMC, Globalfoundries |

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Want to learn how these work

By programming these

| | Microprocessors | FPGAs | |
|---|---|---|---|
| |  |  | |
| **In short** | Common building block of computers | Reconfigurable hardware, flexible | mize ything |
| **Program Development Time** | minutes | days | months |
| **Performance** | o | Using this language | |
| **Good for** | Ubiquitous Simple to use | Prototyping Small volume | Mass production, Max performance |
| **Programming** | Executable file | Bit file | Design masks |
| **Languages** | C/C++/Java/… | **Verilog/VHDL** | **Verilog/VHDL** |
| **Main Companies** | Intel, ARM, AMD, Apple, | Xilinx, Altera | TSMC, Globalfoundries |

# All Computers are Built Upon the Same Building Blocks

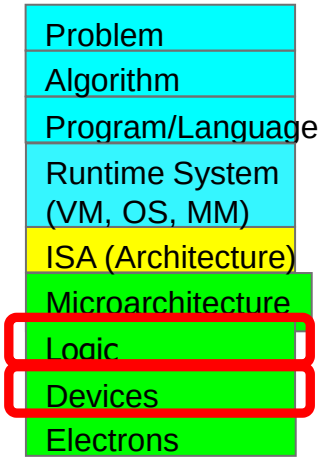
Building Blocks of Modern Computers

# Transistors

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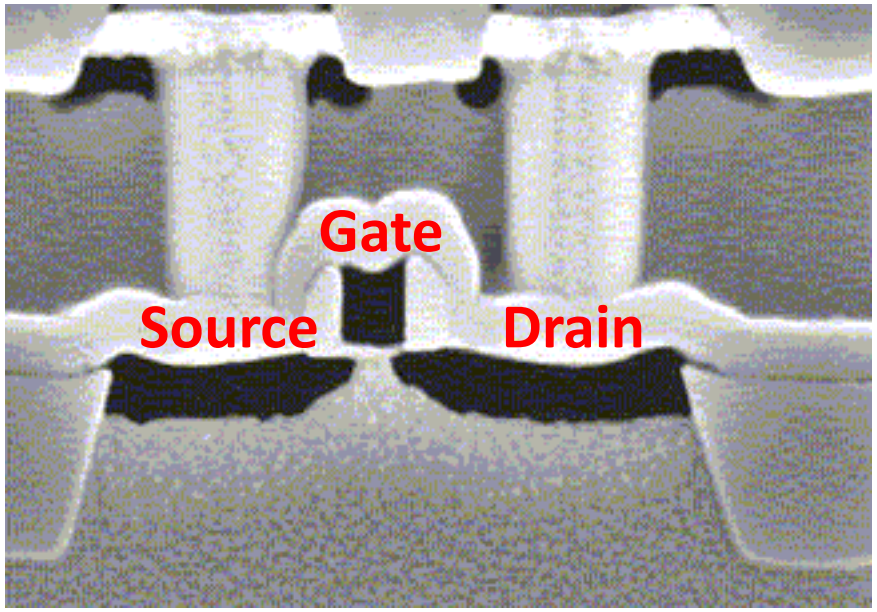
# Transistors

- **Computers are built from very large numbers of very small (and relatively simple) structures: transistors**
  - Intel 4004, in 1971, had 2300 MOS transistors
  - Intel's Pentium IV microprocessor, 2000, was made up of more than 42 **M**illion MOS transistors
  - Apple's M2 Max, offered for sale in 2022, is made up of more than 67 **B**illion MOS transistors
  - **This lecture**
  - How the MOS transistor works (as a logic element)
  - How these transistors are connected to form logic gates
  - How logic gates are interconnected to form larger units that are needed to construct a computer



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# MOS Transistor




- By combining
  - Conductors (**M**etal)
  - Insulators (**O**xide)
  - **S**emiconductors
- We get a Transistor (MOS)
- Why is this useful?
  - We can combine many of these to realize simple logic gates
- The electrical properties of metal-oxide semiconductors are well beyond the scope of what we want to understand in this course
  - They are below our lowest level of abstraction

Sections 1.6 and 1.7 in Harris and Harris provide some lower-level explanations

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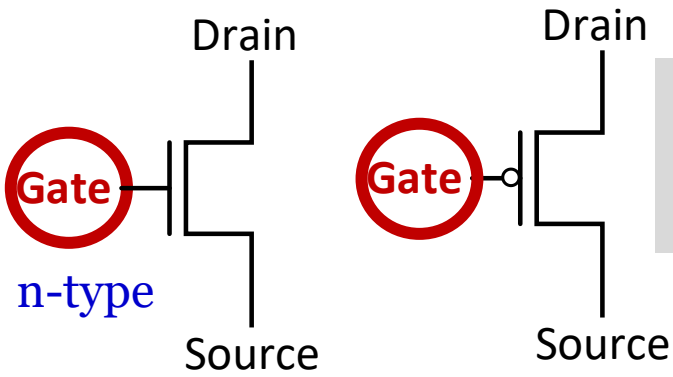
# Different Types of MOS Transistors

› There are two types of MOS transistors: n-type and p-type




› They both operate "logically," very similar to the way wall switches work

The circuit is closed when the gate is supplied with 3V





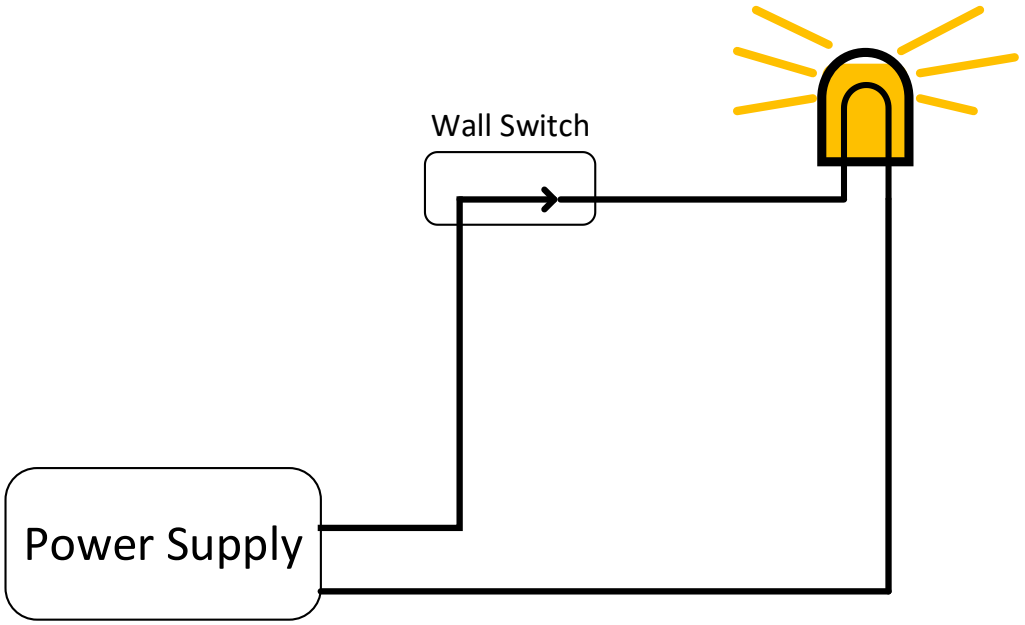


The circuit is closed when the gate is supplied with 0V

p-type

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# How Does a Wall Switch Work?

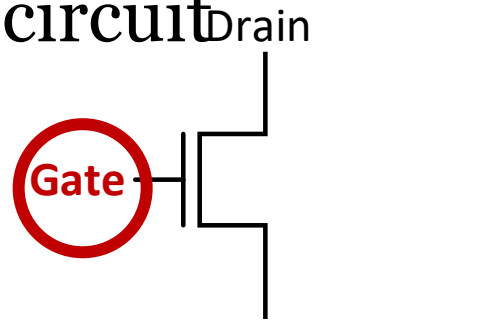



- In order for the lamp to glow, electrons must flow
- In order for electrons to flow, there must be a closed circuit from the power supply to the lamp and back to the power supply
- The lamp can be turned on and off by simply manipulating the wall switch to make or break the closed circuit

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# Transistor Works As a Switch

› Instead of the wall switch, we could use an n-type or a p-type MOS transistor to make or break the closed circuit



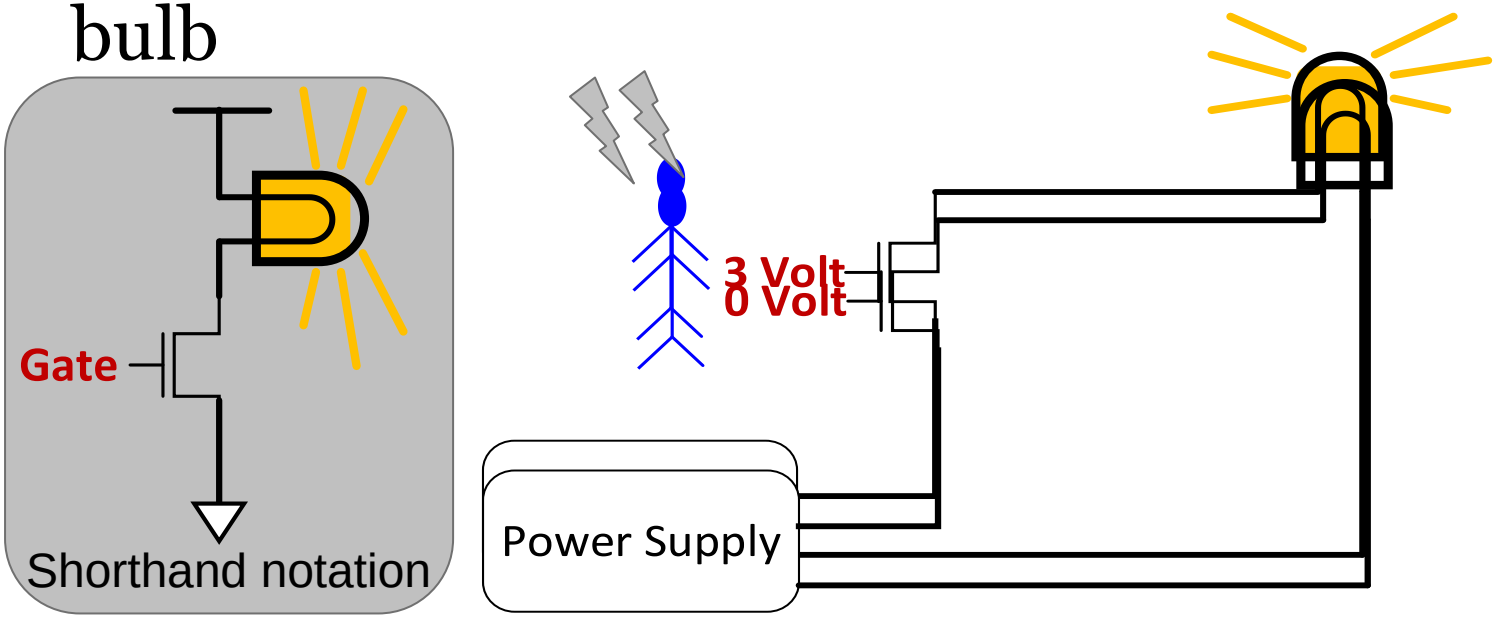
Schematic of an n-type MOS transistor

If the gate of an n-type transistor is supplied with a **high** voltage, the connection from source to drain acts like a piece of wire (i.e., the circuit is closed)

Depending on the technology, high voltage can range from 0.3V to 3V

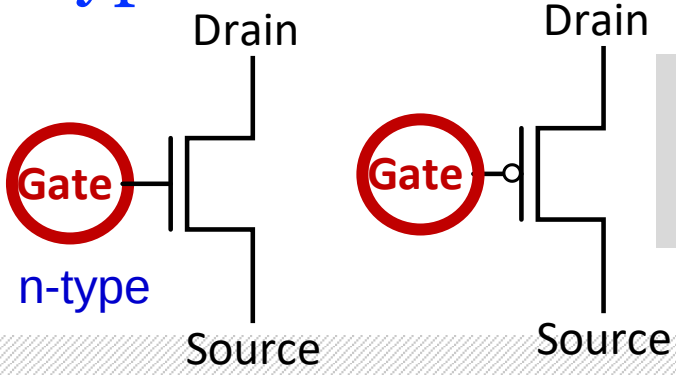
If the gate of the n-type transistor is supplied with **zero** voltage, the connection between the source and drain is broken (i.e., the circuit is open)

- n-type transistor in a circuit with a battery and a bulb




- p-type transistor works in exactly the opposite fashion from n-type transistor

The circuit is closed when the gate is supplied with 3V







The circuit is closed when the gate is supplied with 0V

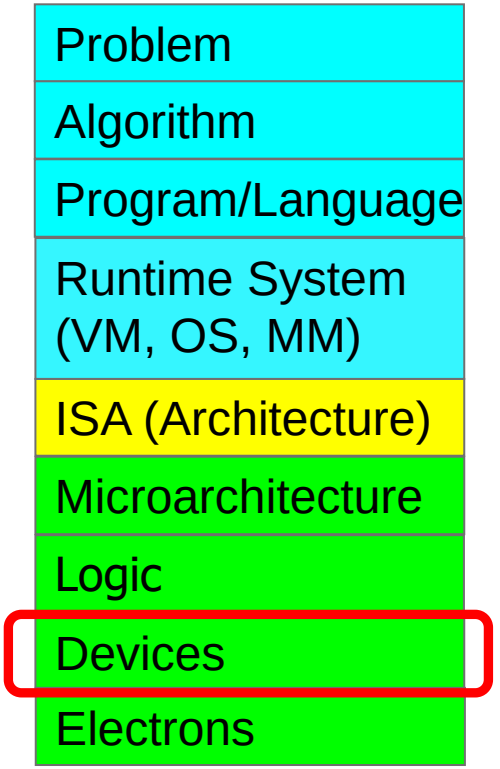
p-type

# Logic Gates

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# One Level Higher in the Abstraction

- **Now, we know how a MOS transistor works**
- How do we build logic structures out of MOS transistors?
  - We construct basic logical units out of individual MOS transistors
  - These logical units are called logic gates
    - They implement simple Boolean functions




George Boole, “The Mathematical Analysis of Logic,” 1847.

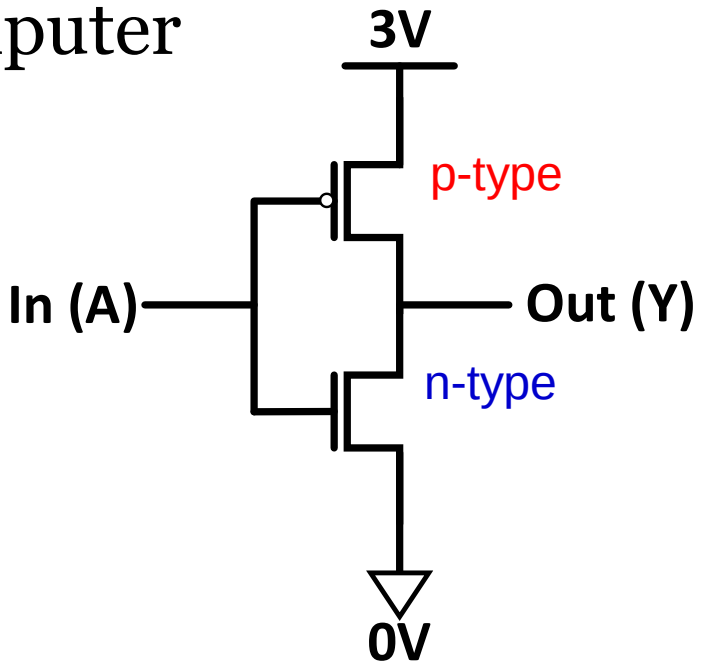
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# Making Logic Blocks Using CMOS Technology

- Modern computers use both n-type and p-type transistors, i.e. Complementary MOS (CMOS) technology

**nMOS + pMOS = CMOS**

- The simplest logic structure that exists in a modern computer

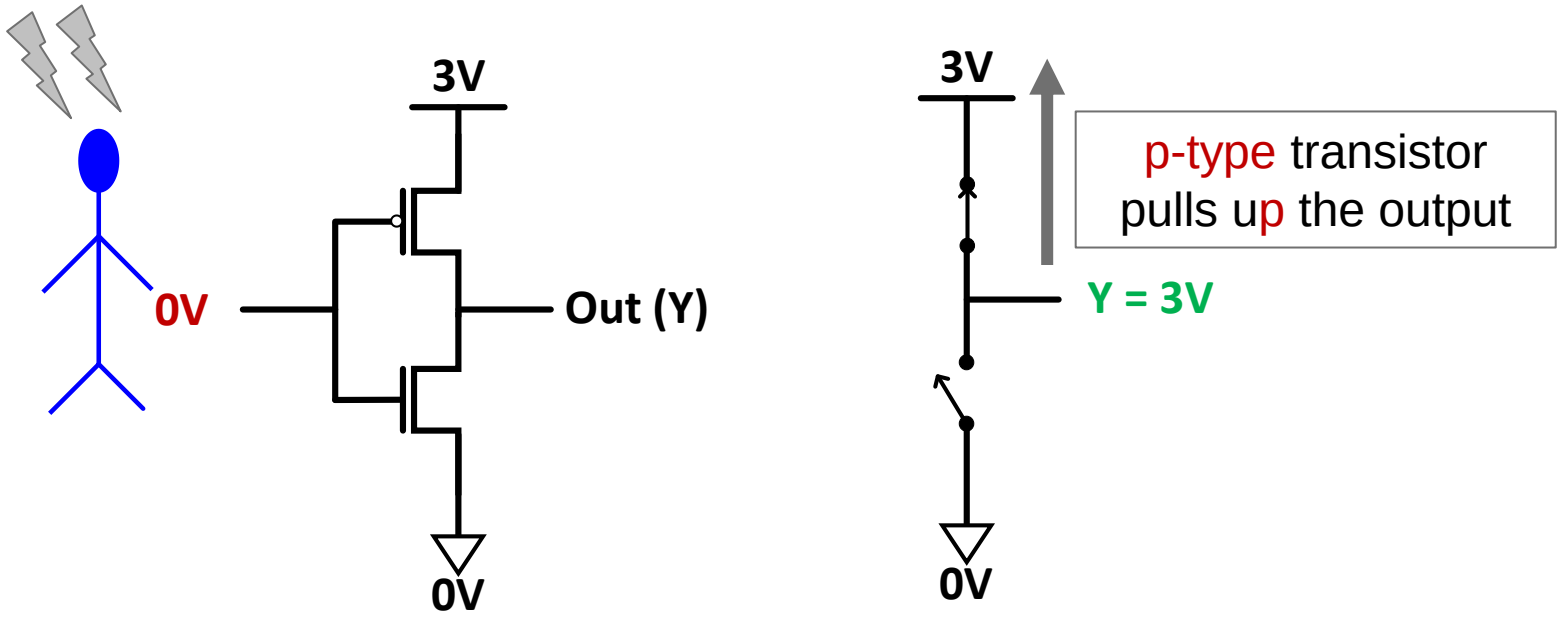



What does this circuit do?

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# Functionality of CMOS Circuit

What happens when the input is connected to 0V?

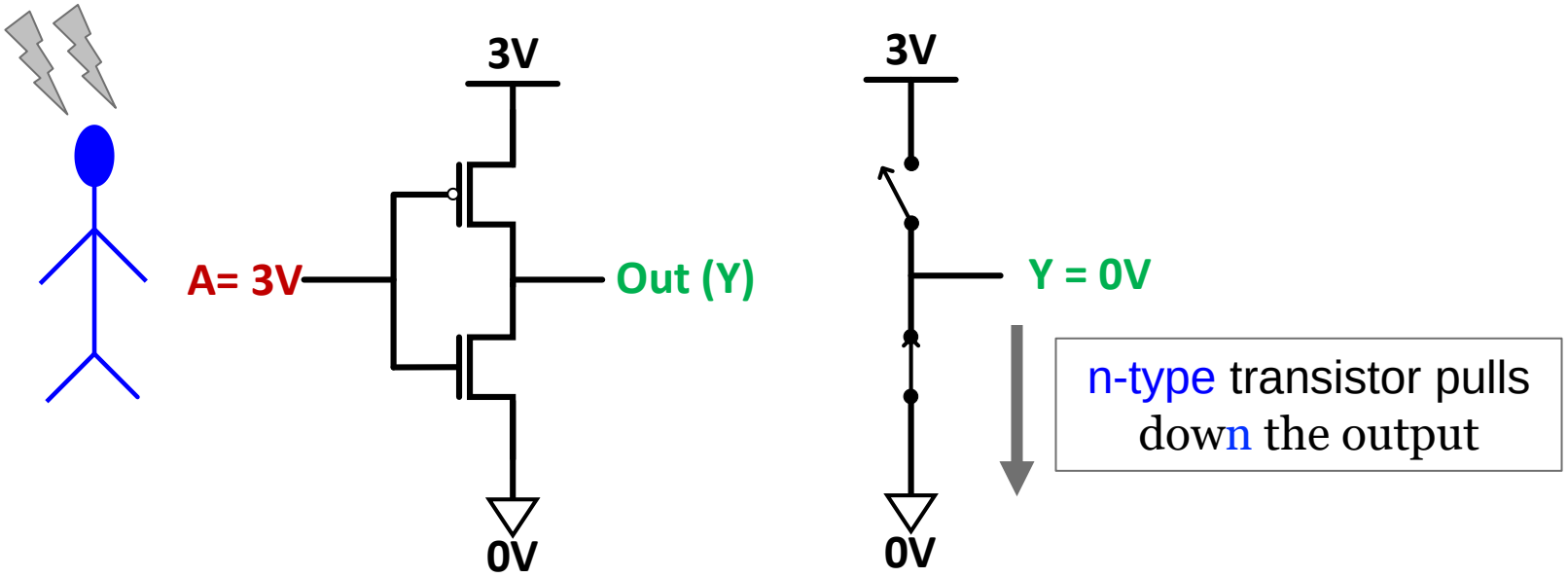



**p**-type transistors are good at pulling u**p** the voltage

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# Functionality of CMOS Circuit

What happens when the input is connected to 3V?

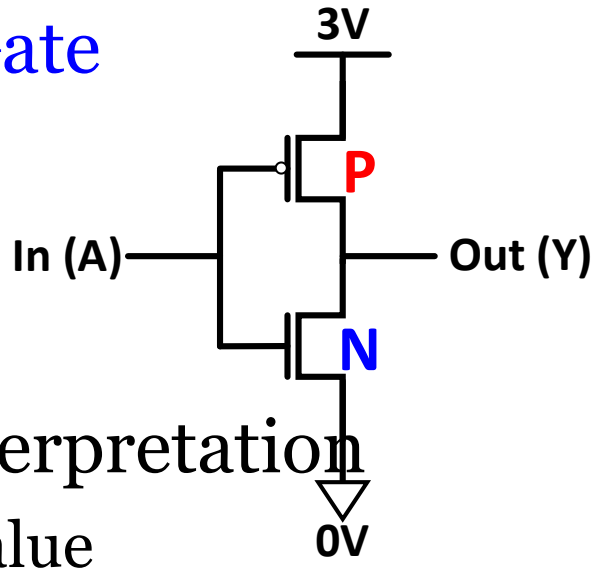



**n**-type transistors are good at pulling down the voltage

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# CMOS NOT Gate (Inverter)

- This is actually the CMOS NOT Gate
- Why do we call it NOT?
  - If A = 0V then Y = 3V
  - If A = 3V then Y = 0V
- **Digital circuit:** one possible interpretation
  - Interpret 0V as logical (binary) 0 value
  - Interpret 3V as logical (binary) 1 value



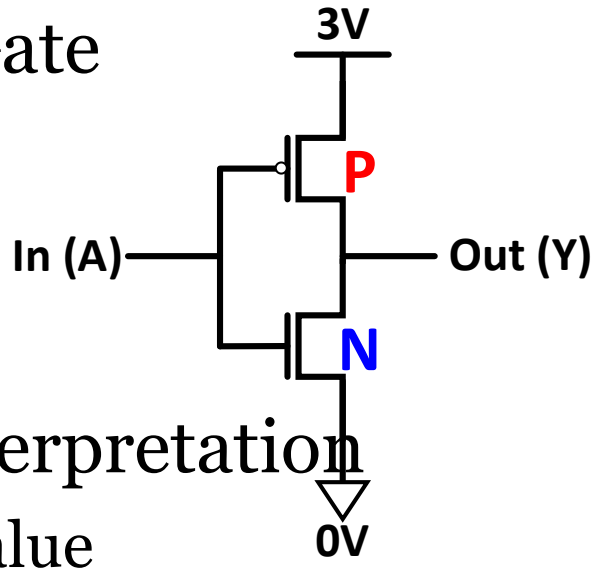

| A | P | N | Y |
|---|---|---|---|
| 0 | **ON** | OFF | 1 |
| 1 | OFF | **ON** | 0 |

$$Y = \bar{A}$$

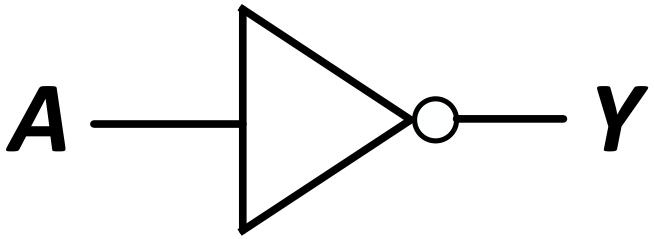
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  - Interpret 3V as logical (binary) 1 value




$$Y = \overline{A}$$




We call this a NOT gate
or an inverter
(bubble indicates inversion)

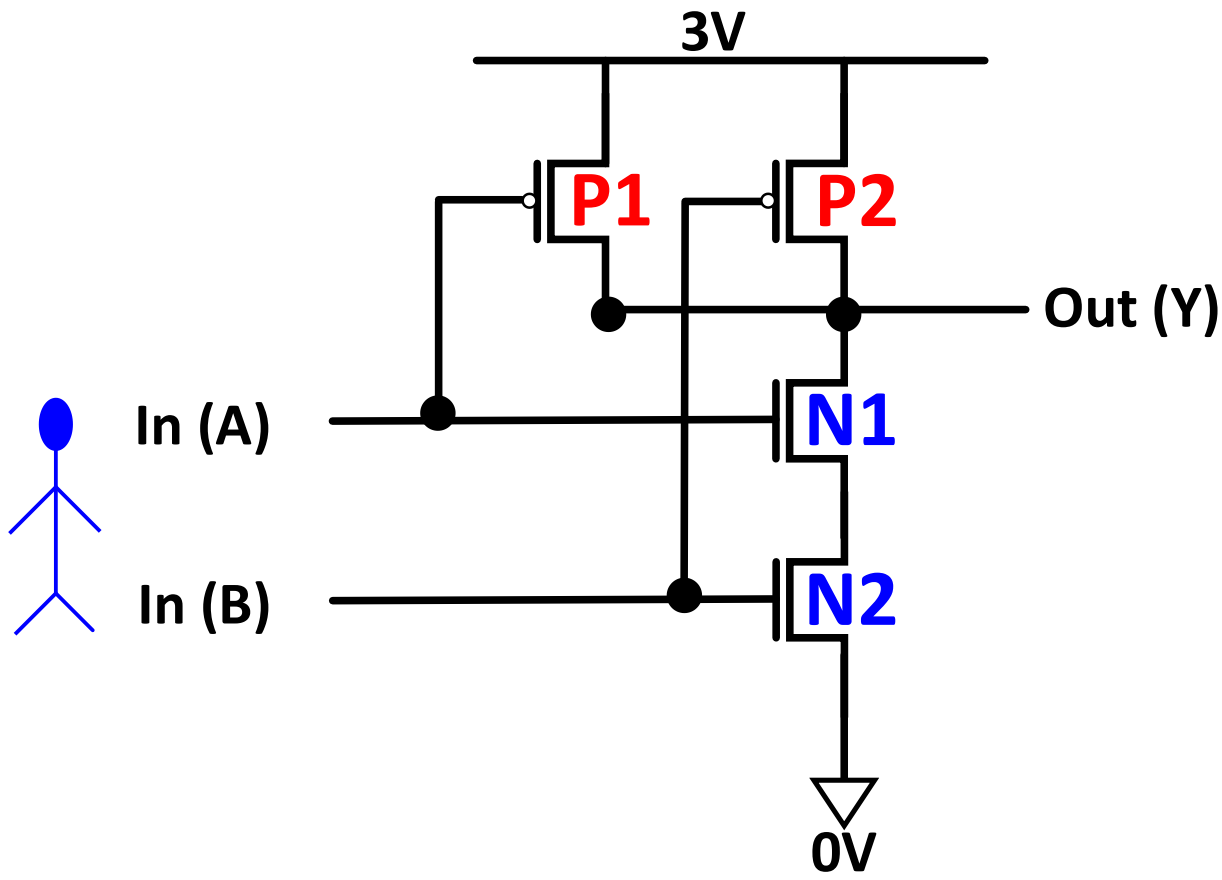
**Truth table:** shows the logical output of the circuit for each possible input

| A | Y |
|---|---|
| 0 | 1 |
| 1 | 0 |

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# Another CMOS Gate: What Is This?

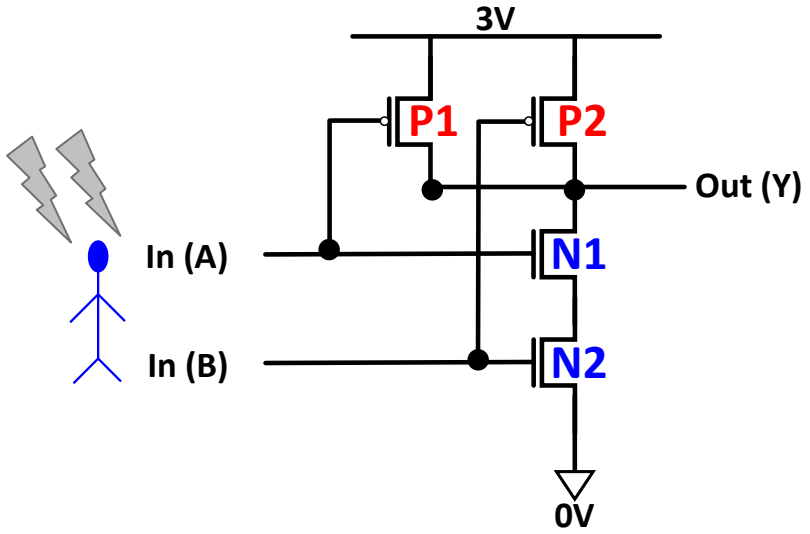
› Let's build more complex gates!



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# CMOS NAND Gate

› Let's build more complex gates!




$$Y = \overline{A \cdot B} = \overline{AB}$$

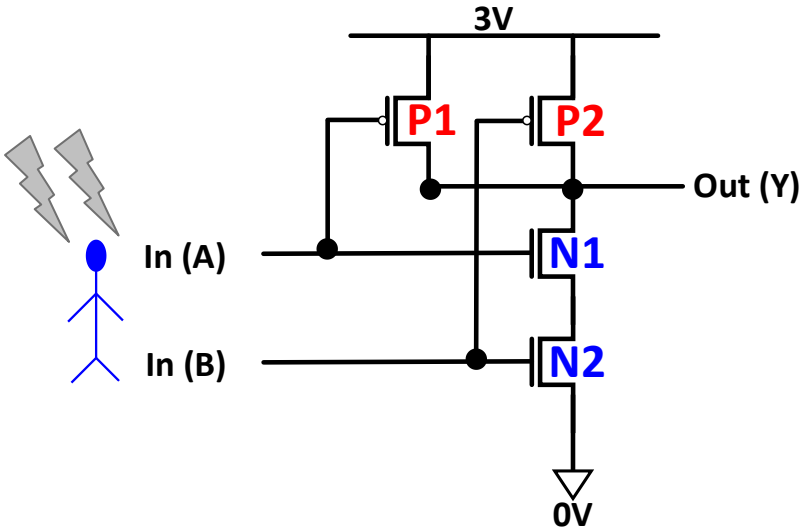
| A | B | P1 | P2 | N1 | N2 | Y |
|---|---|---|---|---|---|---|
| 0 | 0 | **ON** | **ON** | OFF | OFF | 1 |
| 0 | 1 | **ON** | OFF | OFF | **ON** | 1 |
| 1 | 0 | OFF | **ON** | **ON** | OFF | 1 |
| 1 | 1 | OFF | OFF | **ON** | **ON** | 0 |

- P1 and P2 are in parallel; **only one must be ON to pull up the output to 3V**
- N1 and N2 are connected in series; **both must be ON to pull down the output to 0V**

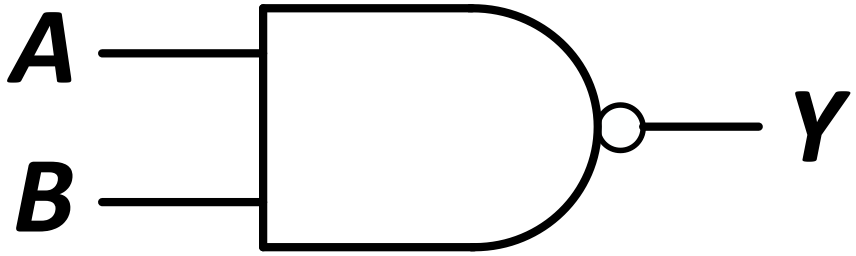
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# CMOS NAND Gate

› Let's build more complex gates!




$$Y = \overline{A \cdot B} = \overline{AB}$$




| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

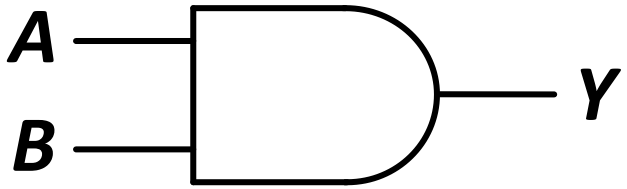
We call this a NAND gate
(bubble indicates inversion)

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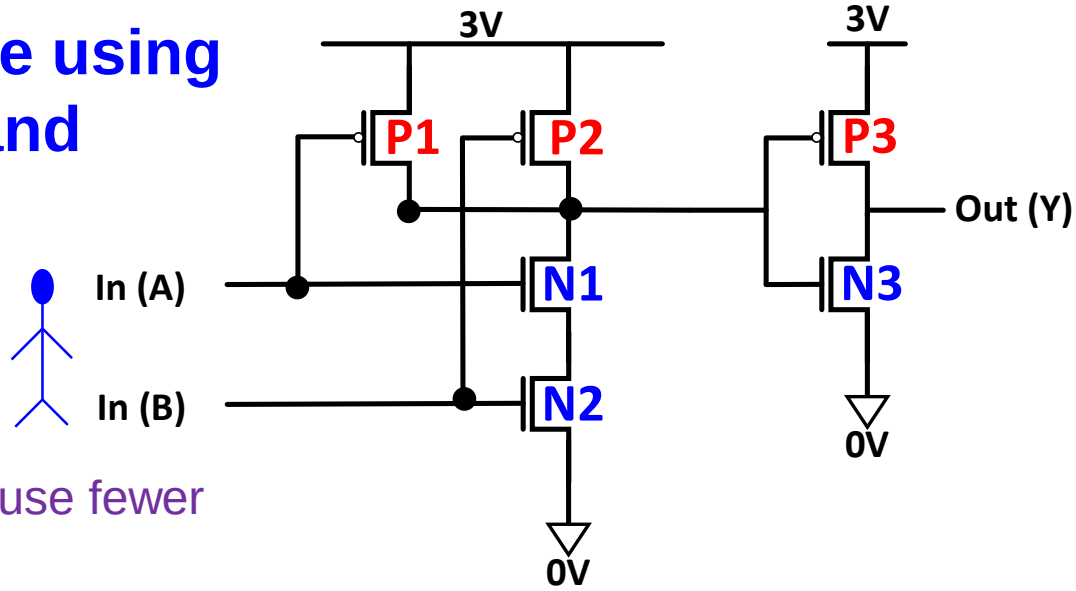
# CMOS AND Gate

› How can we make an AND gate? $Y = A \cdot B = AB$

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |



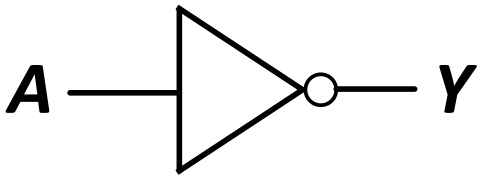

**We make an AND gate using one NAND gate and one NOT gate**



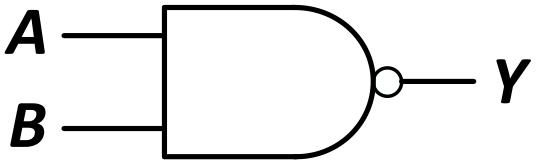

Food for thought: Can we not use fewer transistors for the AND gate?

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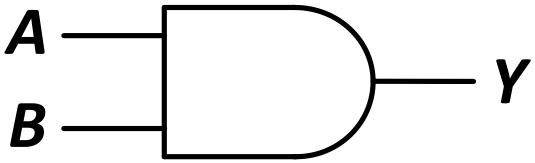
# CMOS NOT, NAND, AND Gates



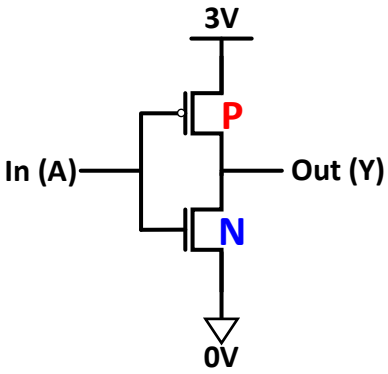

| A | Y |
|---|---|
| 0 | 1 |
| 1 | 0 |

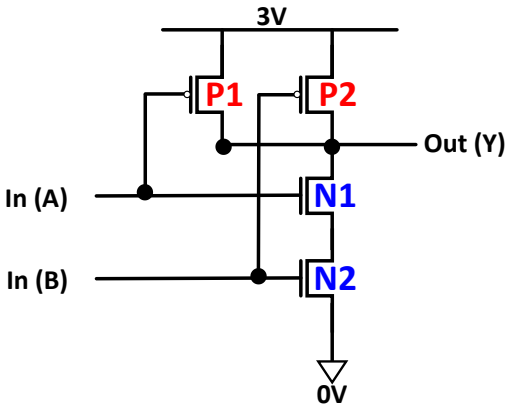


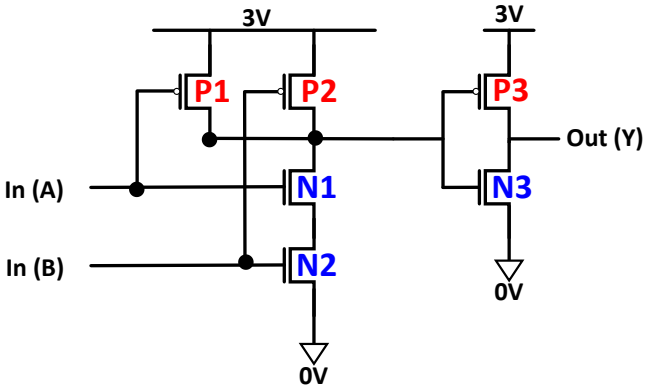

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |




| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |



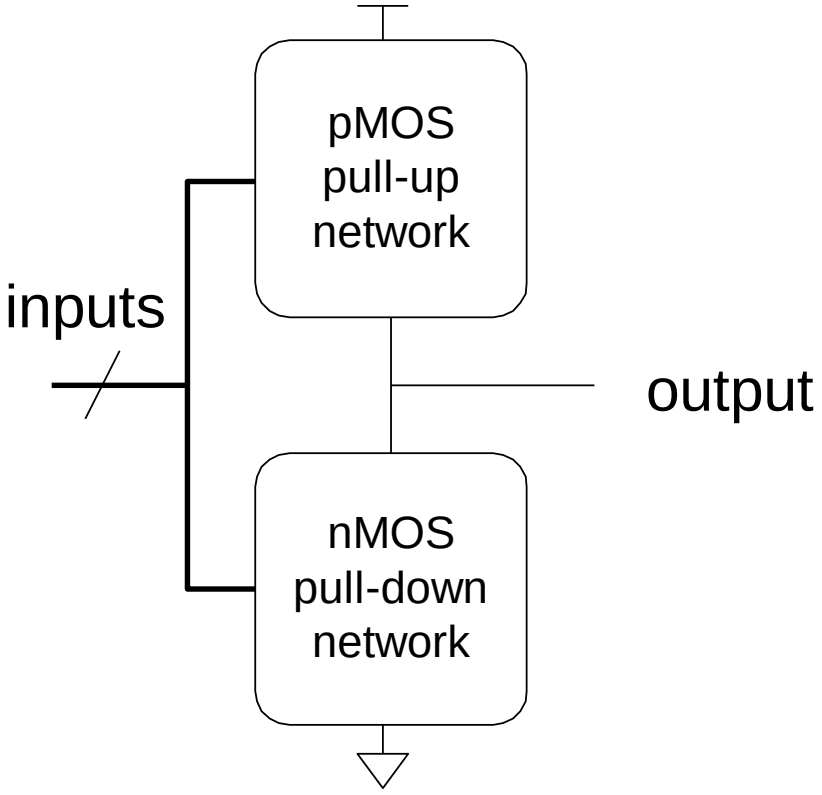



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# General CMOS Gate Structure

- The general form used to construct any inverting logic gate, such as: NOT, NAND, or NOR
  - The networks may consist of transistors in series or in parallel
  - When transistors are in parallel, the network is **ON** if one of the transistors is **ON**
  - When transistors are in series, the network is **ON** only if all transistors are **ON**

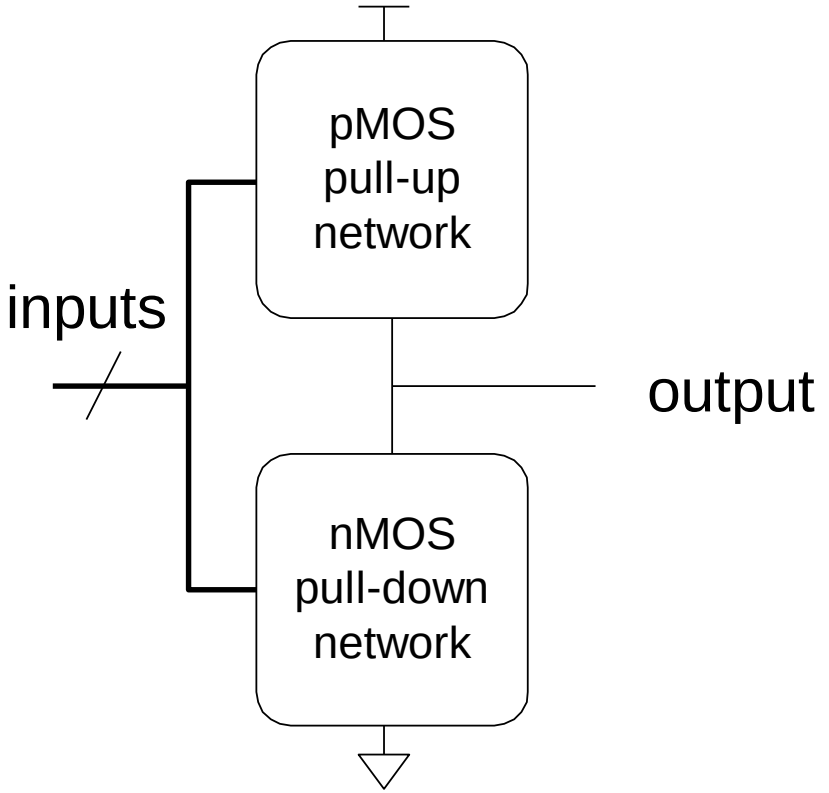



**p**MOS transistors are used for pull-u**p**
**n**MOS transistors are used for pull-dow**n**

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# General CMOS Gate Structure (II)

- Exactly one network should be ON, and the other network should be OFF at any given time
  - If both networks are ON at the same time, there is a short circuit → likely incorrect operation
  - If both networks are OFF at the same time, the output is floating → undefined




**p**MOS transistors are used for pull-u**p**
**n**MOS transistors are used for pull-dow**n**

# Organizational Issues

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# Course structure (2023-2024)

- **Sixteen** (16) **lectures**
  - Roughly following Chapters 1 to 8 in the book,
  - Some additional information not covered in the book
  - Final review and Q&A lecture on Tuesday, January 12, 2027
- **Seven** (7) **tutorials**
  - Every Wednesday 15:00-17:00 from November 18, 2026
  - For receiving guidance by the TAs
- **Seven** (7) **lab sessions** (Thursday)
  - For receiving feedback from the TAs
- **Seven** (3) **bonus homework assignments**
- Slides on Brightspace
  - New relevant material will be added when necessary

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# Assignment delivery

- Submit your assignments in PDF format
- Solutions are to be uploaded in time
  - Schedule is communicated in the “Welcome Email”
- Grading results will be on Brightspace
  - feedback is also provided on **Brightspace**
- 1 bonus point for each assingnment
  - Assignment 1 – 60% passing
  - Assignment 2 – 60% passing
  - Assignment 3 – 40% passing
  - You can potentially core 13 out of 10 – however, the final grade is to be capped at 10.

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# Assignment delivery (continued)

- What constitutes *unacceptable* delivery?
  - Delivery **after** the deadline
  - **Plagiarism** of another student answers (penalty applies to **all students involved**, disputes to be resolved by BoE)
  - The submission is **unreadable** (e.g., as a failing PDF)
  - The submission documents contain any items that are **scanned from notes or photographed**, e.g., no handwritten solutions will be accepted (use Word/LaTeX!)

# Corner cases

- Exam/lab grades from the last year **can't be** used
  - **Repeating students** should perform all homework as everybody else
  - Please contact me if you have questions about this
- Assignment grades are valid for the same year's resit

# Questions about the organization?

